How to Use AI with SQL Databases

Using ChatGPT with **Oracle 26ai** and SELECT AI

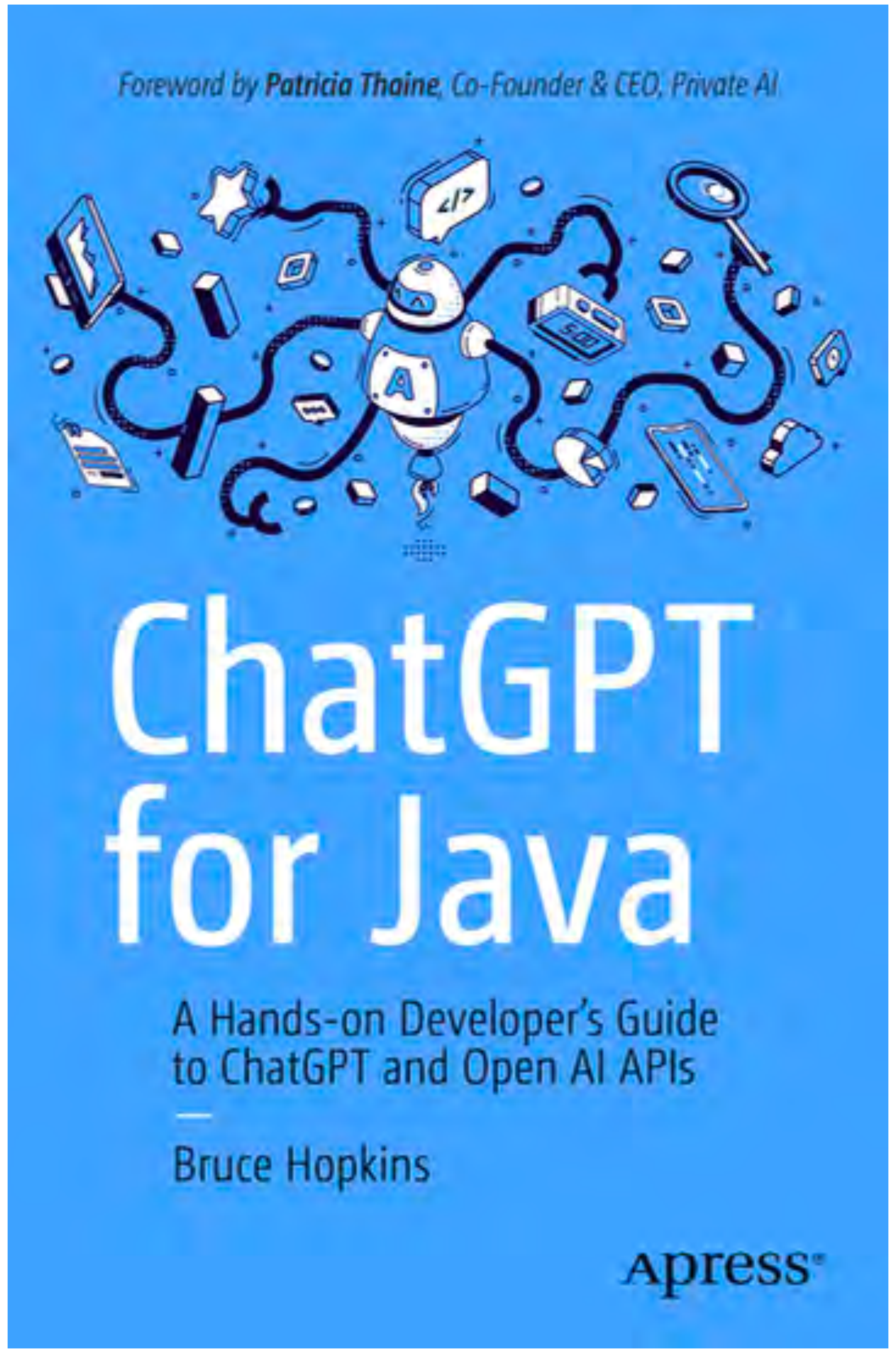
Bruce Hopkins

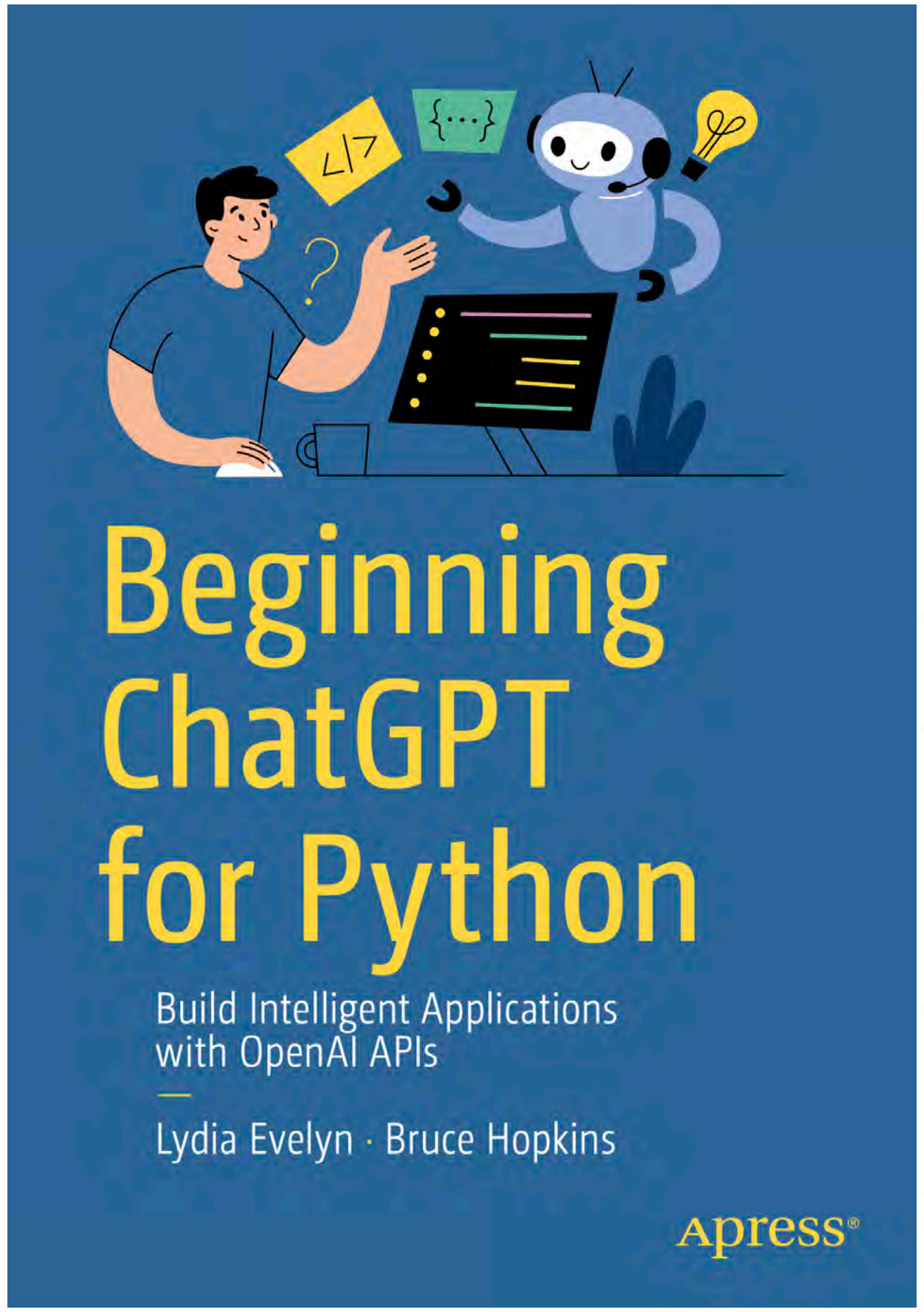
Author, “ChatGPT for Java”, 2024

Pearson

About the presenter

- Intel Software Innovator for AI
- **Oracle** Java Champion
- Book Author





OpenAI Forum

Java Champions

Intel Software Innovator

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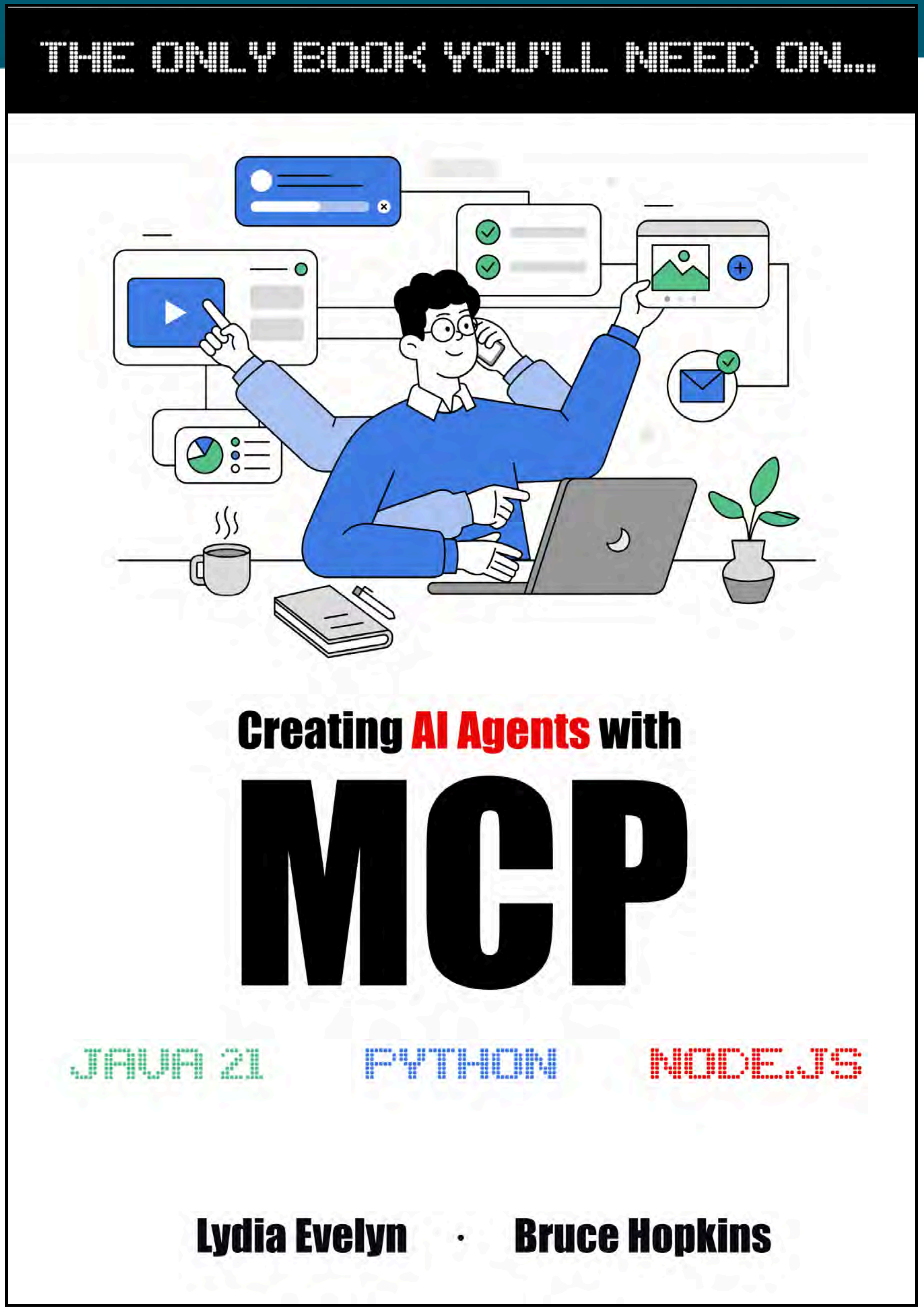
Next Book



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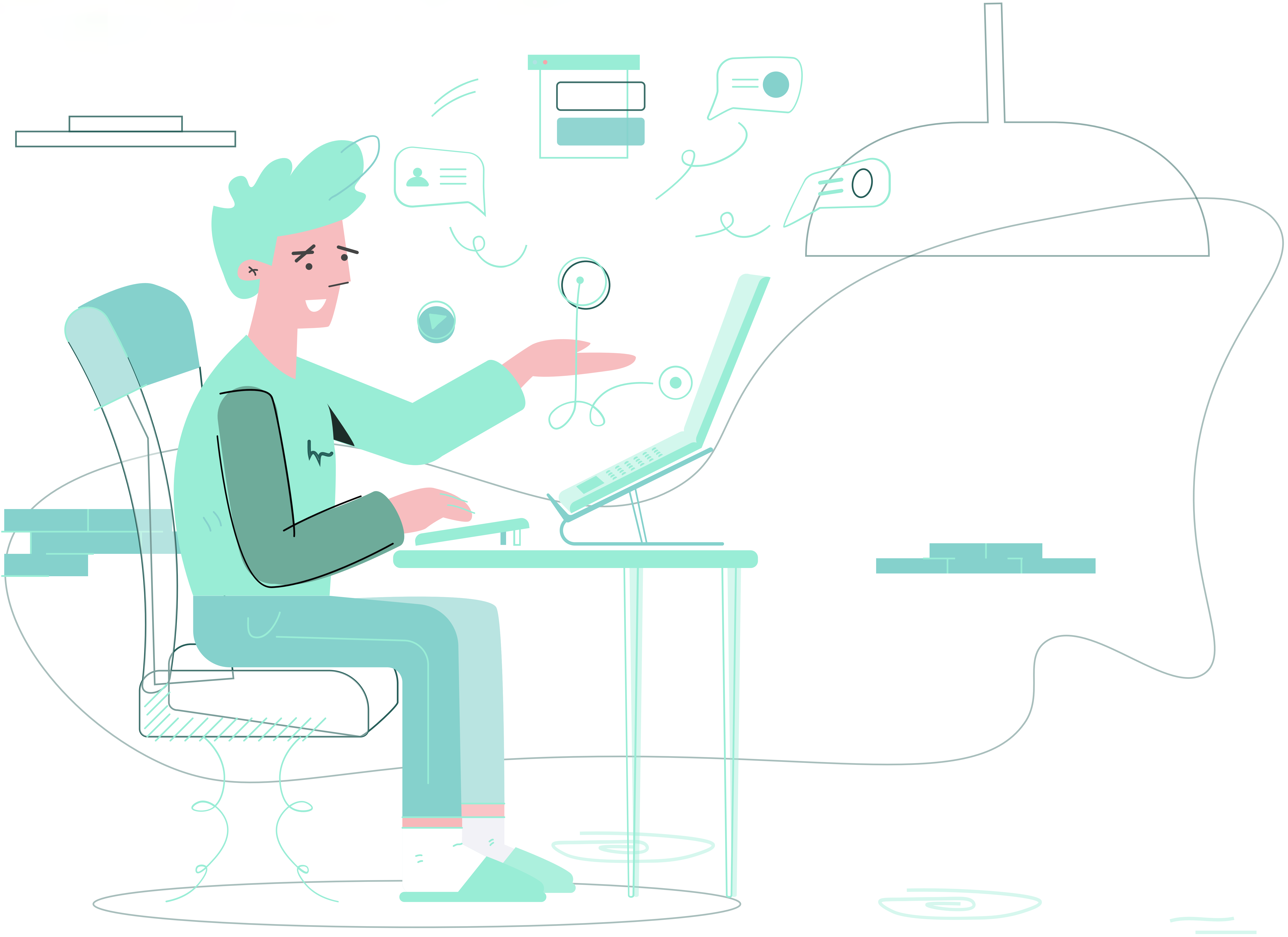
GitHub Repo

https://github.com/BruceTraining/Generative-AI-Databases-Oracle-23ai-SELECT-AI

Audience Poll Question #1



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Audience Poll Question #1

Which category best suits your industry? (Single response)

- Technology and IT
- Manufacturing
- E-commerce and retail
- Finance
- Government/education
- Personal/other

PART 1

Introducing Generative AI for SQL Databases

Course Overview

- 1: Getting Started with Oracle ADB 26ai
- 2: Automating Schema Creation with AI and ChatGPT
- 3: Working with structured data in the enterprise: ETL flows with AI

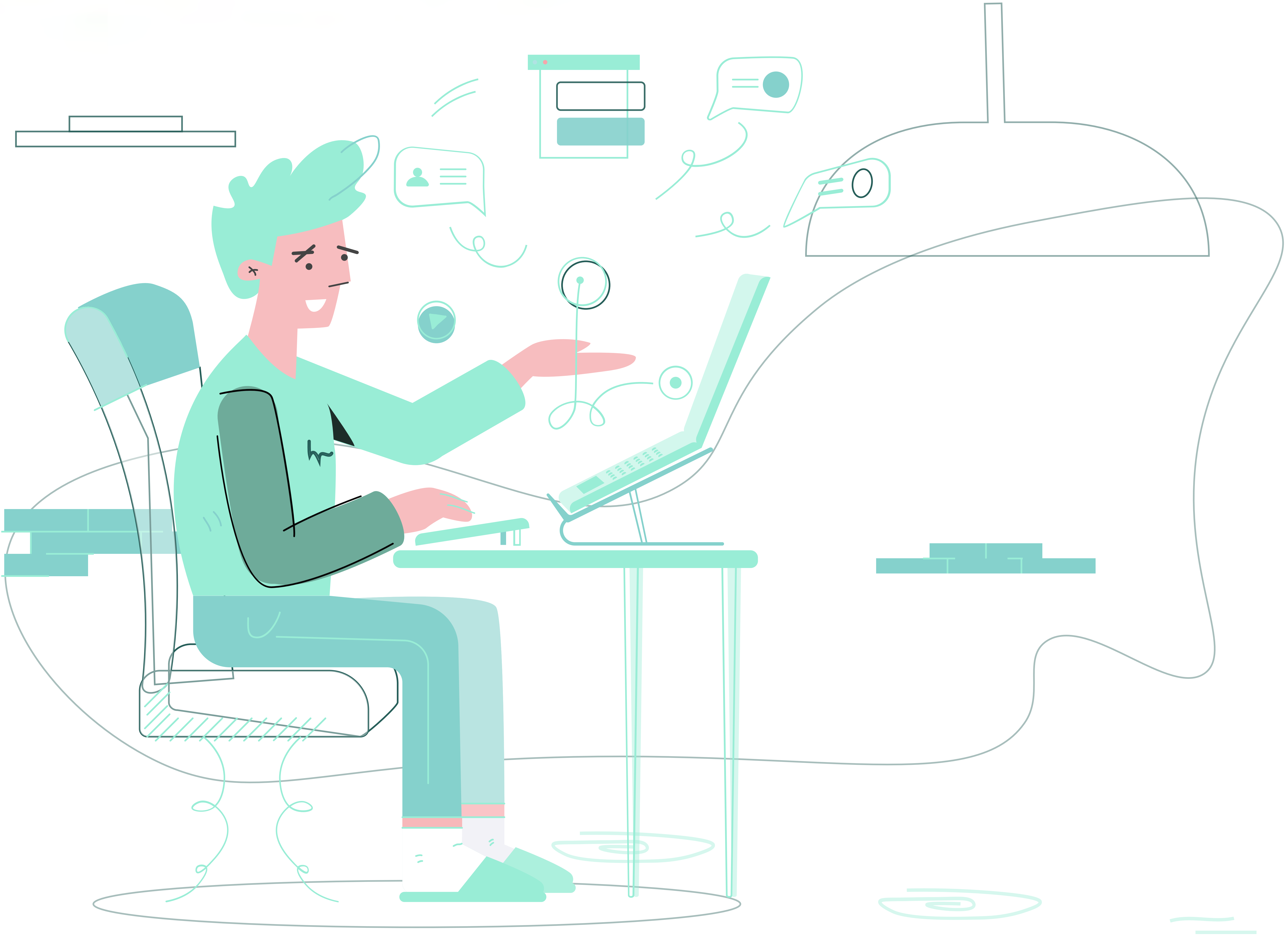
Course Overview

- 4: Managing Unstructured Data: LOB datatypes
- 5: Getting started with SELECT AI in Oracle Autonomous Database 26ai
- 6: Hands-on Enabling AI Features in Oracle Autonomous Database 26ai

Focus

Oracle ADB 26ai

Audience Poll Question #2



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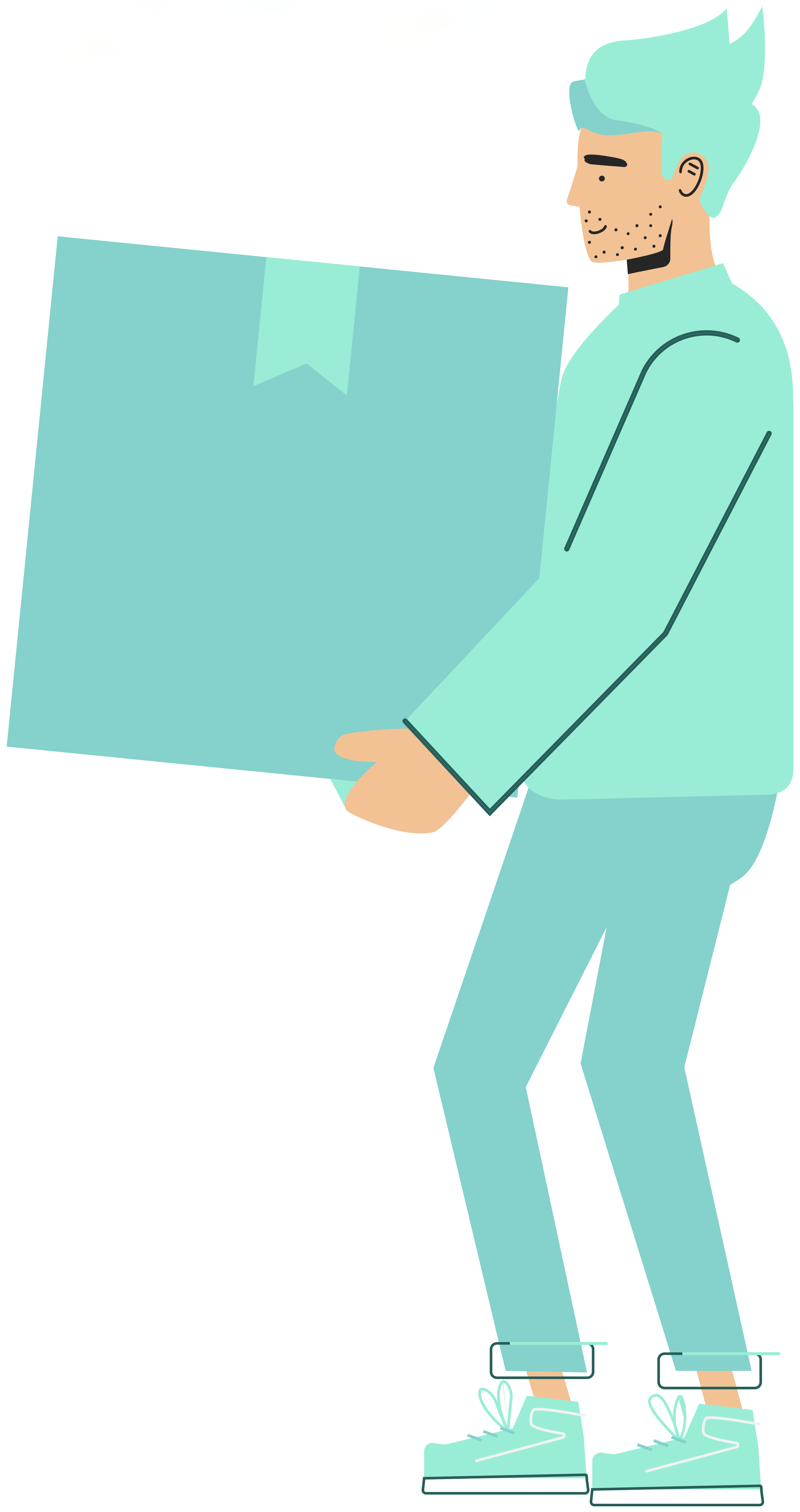
Audience Poll Question #2

What is your preferred database? (Single response)

- Oracle
- PostgreSQL
- MS SQL Server
- Mongo DB
- Maria DB
- Other

What are the Key Takeaways from This Course?



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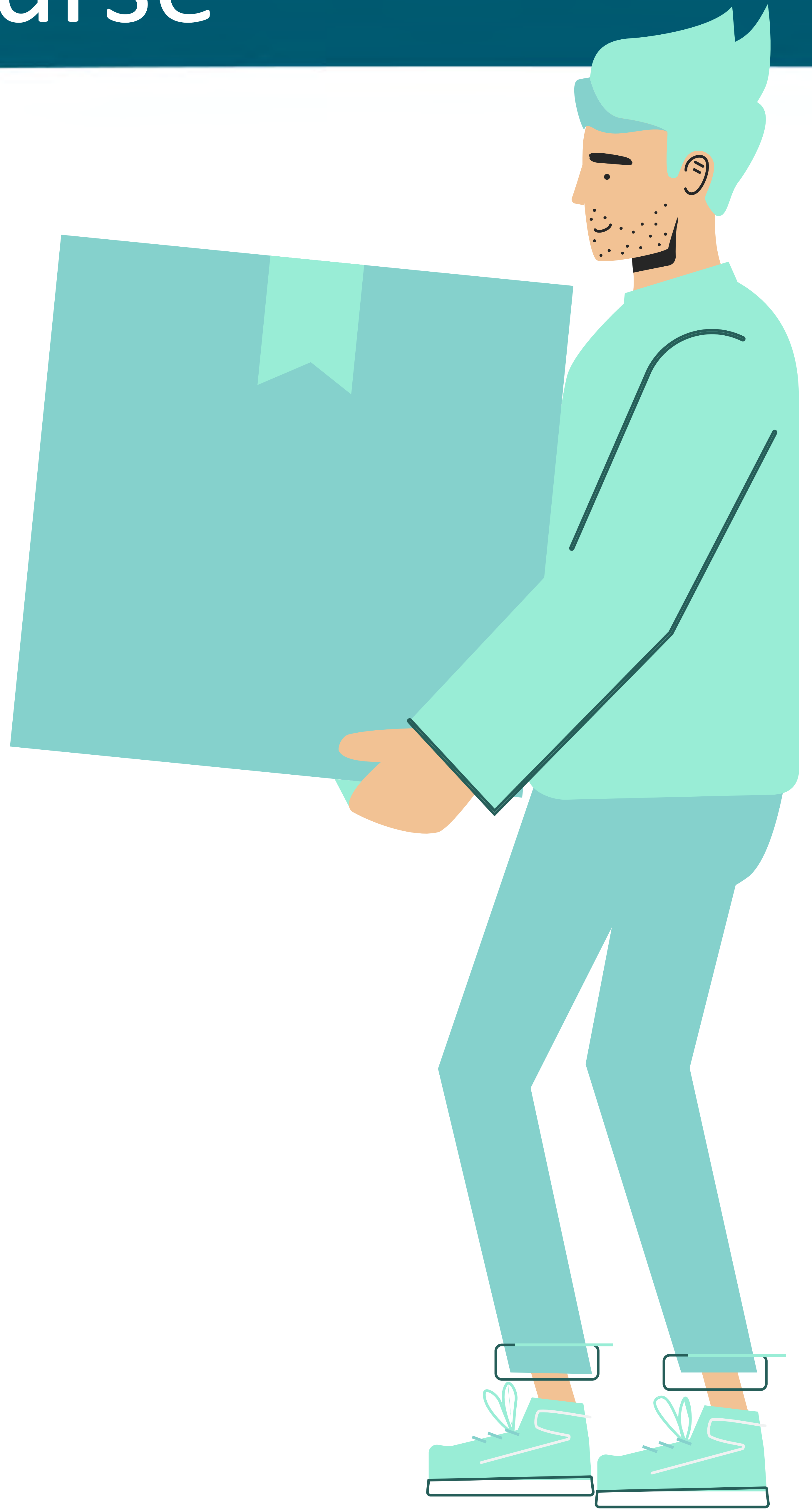
Key Takeaways from This Course

PART 1

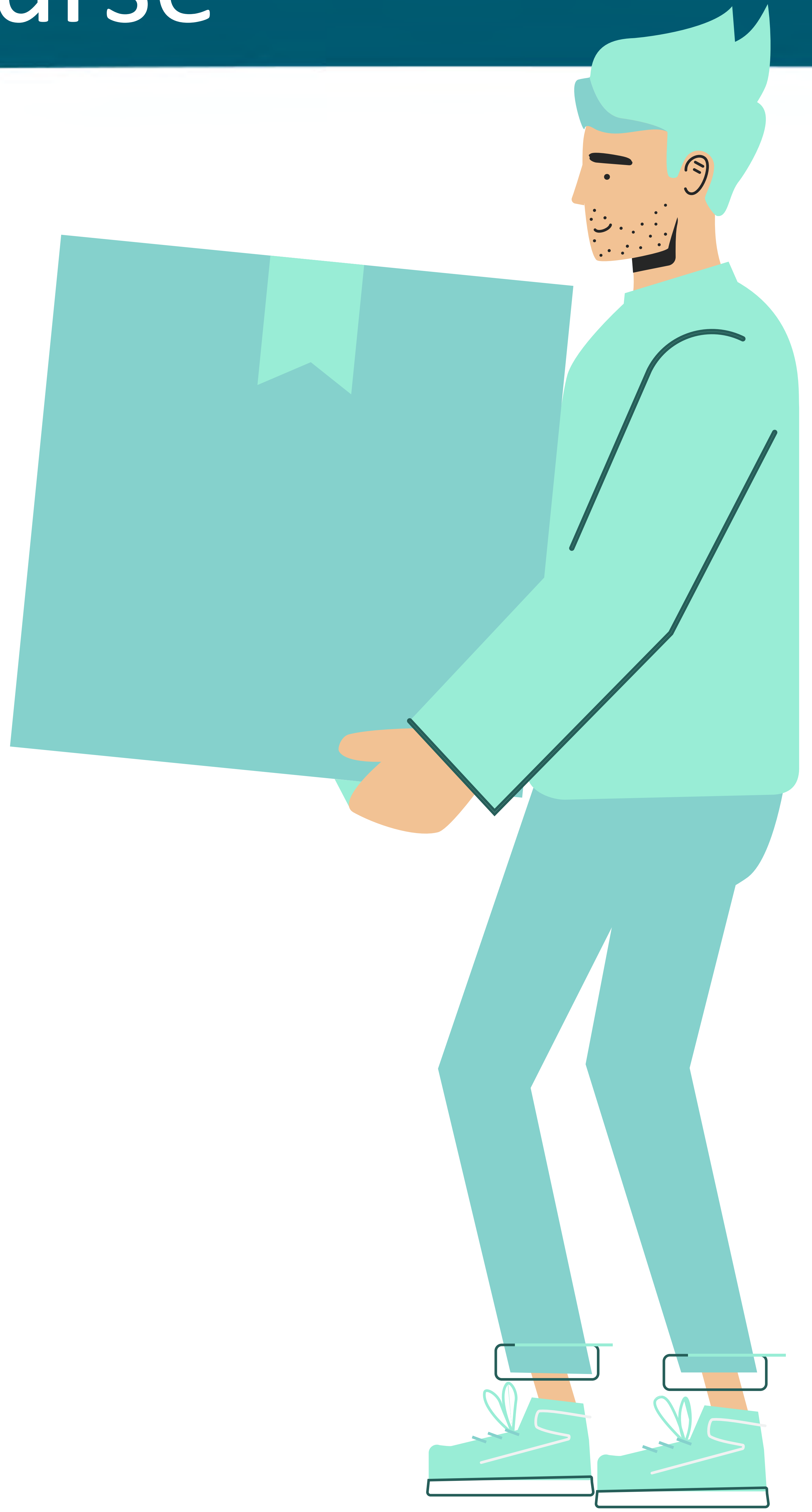
You will be get familiar with the features and capabilities of the Oracle Autonomous Database 23 ai



Key Takeaways from This Course

PART 2

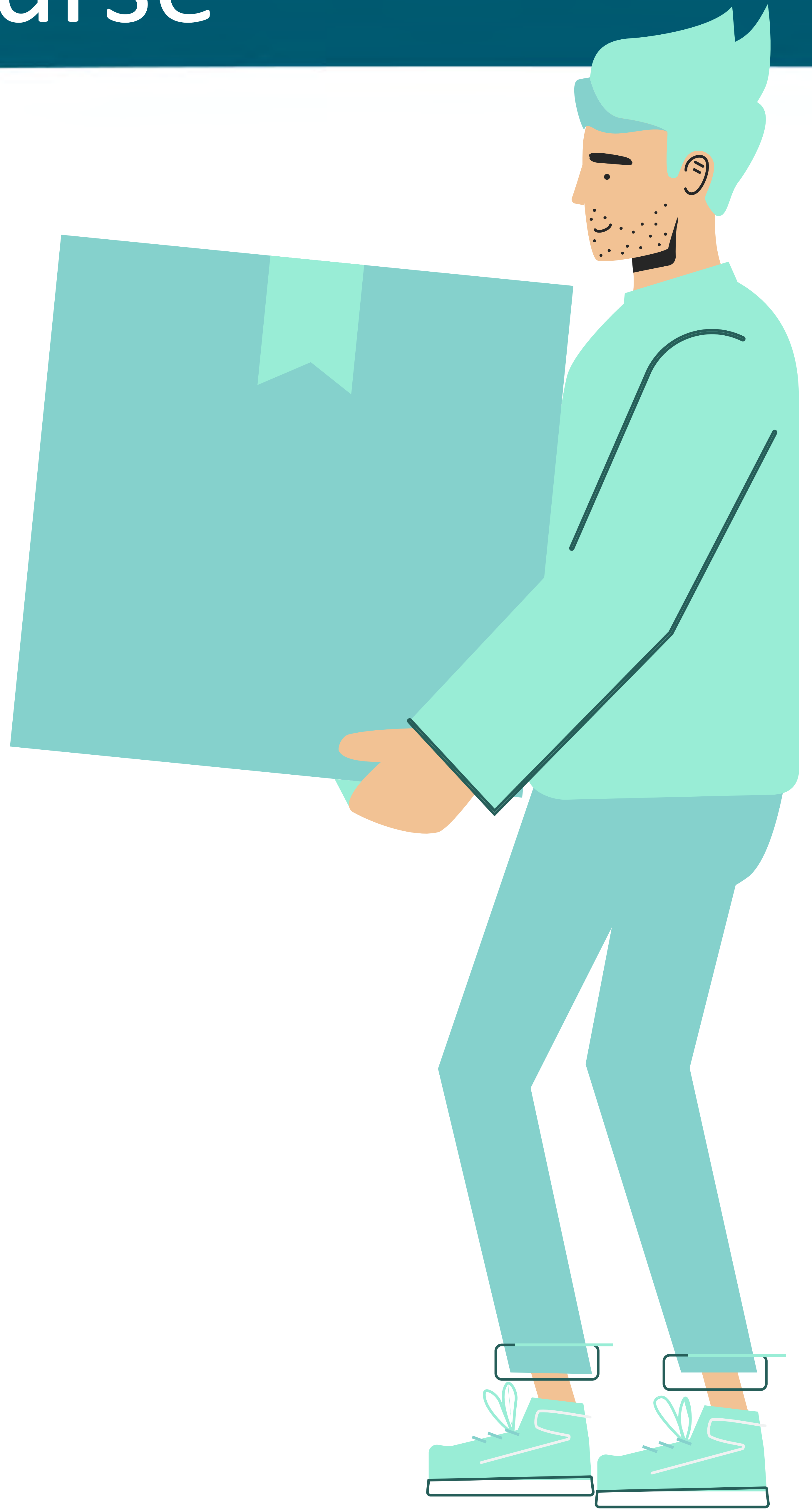
You will learn how to automatically create database schemas for POC type applications



Key Takeaways from This Course

PART 3

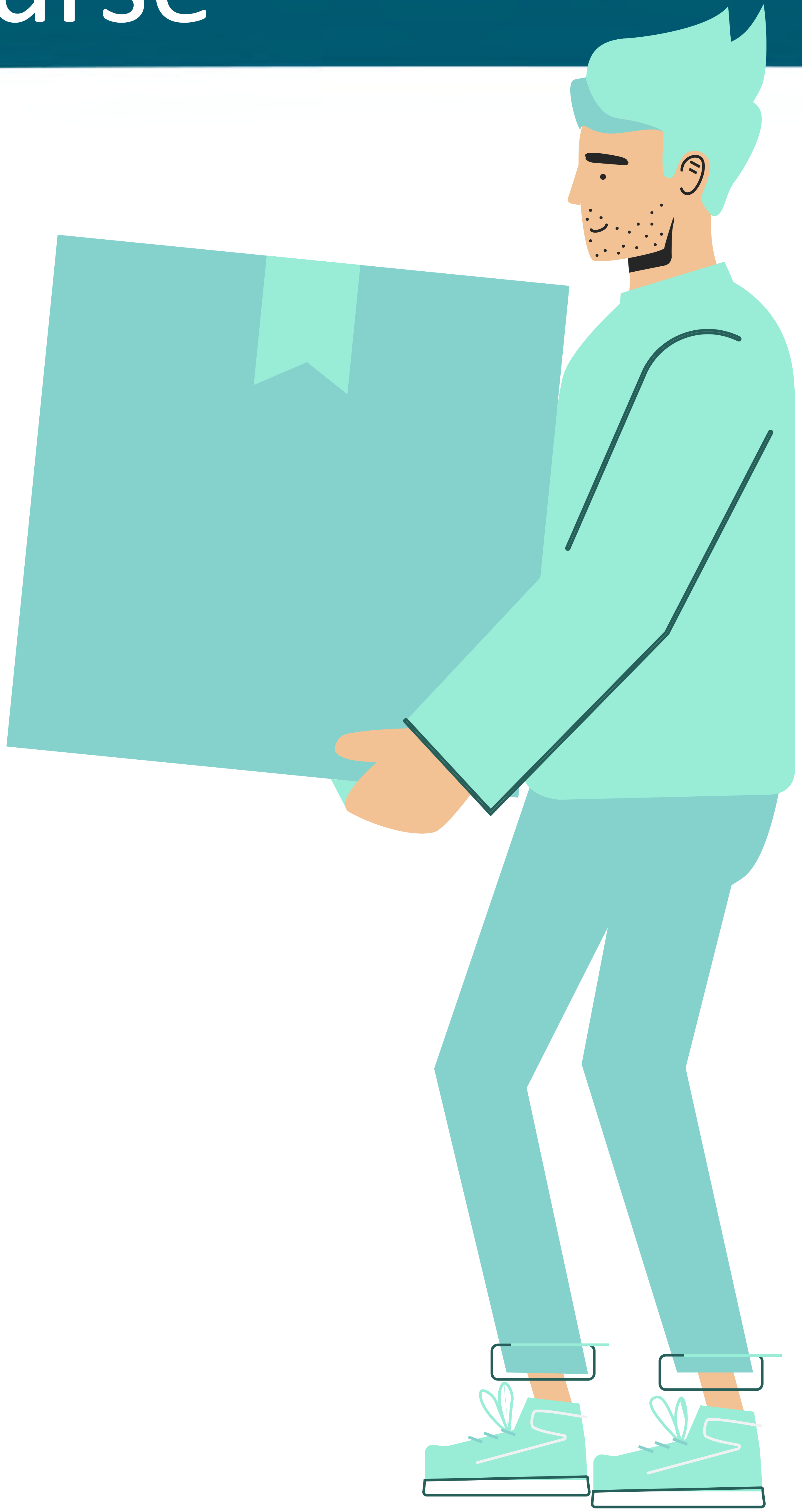
You will be learn how to use AI and ChatGPT effectively in order to extract, transform, and load (ETL) data in your database



Key Takeaways from This Course

PART 4

You will learn how to extract structured data from LOB (Large Object) datatypes



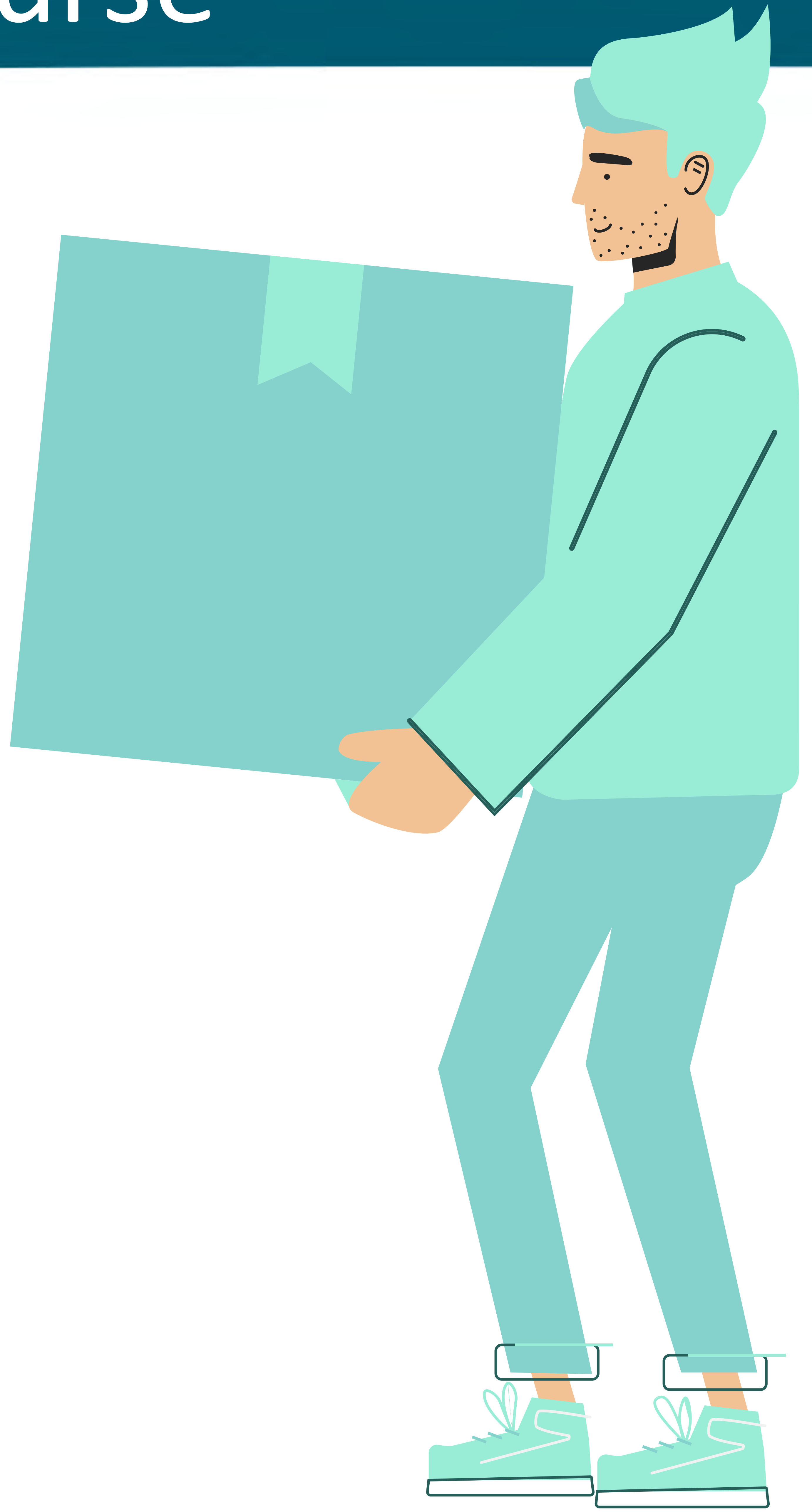
Key Takeaways from This Course

PART 5

You will learn about a powerful new feature of the Oracle ADB 26ai

SELECT AI

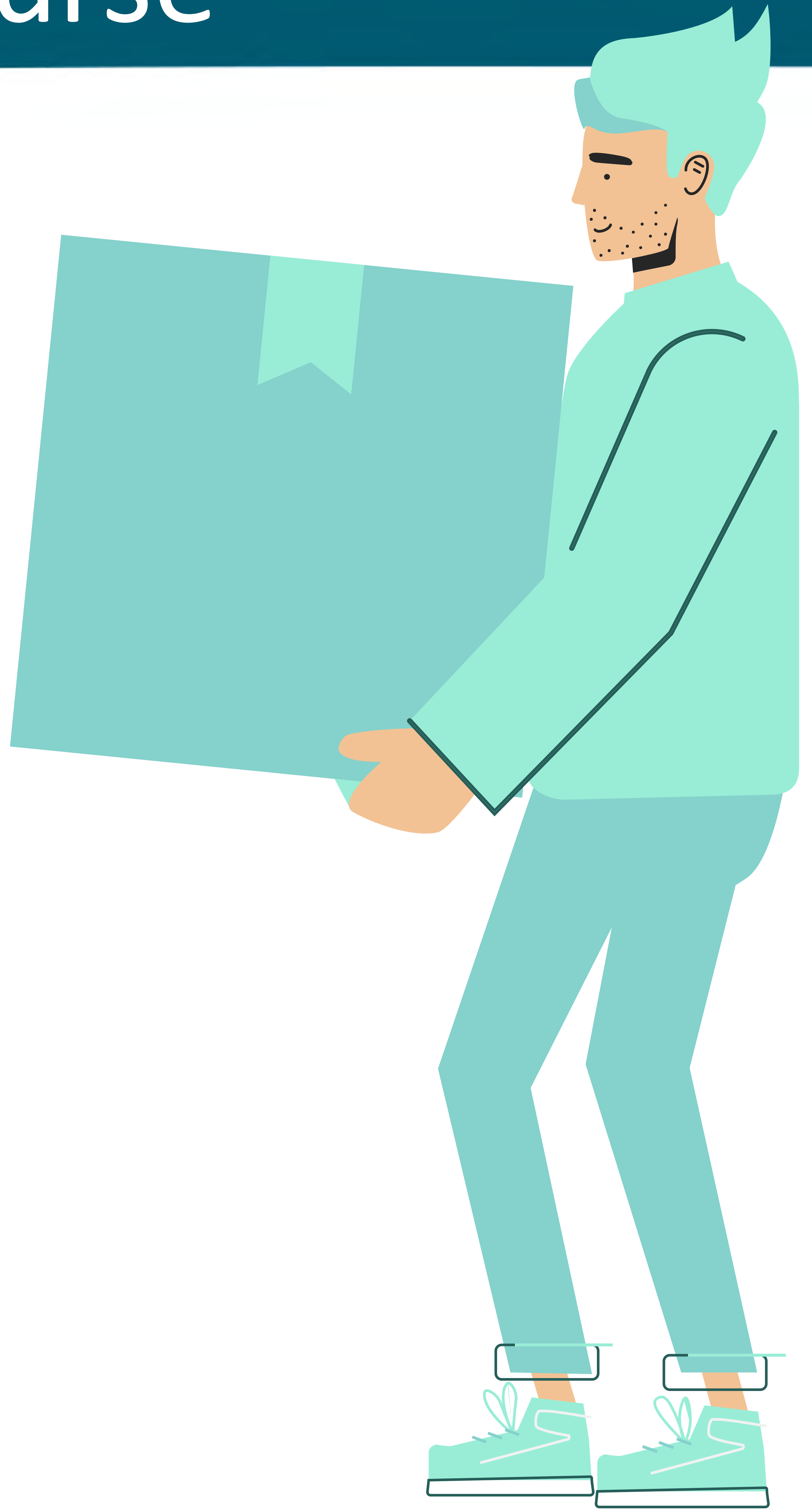
This allows you to execute queries in your database in Natural Language



Key Takeaways from This Course

PART 6

You will learn all the steps necessary to enable SELECT AI within your own database using your preferred AI provider



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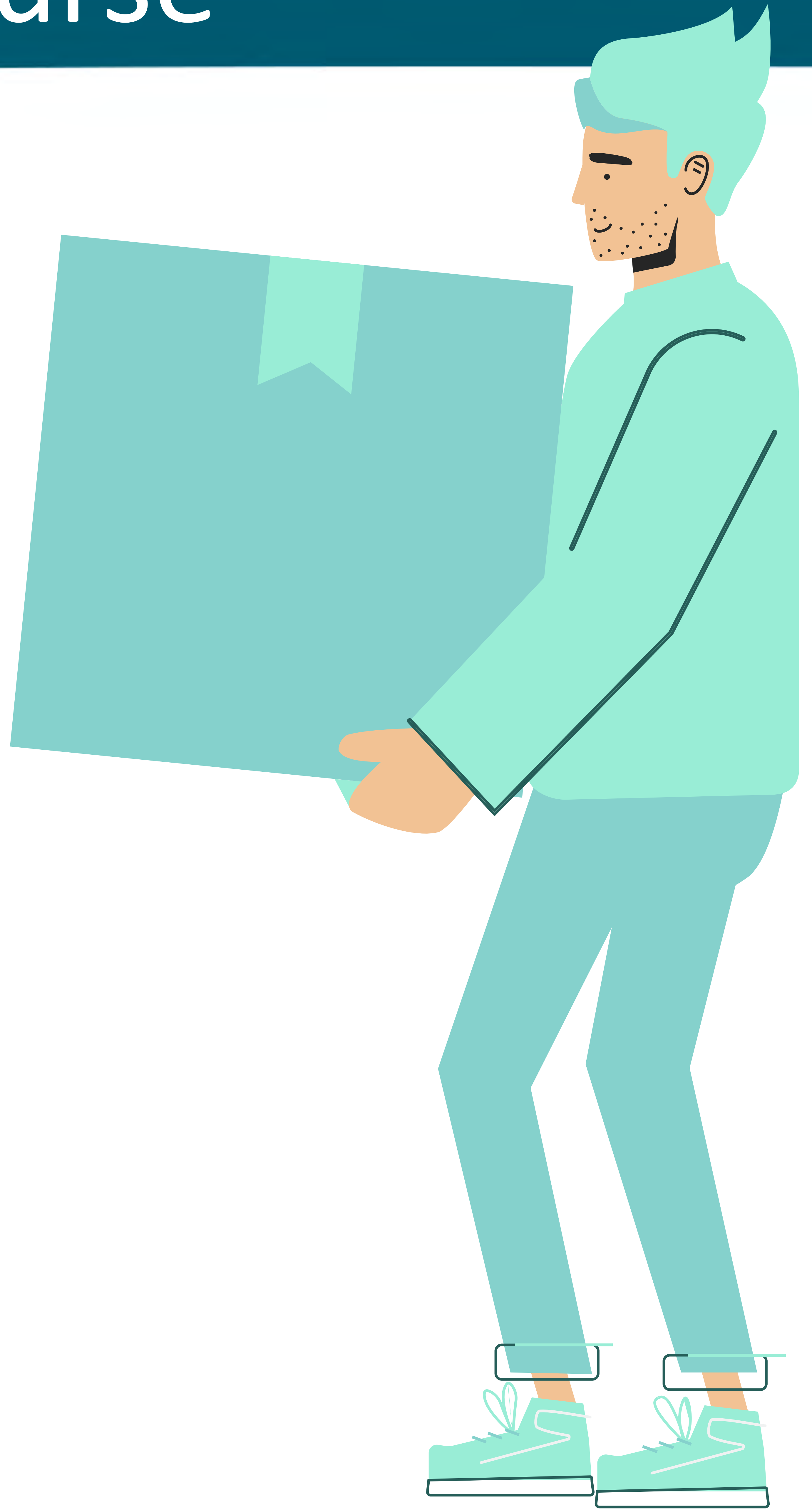
Key Takeaways from This Course

PART 7

You will learn all use MCP to have a natural language conversation with your database



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Type of Open AI Account

Free or Plus?

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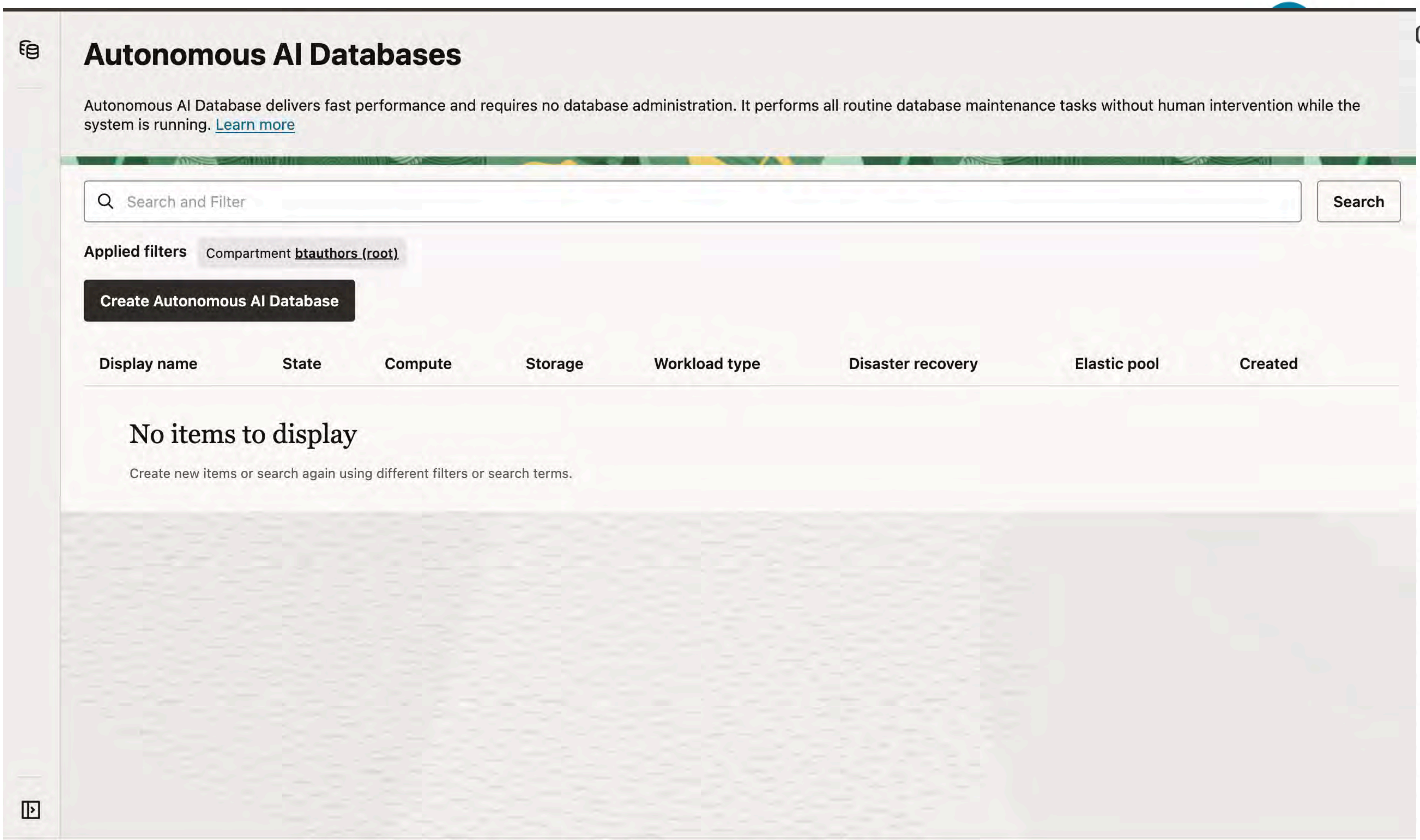
DEMO 1

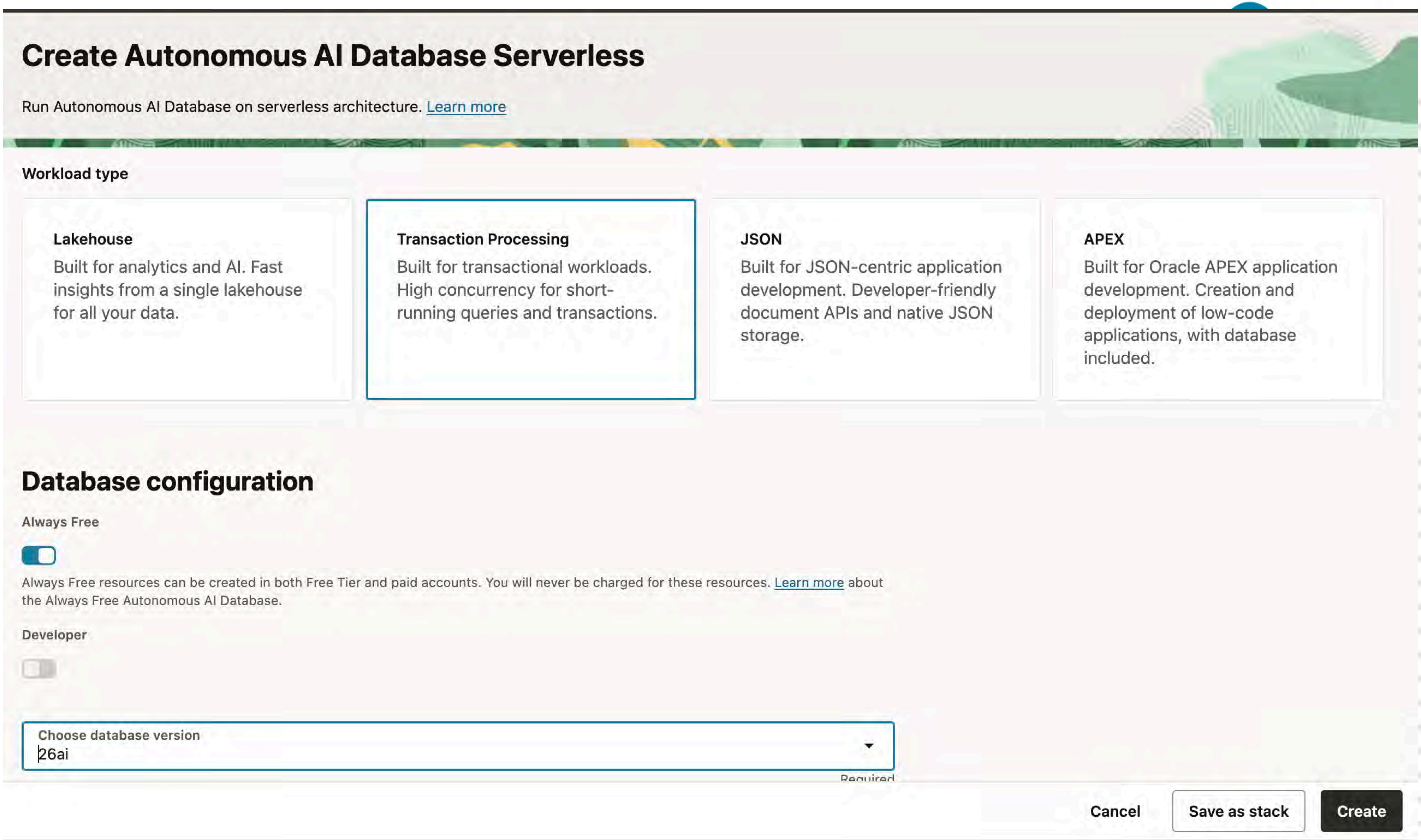
DEMO 1

GOAL:

1. *Login to Oracle Cloud*
2. *Create DW Database*
3. *SQL Worksheet > “Select sysdate from DUAL;”*

on


Autonomous AI Databases
Autonomous AI Database delivers fast performance and requires no database administration. It performs all routine database maintenance tasks without human intervention while the system is running. Learn more
Search and Filter
Search
Applied filters
Compartment btauthors (root)
Create Autonomous AI Database
Display name
State
Compute
Storage
Workload type
Disaster recovery
Elastic pool
Created
No items to display
Create new items or search again using different filters or search terms.


Create Autonomous AI Database Serverless
Run Autonomous AI Database on serverless architecture. Learn more
Workload type
Lakehouse
Built for analytics and AI. Fast insights from a single lakehouse for all your data.
Transaction Processing
Built for transactional workloads. High concurrency for short-running queries and transactions.
JSON
Built for JSON-centric application development. Developer-friendly document APIs and native JSON storage.
APEX
Built for Oracle APEX application development. Creation and deployment of low-code applications, with database included.
Database configuration
Always Free
Always Free resources can be created in both Free Tier and paid accounts. You will never be charged for these resources. Learn more about the Always Free Autonomous AI Database.
Developer
Choose database version
26ai
Required
Cancel
Save as stack
Create

Create Autonomous AI Database Serverless

Run Autonomous AI Database on serverless architecture. Learn more

Network access

Access type
The virtual cloud network option is not available for OCI Free Tier accounts.

Secure access from everywhere
Allow users with database credentials to access the database from the internet.

Secure access from allowed IPs and VCNs only
Restrict access to specified IP addresses and VCNs.

Private endpoint access only
Restrict access to a private endpoint within an OCI VCN.

Require mutual TLS (mTLS) authentication

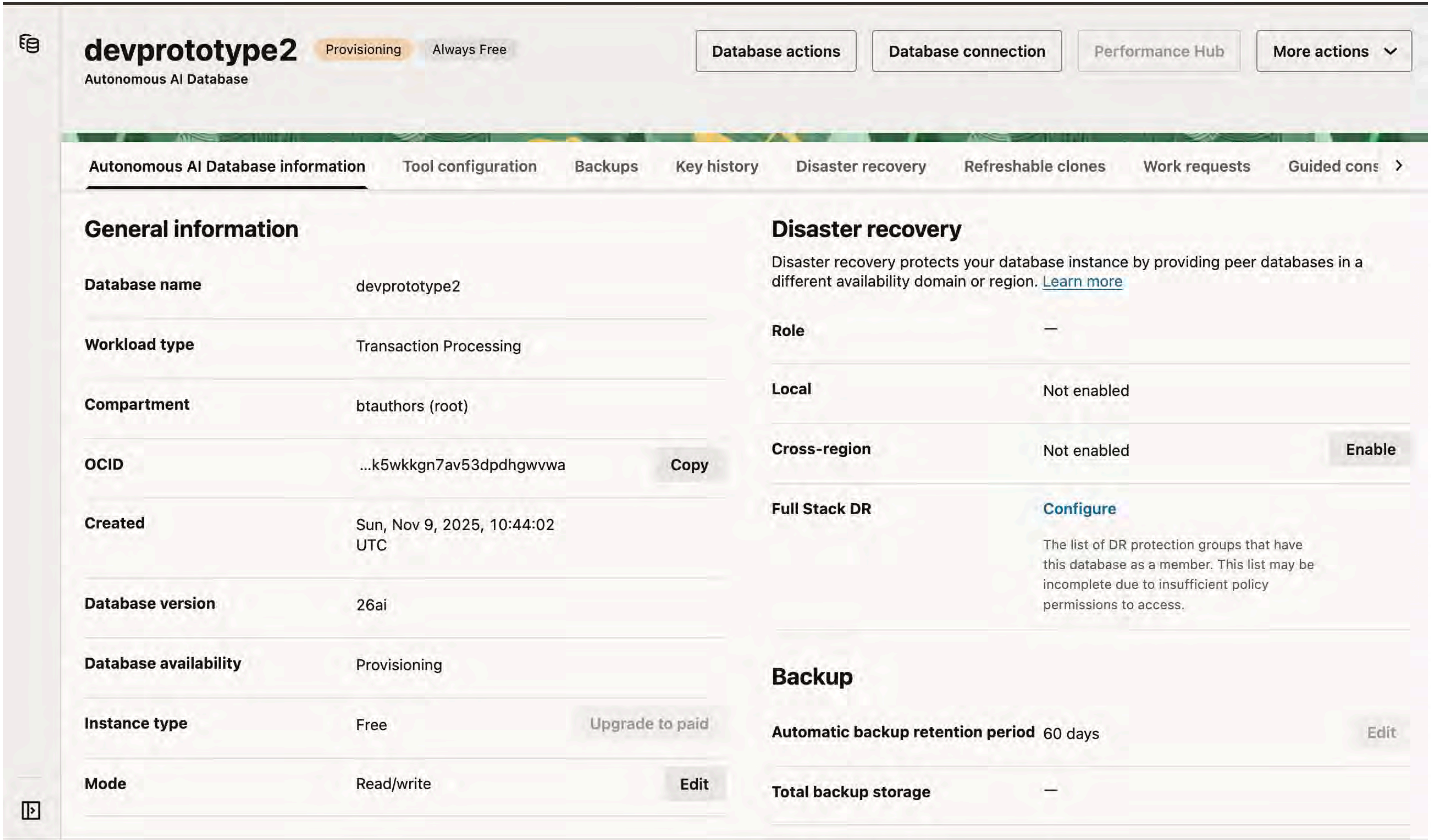
Contacts for operational notifications and announcements

Contacts in this list receive operational notices and announcements as well as unplanned maintenance notifications. Learn more

Contact email

Add customer contact

Cancel | Save as stack | Create


devprototype2
Provisioning
Always Free
Autonomous AI Database
Database actions
Database connection
Performance Hub
More actions
Autonomous AI Database information
Tool configuration
Backups
Key history
Disaster recovery
Refreshable clones
Work requests
Guided cons
General information
Database name
devprototype2
Workload type
Transaction Processing
Compartment
btauthors (root)
OCID
...k5wkkgn7av53dpdhgwvwa
Copy
Created
Sun, Nov 9, 2025, 10:44:02 UTC
Database version
26ai
Database availability
Provisioning
Instance type
Free
Upgrade to paid
Mode
Read/write
Edit
Disaster recovery
Disaster recovery protects your database instance by providing peer databases in a different availability domain or region. Learn more
Role
—
Local
Not enabled
Cross-region
Not enabled
Enable
Full Stack DR
Configure
The list of DR protection groups that have this database as a member. This list may be incomplete due to insufficient policy permissions to access.
Backup
Automatic backup retention period
60 days
Edit
Total backup storage
—

PART 2

Automating Schema Creation with AI and ChatGPT

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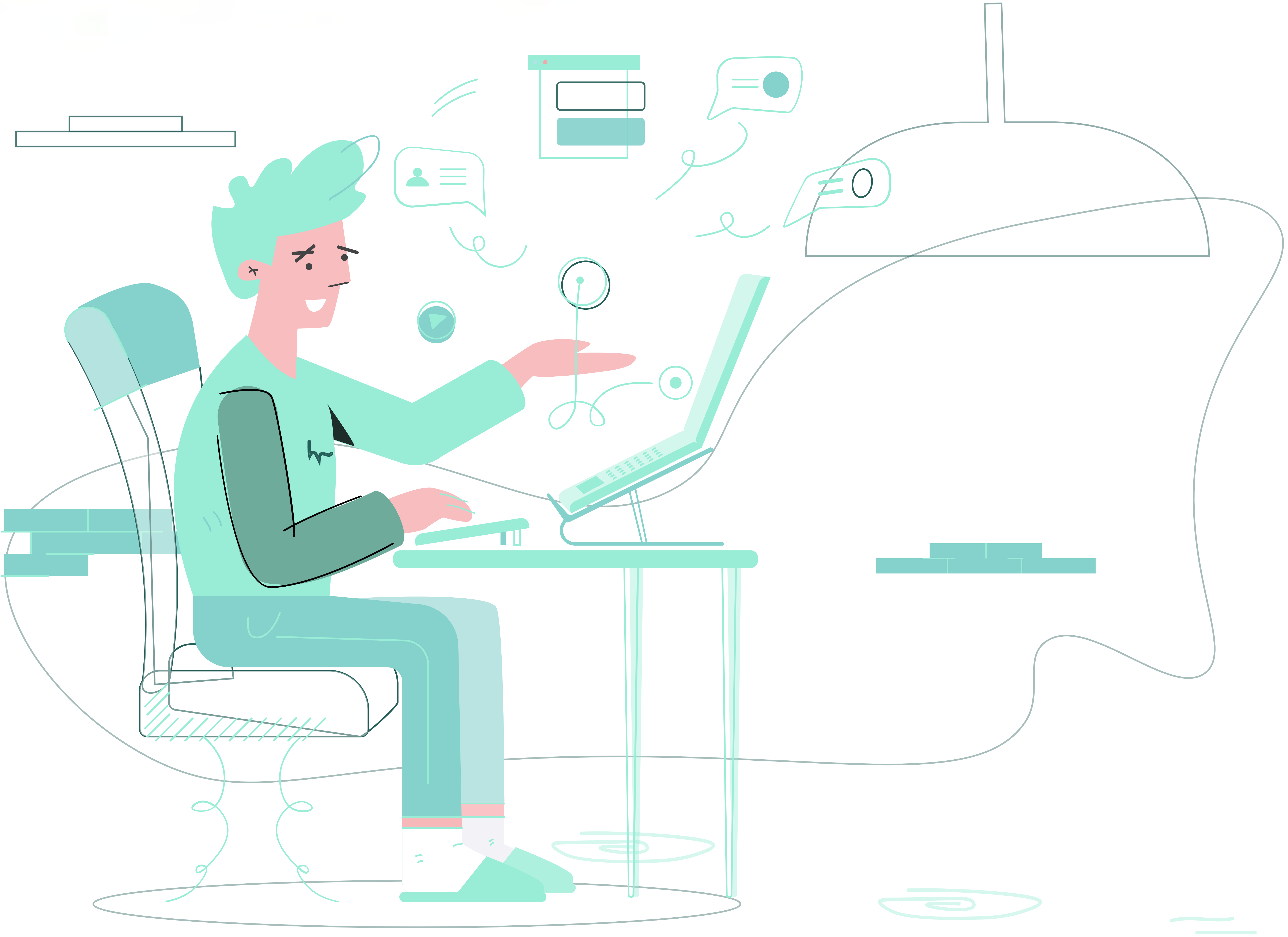
Audience Poll Question #3



Audience Poll Question #3

What best describes your day-to-day tasks? (Single response)

- DBA or DA
- IT Manager
- Software developer
- Executive
- Other

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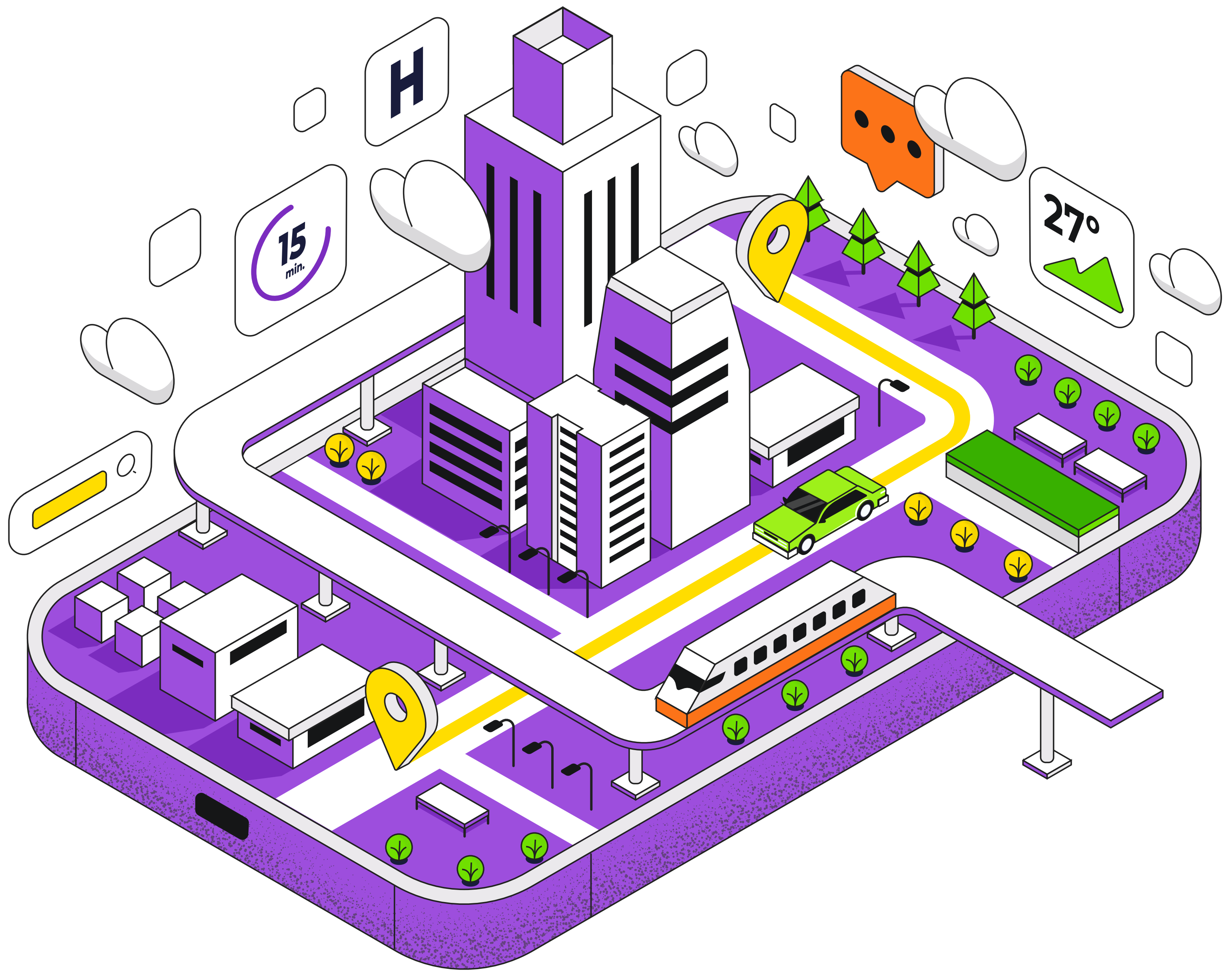
Problem: Need to Launch an App ASAP

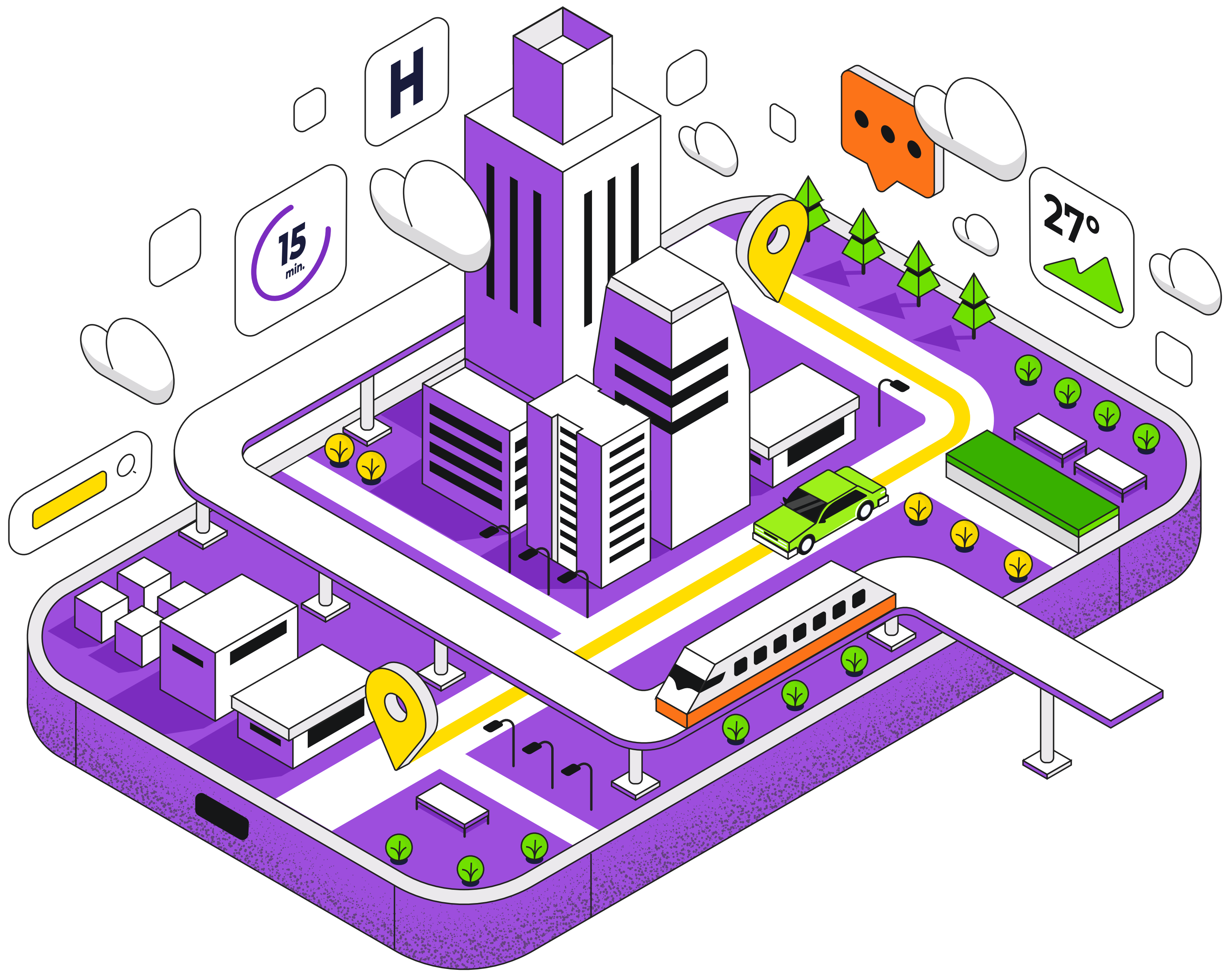
- You're a DBA and you work for a startup



Problem: Need to Launch an App ASAP

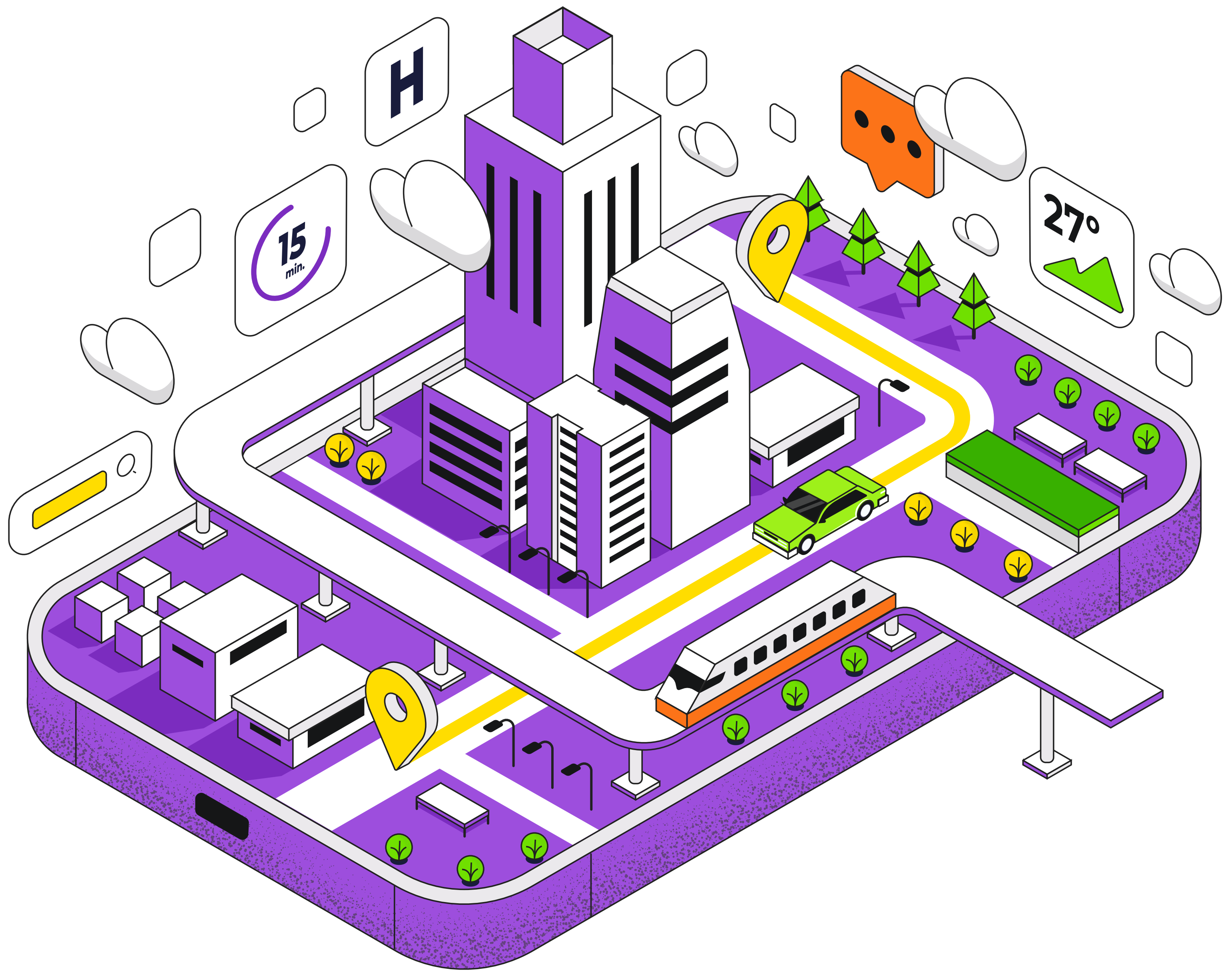
- You're a DBA and you work for a startup

- The startup works as a ride-sharing service



Problem: Need to Launch an App ASAP

- You're a DBA and you work for a startup
- The startup works as a ride-sharing service
- They want to launch a new business with food delivery



Constraints

- Management isn't sure if this is a viable business

Constraints

- Management isn't sure if this is a viable business
- Therefore, they only want to develop a POC

Constraints

- Management isn't sure if this is a viable business
- Therefore, they only want to develop a POC
- The system will NOT be integrated with the rest of the enterprise data

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User Interface

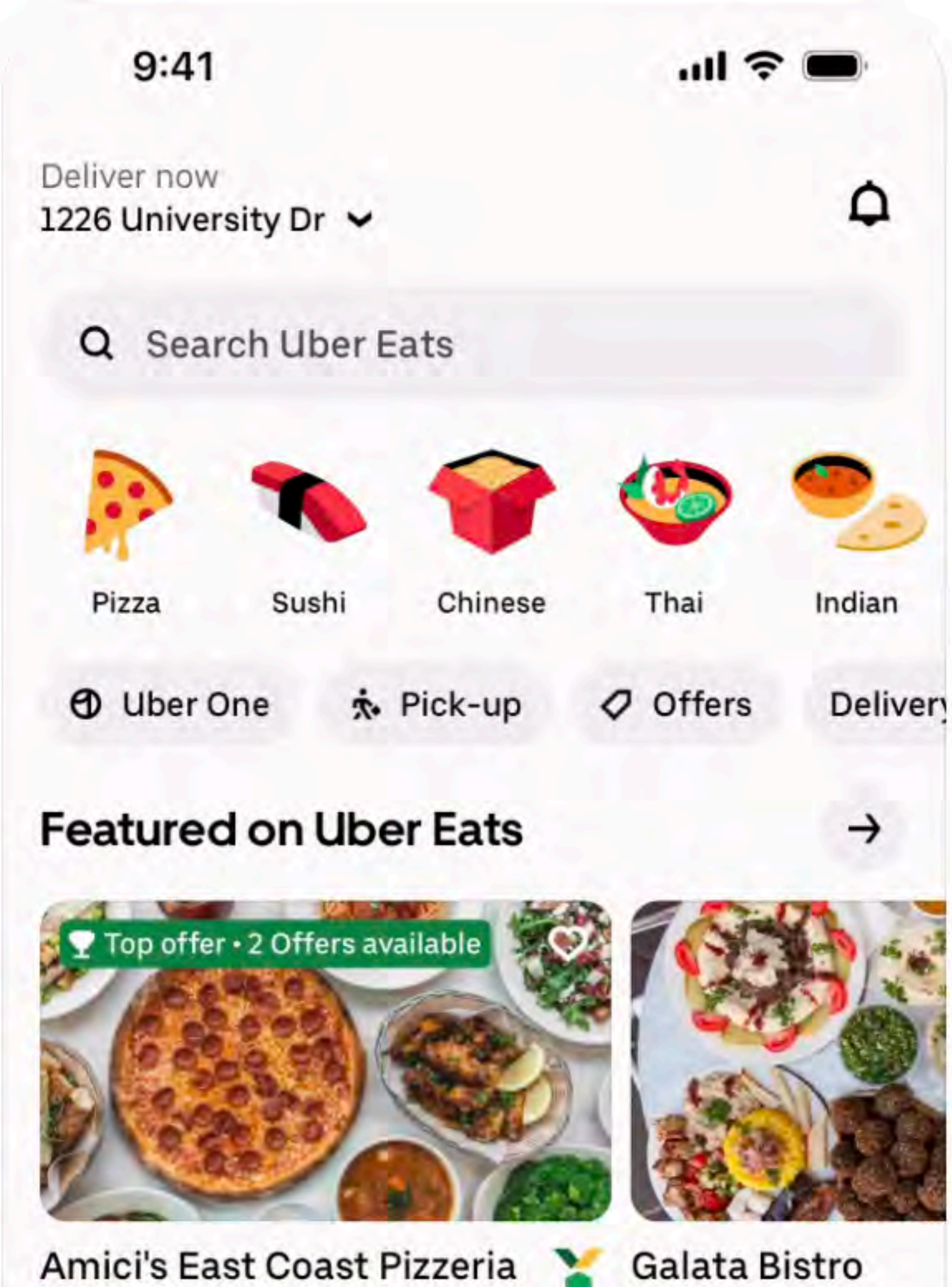
- The original business is Uber

User Interface

- The original business is Uber
- We're making Uber Eats

User Interface

- The original business is Uber
- We're making Uber Eats



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Typical Product Development Cycle

- Product managers gather the requirements

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Typical Product Development Cycle

- Product managers gather the requirements
- UI design team creates the wire frames and UI mockups

Typical Product Development Cycle

- Product managers gather the requirements
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- Development team implements the code on the mobile app and the backend

Typical Product Development Cycle

- Product managers gather the requirements
- UI design team creates the wire frames and UI mockups
- Development team implements the code on the mobile app and the backend
- The DBAs create the database and implement the data model

Can ChatGPT Reliably Create DB Schemas?

DEMO 2

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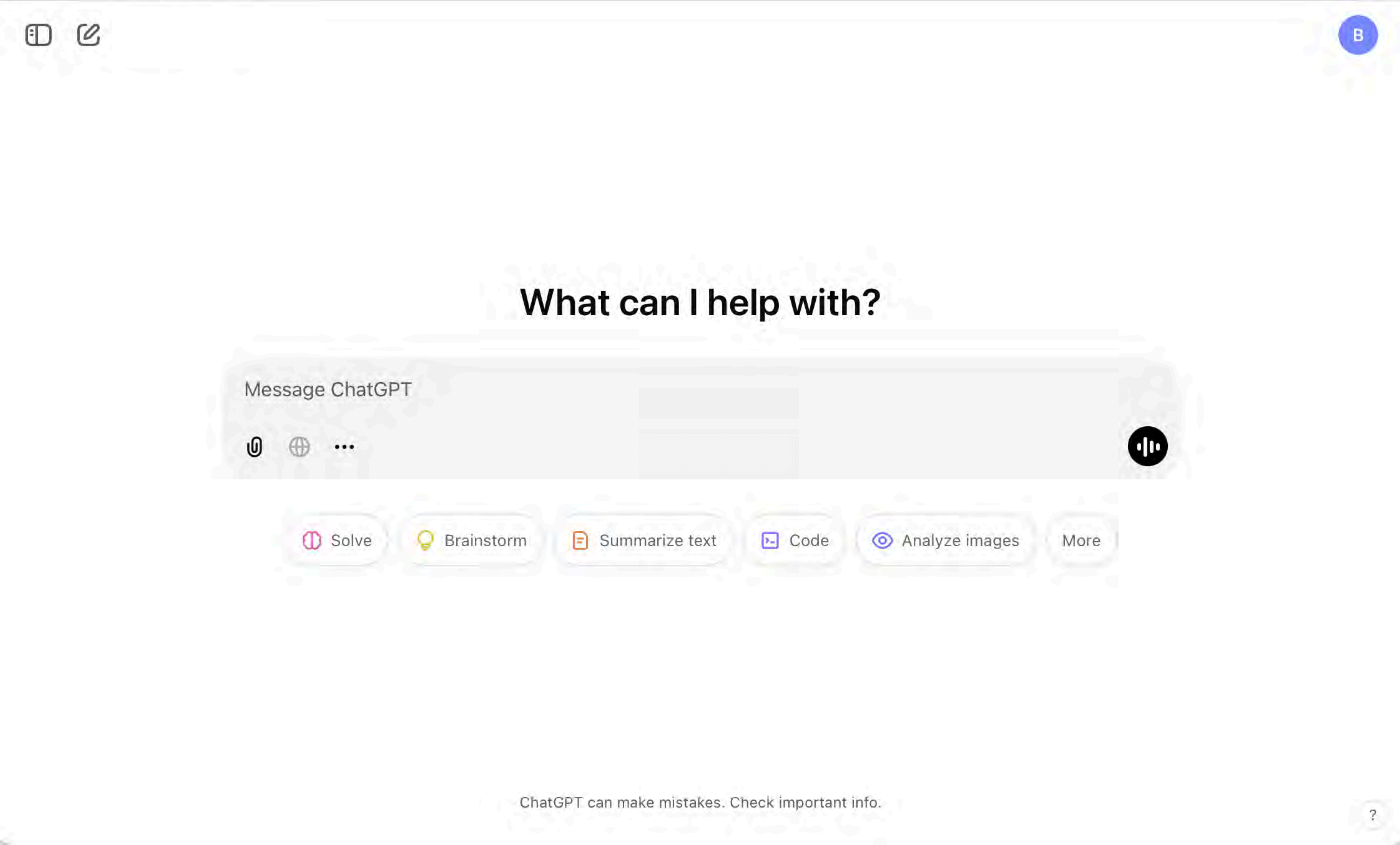

B
What can I help with?
Message ChatGPT
Solve
Brainstorm
Summarize text
Code
Analyze images
More
ChatGPT can make mistakes. Check important info.
?

Can ChatGPT Reliably Create DB Schemas?

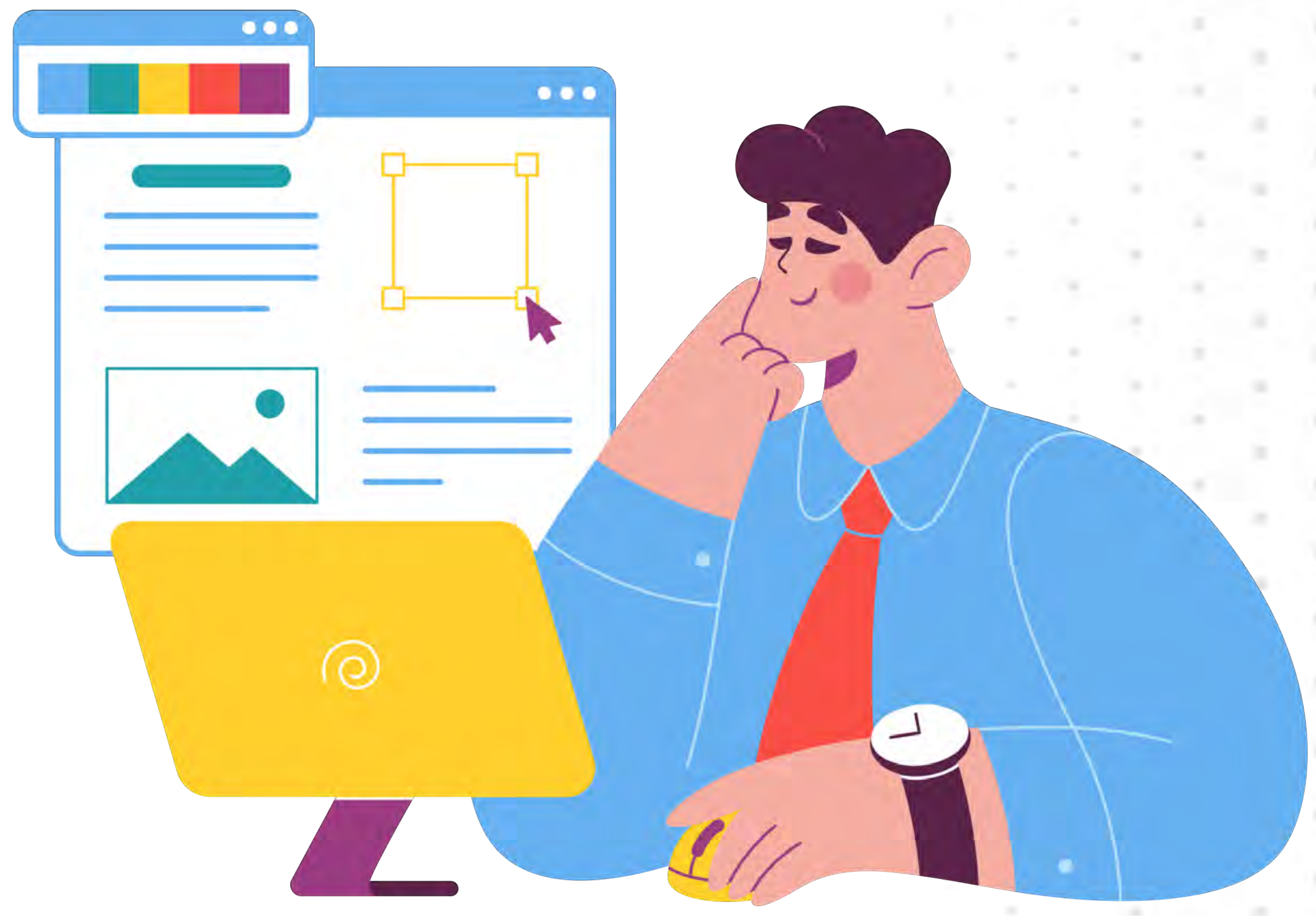
DEMO 2

PROMPT:

Create the DB schema for a new app for food delivery

Let's learn a few terms for now

- Models
- System Instructions

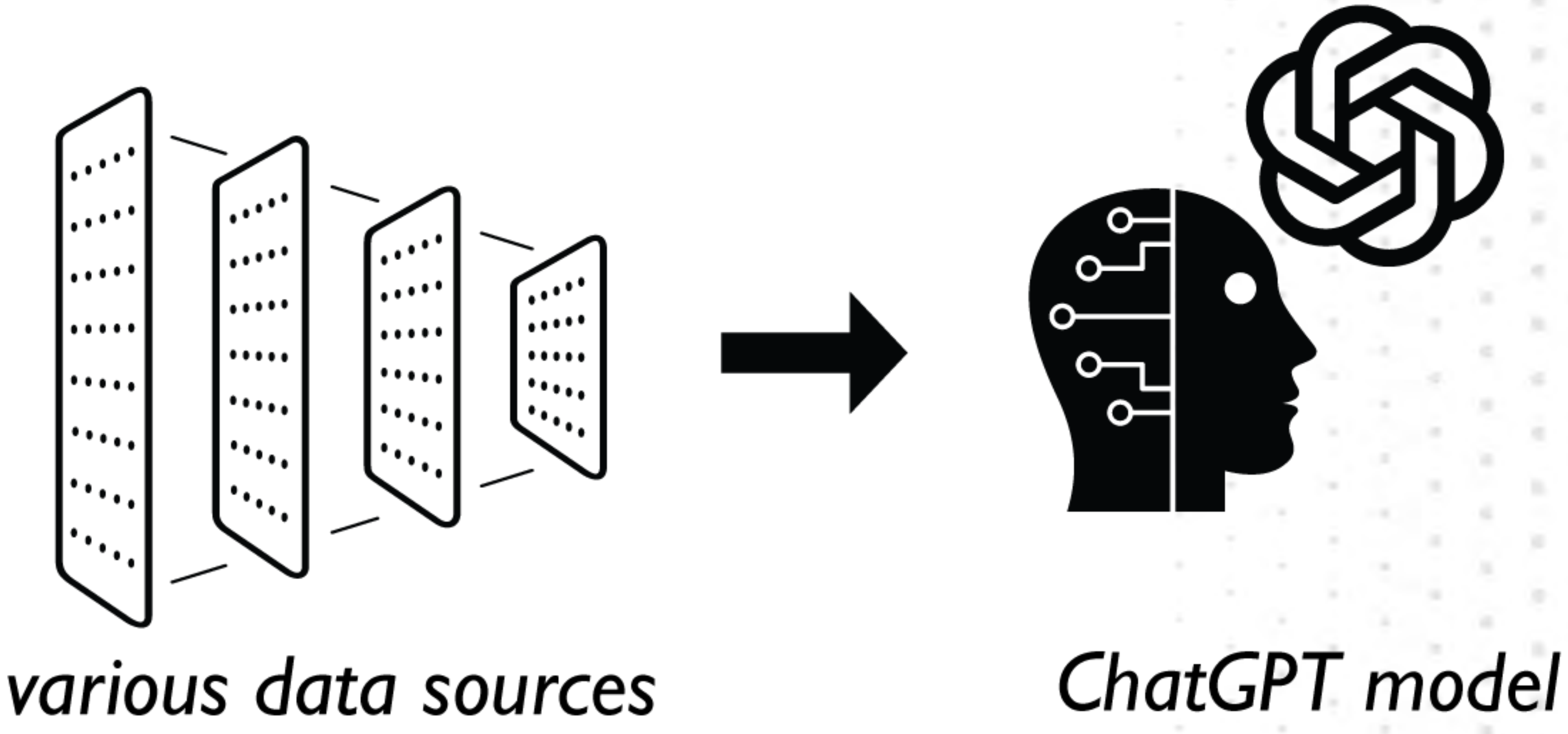


Models

- As a DBA, get rid of any preconceived notion of a **data model**

Models

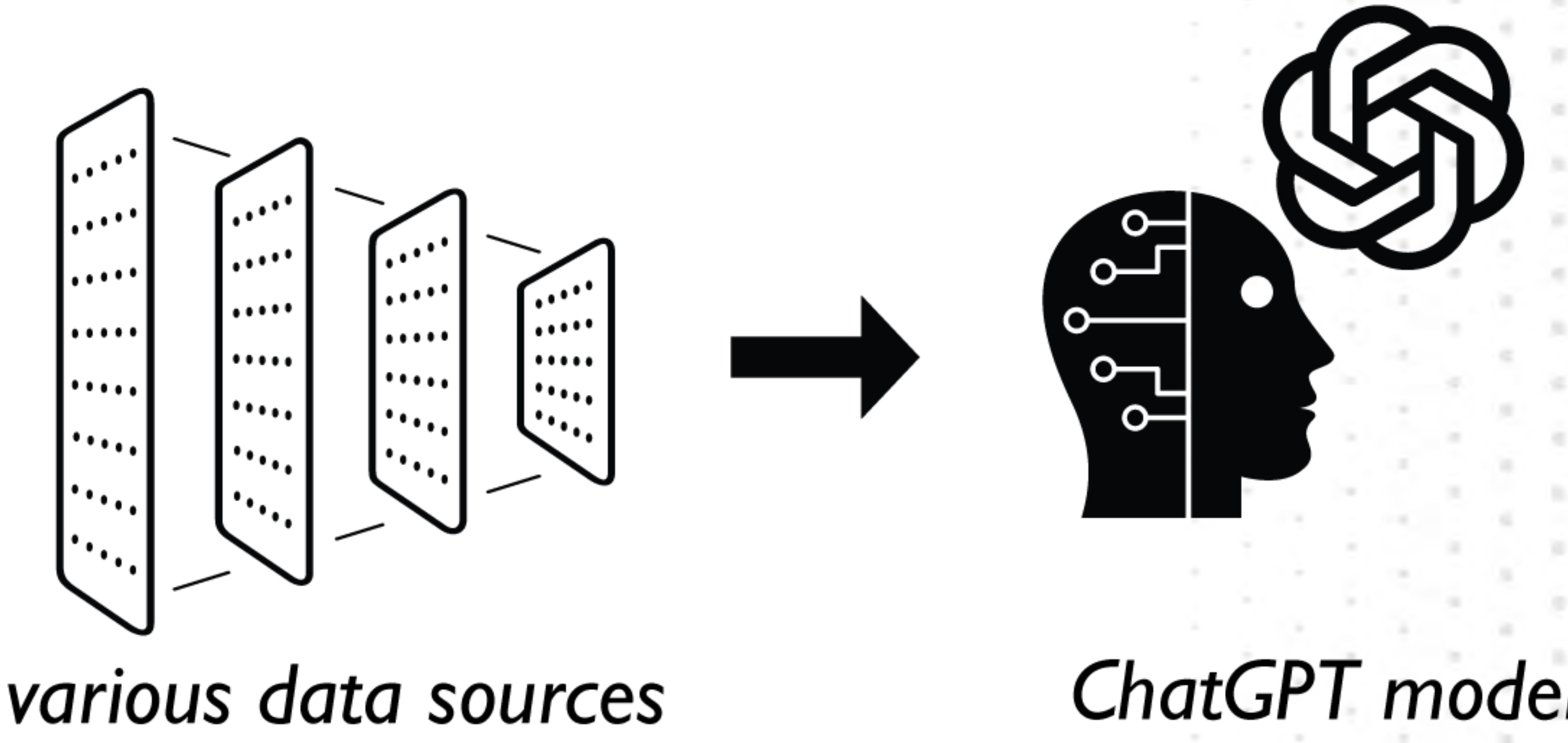
- A **pre-trained model** refers to a neural network that has been trained on a specific task or dataset



Models

- A **pre-trained model** refers to a neural network that has been trained on a specific task or dataset

- This training process involves exposing the model to large amounts of labeled and categorized (also called, “annotated”) data and adjusting its internal parameters to optimize its performance on the given task



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System Instructions

- Instructions

System Instructions

- Instructions
  - **This is how you explain to ChatGPT its role in the conversation**

System Instructions

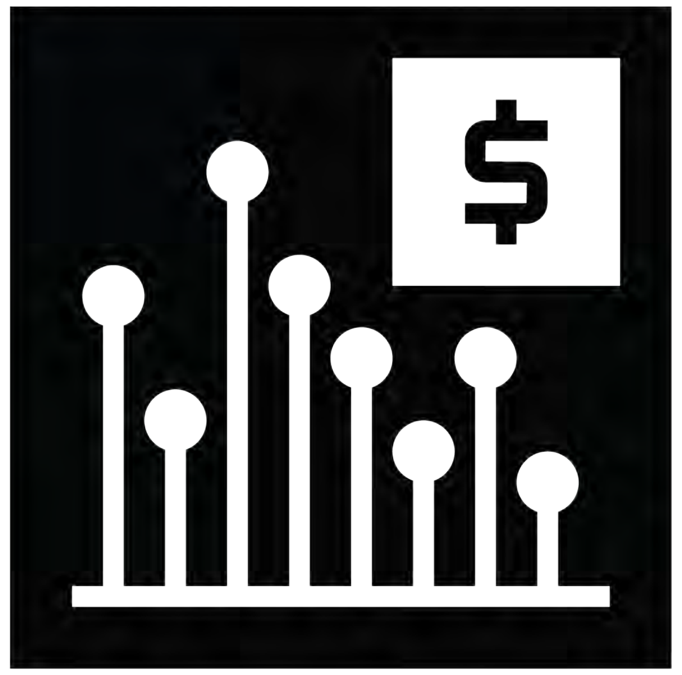
- Instructions
  - **This is how you explain to ChatGPT its role in the conversation**
  - **This is a vitally important prompt**

System Instructions

- Instructions
  - **This is how you explain to ChatGPT its role in the conversation**
  - **This is a vitally important prompt**
  - **You can only do this when creating a project**

System Instructions

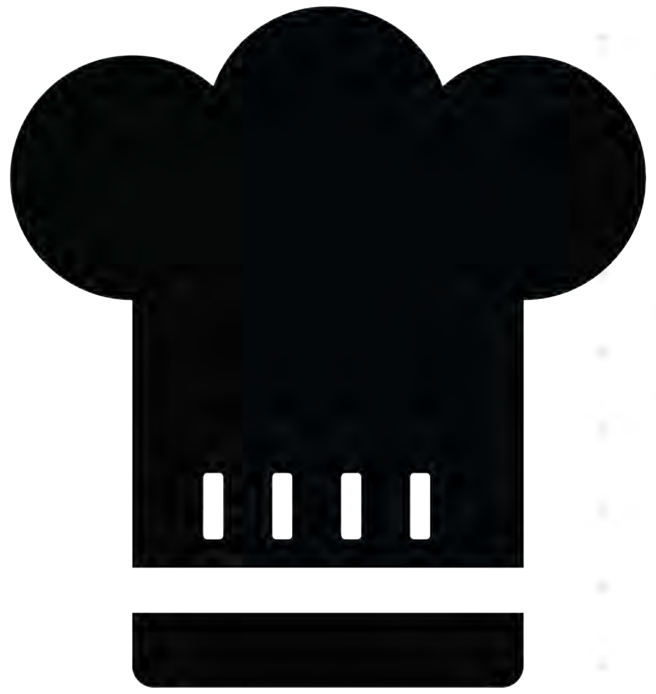
System: " `You are a...`"



"`economist`"

"`blogger`"

"`15th century poet`"

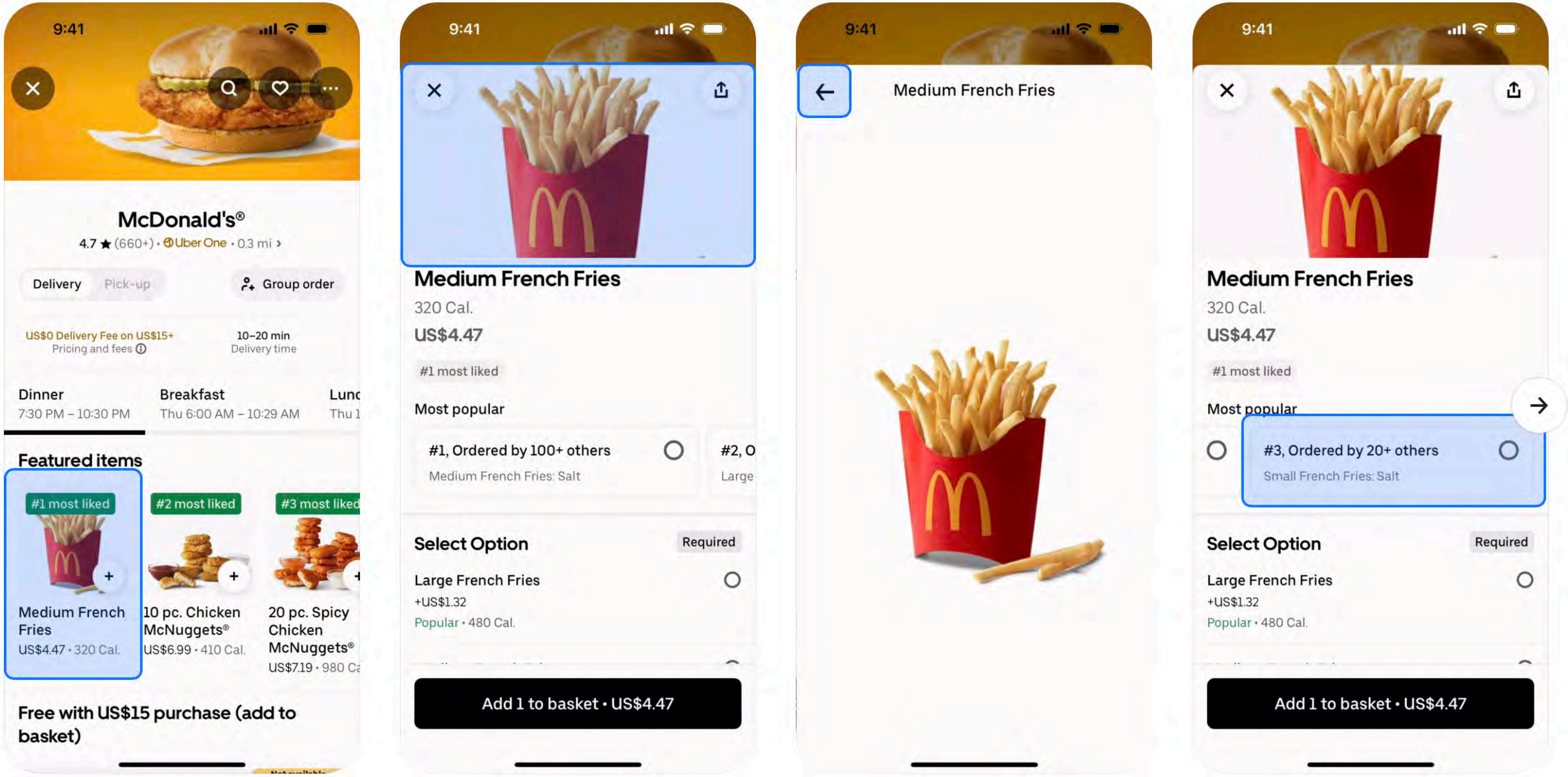


"`Chef`"

Giving ChatGPT More Context

DEMO 2.1

Giving ChatGPT More Context



Giving ChatGPT More Context

DEMO 2.1

GOAL:

1. *Create schema from UI wireframe*
2. *Refine based upon observations*
3. *Create sample data*
4. *SQL Worksheet > "Select * from RESTAURANTS;"*

DEMO 2 - Prompts Used

<u>SYSTEM MESSAGE:</u>

You are a database administrator for Uber eats.

Included in this conversation are screenshots for the new Uber eats mobile application for iOS.

When prompted, your role in this conversation is to produce a database schema that will enable an Oracle DBA to execute in order to create the tables necessary to store the information from the mobile app, or you will create example data that adheres to a database schema.

DEMO 2 - Prompts Used

PROMPT:

Attached to this conversation are four screenshots from the new iOS application.

Examine each screen and create a database schema for Oracle 26 that's fully commented that is necessary to persist all the data and any other necessary data in order for the new mobile application to work.

If you have any questions about the screenshots or the task, then ask me.

DEMO 2 - Prompts Used

PROMPT:

Good job. Now create a script to drop all the tables in the create script

DEMO 2 - Prompts Used

PROMPT:

Ok, using the schema above, and the screenshots I provided you at the beginning of this conversation, create the SQL needed to create 10 restaurants in the database, and ensure that each restaurant has a least 10 menu items to select

This will be used for a live demo for a proof of concept, so I need you to use real world restaurant names and menu items. I want the full script to execute

Ramifications

Ramifications

- You are able to create a **working DB** for a POC within a few minutes

Ramifications

- You are able to create a **working DB** for a POC within a few minutes
- You are able to create **REALISTIC demo data** for the POC for testing purposes

Questions?

7 min Break

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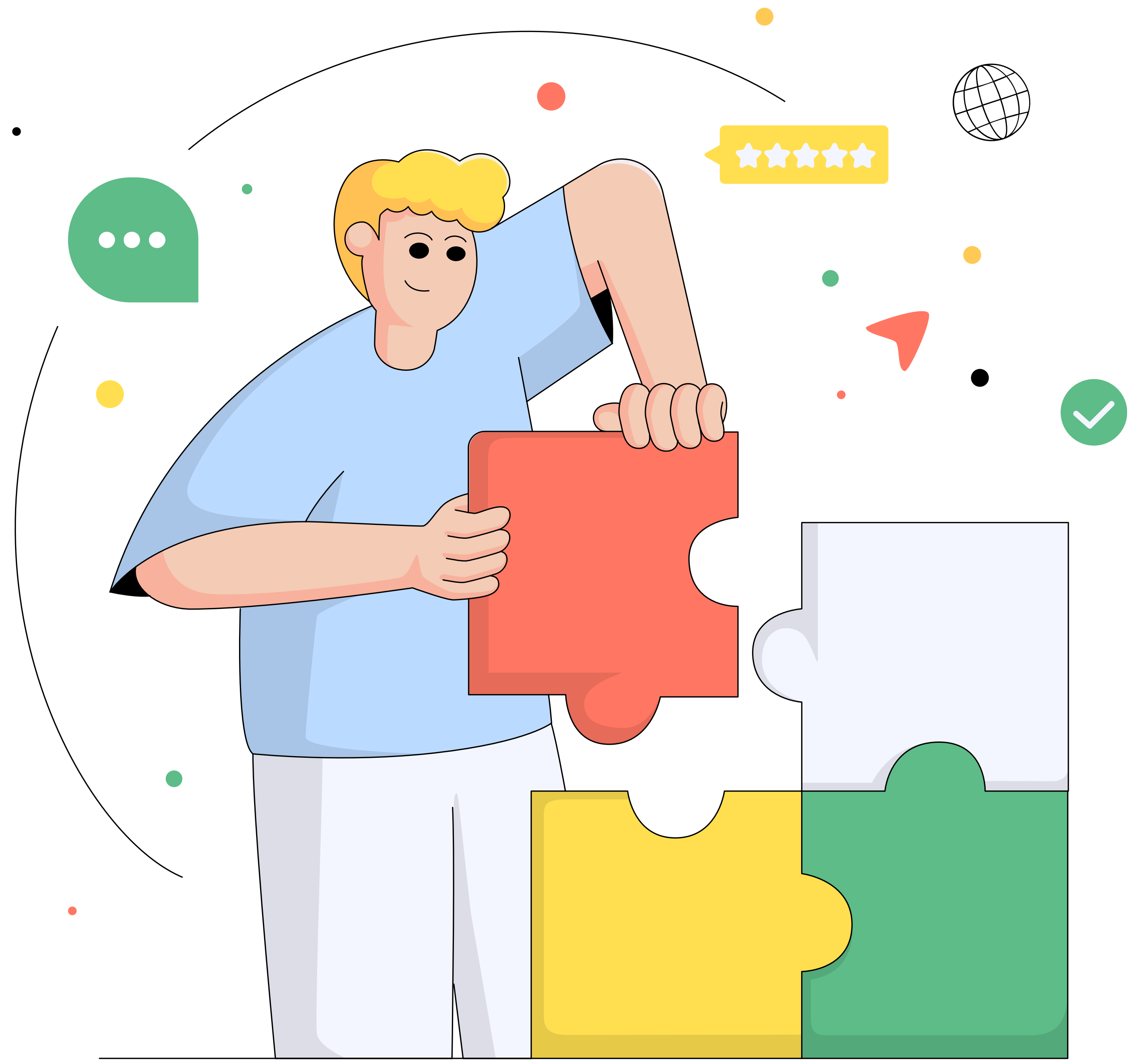
PART 3

Working with structured data in the enterprise: ETL flows with AI

What have we accomplished so far?

- We understand basic terms and terminology for ChatGPT
- We learned best practices on how to generate schemas for a POC
- We learned how to generate realistic demo data

What can be improved upon?



- What happens if we're in a situation where we are given **dirty data**?

Audience Poll Question #4

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Audience Poll Question #4

Have you encountered an ETL problem with dirty data? (Single response)

- Yes
- No
- No, I live a sheltered life

Problem: Two Divisions are Merging

Problem: Two Divisions are Merging

- You work for a globally recognized international shoe brand

Problem: Two Divisions are Merging

- You work for a globally recognized international shoe brand
- The company has two divisions: **online** and **retail**

Problem: Two Divisions are Merging

- You work for a globally recognized international shoe brand
- The company has two divisions: **online** and **retail**
- You have just been informed that the divisions are **merging**

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The Online Division

The Online Division

- All the databases are clean, pristine, clustered, sharded, with hourly and daily backups

The Online Division

- All the databases are clean, pristine, clustered, sharded, with hourly and daily backups

- Your team is prepped and ready to support any in all web and mobile applications

Pearson

The Online Division

- All the databases are clean, pristine, clustered, sharded, with hourly and daily backups

- Your team is prepped and ready to support any in all web and mobile applications

- The reason why?
  You're part of the team that accomplished this

The Retail Division

The Retail Division

- There are no databases

The Retail Division

- There are no databases
- The system is literally a POS

The Retail Division

- There are no databases
- The system is literally a POS
- Point of Sale

The Retail Division

- There are no databases
- The system is literally a POS
- Point of Sale
- No databases, just **RAW** transaction records based upon a 3rd party loyalty card

The Retail Division

- The POS system is 30 years old

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The Retail Division

- The POS system is 30 years old
- A transaction record is an ASCII formatted text file

The Retail Division

- The POS system is 30 years old
- A transaction record is an ASCII formatted text file
- There are millions of files dating back to 1990

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The Retail Division

- The POS system is 30 years old
- A transaction record is an ASCII formatted text file
- There are millions of files dating back to 1990
- Each register produces a .TXR

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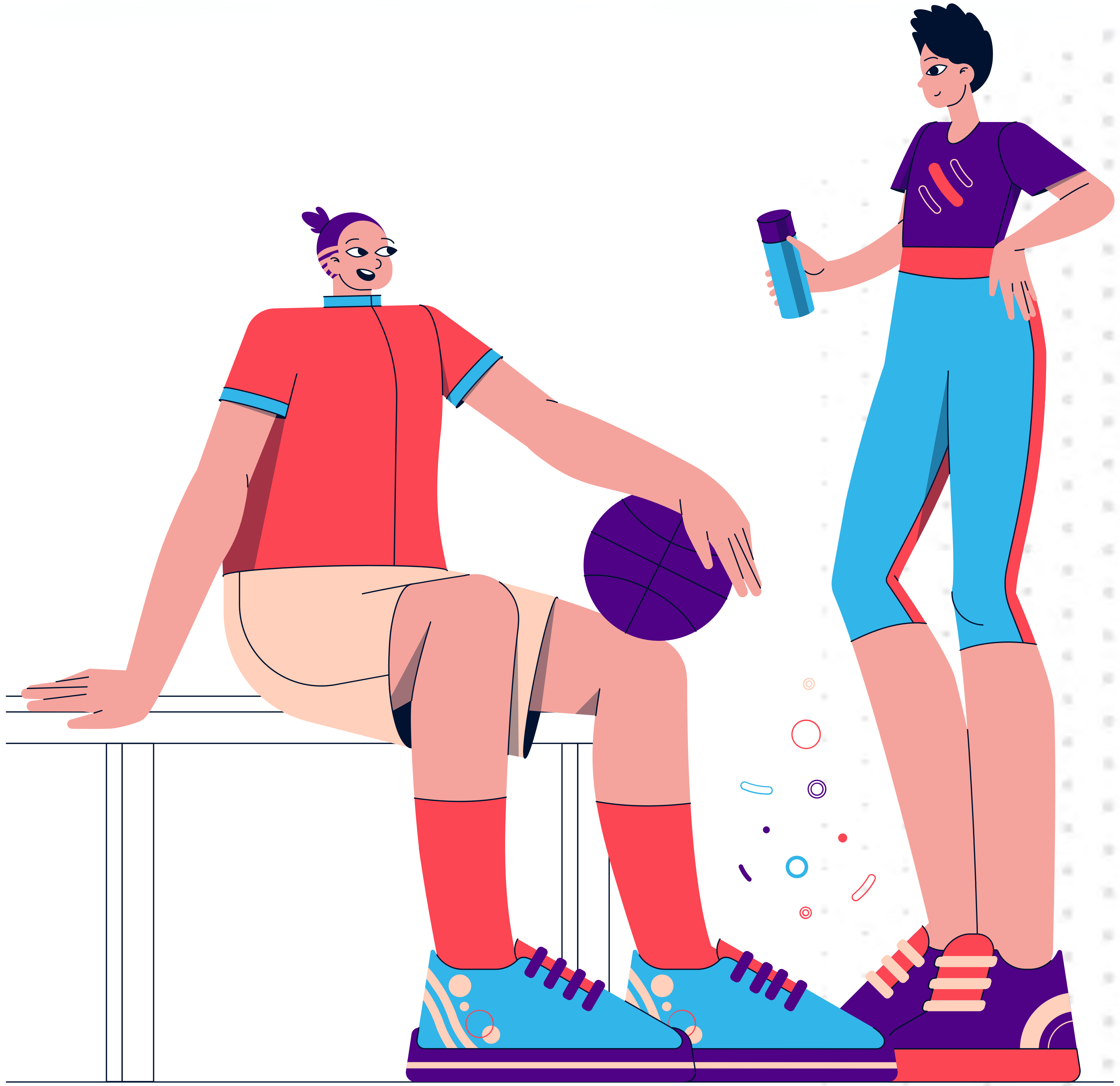
Management Has a Question

What % of retail customers shop online?



The Online Division - CUSTOMERS table

```
CREATE TABLE CUSTOMERS (
    CUSTOMER_ID           NUMBER,
    FIRSTNAME             VARCHAR2(255),
    LASTNAME              VARCHAR2(255),
    ADDRESS_LINE1         VARCHAR2(255),
    ADDRESS_LINE2         VARCHAR2(255),
    ADDRESS_LINE3         VARCHAR2(255),
    CITY                  VARCHAR2(100),
    STATE                 VARCHAR2(2),
    ZIP_CODE              VARCHAR2(20),
    COUNTRY               VARCHAR2(100),
    PHONE_NUMBER          VARCHAR2(20)
);
```

The Retail Division - .TXR file

- Transaction ID
- Date (yyyy-mm-dd)
- Store ID (SEA01)
- Gender prefix (optional)
- First name
- Middle name (optional)
- Last name
- Full address
- UPC (12 digits)

The Retail Division - .TXR file

L9ERQJEGOS2Z 2010-01-01 SEA01 Dr. Maryam Huda Ali 9599 Market St SW Seattle WA 98103

X5H2EQYN5L93 2010-01-02 SEA01 - Alessandro - Bianchi 6707 Aurora Ave N Seattle WA 98154

R2JX5SE1Z3U 2010-01-03 SEA01 Mrs. Chiara - Ferrari 4527 Yesler Way W Seattle WA 98134

Y4MECRIIC0OY 2010-01-04 SEA01 - Wei Jian Li 310 1st Ave SE Seattle WA 98107

XEUDK9LEBGX9 2010-01-06 SEA01 Dr. Marco Giovanni Ricci 508 Pine St S Apt 119 Seattle WA 98103

…

.TXR file Issue #1 - Delimiter is a " "

L9ERQJEGOS2Z 2010-01-01 SEA01 Dr. Maryam Huda Ali **9599 Market St SW** Seattle WA 98103

X5H2EQYN5L93 2010-01-02 SEA01 - Alessandro - Bianchi 6707 Aurora Ave N Seattle WA 98154

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XEUDK9LEBGX9 2010-01-06 SEA01 Dr. Marco Giovanni Ricci **508 Pine St S Apt 119** Seattle WA 98103

…

.TXR file Issue #2 - Title is optional

L9ERQJEGOS2Z 2010-01-01 SEA01 **Dr.** Maryam Huda Ali 9599 Market St SW Seattle WA 98103

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XEUDK9LEBGX9 2010-01-06 SEA01 **Dr.** Marco Giovanni Ricci 508 Pine St S Apt 119 Seattle WA 98103

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.TXR file Issue #3 - Middle Name is Optional

L9ERQJEGOS2Z 2010-01-01 SEA01 Dr. Maryam **Huda** Ali 9599 Market St SW Seattle WA 98103

X5H2EQYN5L93 2010-01-02 SEA01 - Alessandro **-** Bianchi 6707 Aurora Ave N Seattle WA 98154

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XEUDK9LEBGX9 2010-01-06 SEA01 Dr. Marco **Giovanni** Ricci 508 Pine St S Apt 119 Seattle WA 98103

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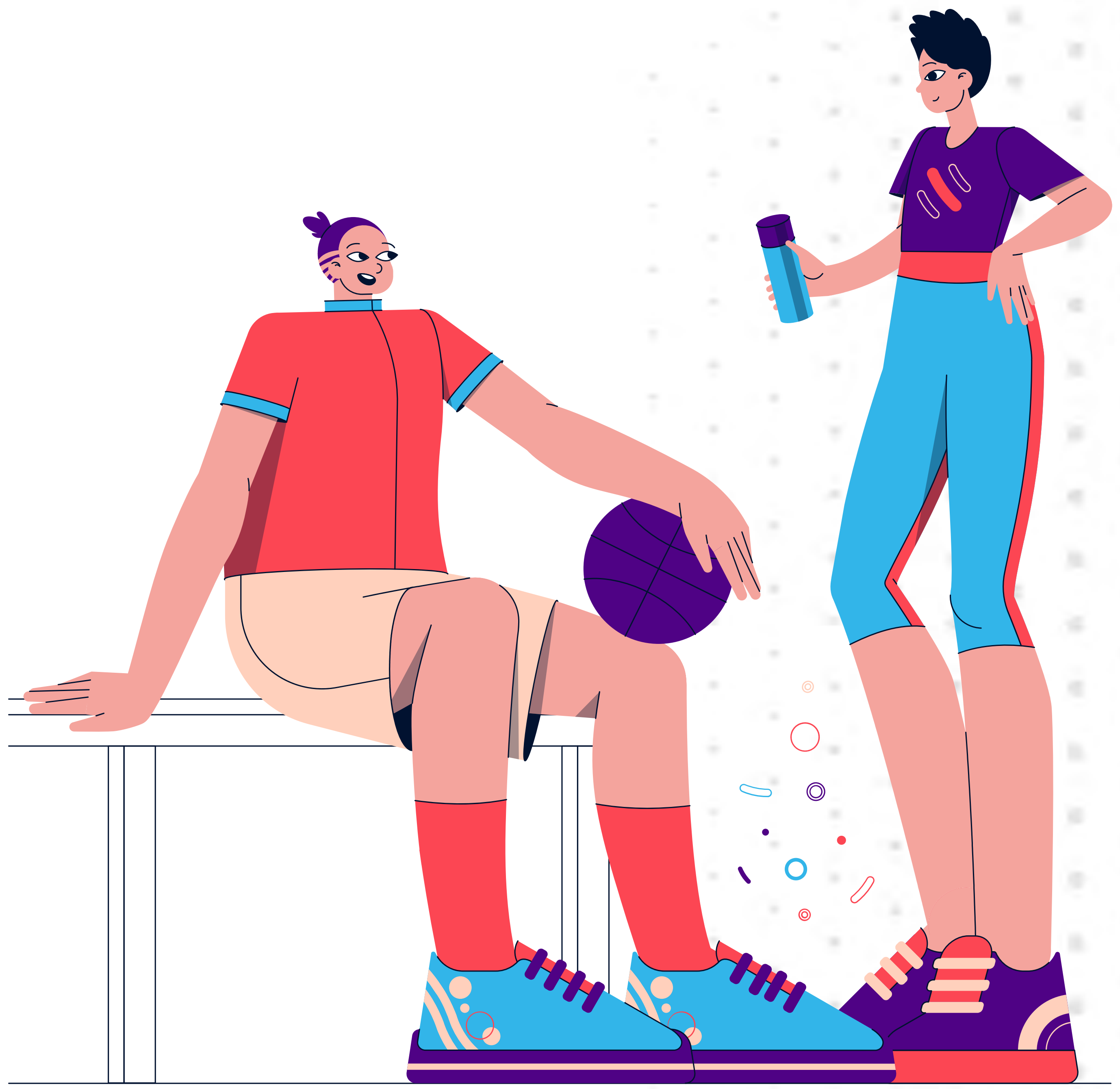
ETL - Cleaning Up Dirty Data

DEMO 3



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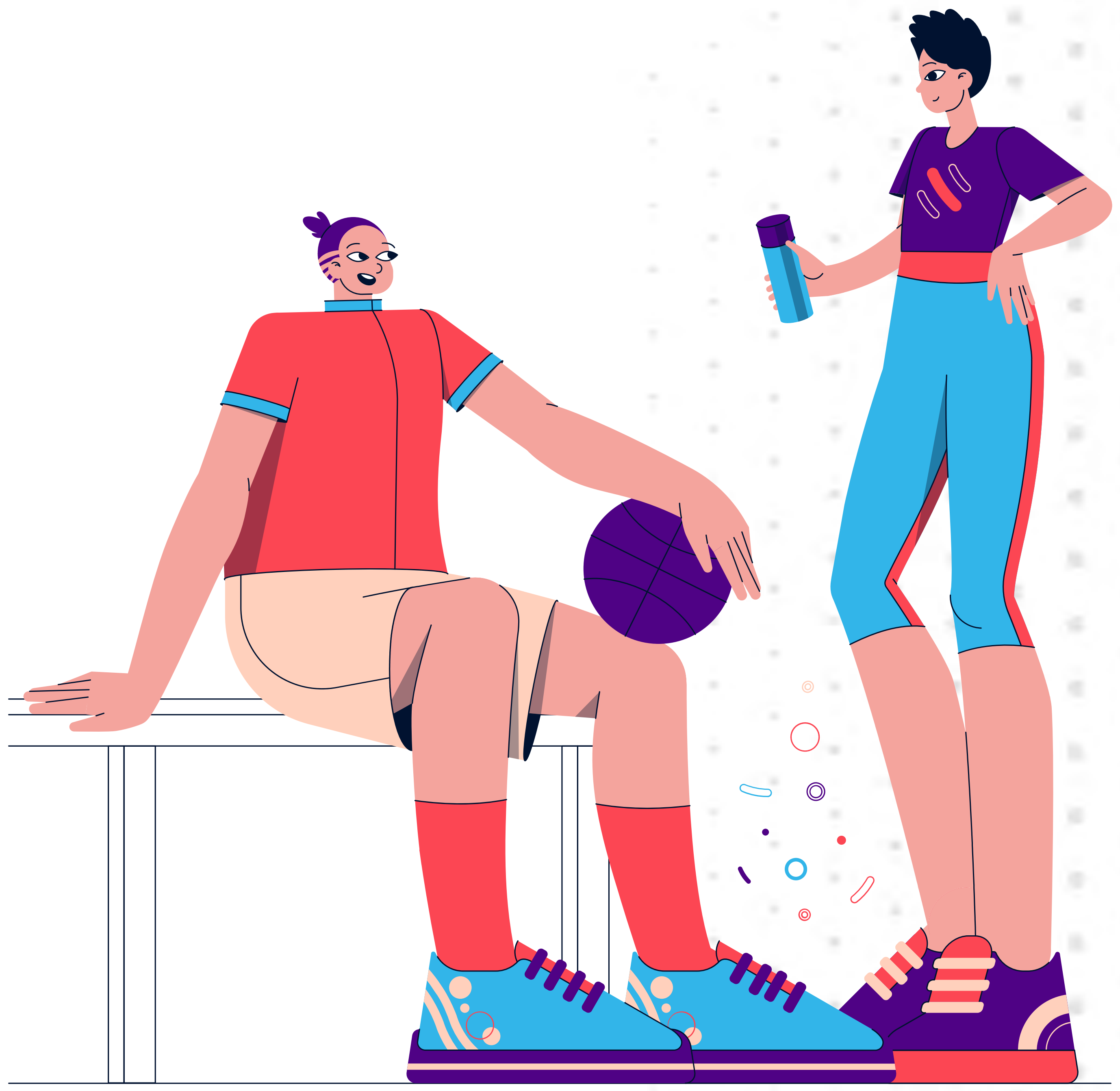
DEMO 3



GOAL:

1. *Give instructions to ChatGPT*
2. *Give schema and raw data to ChatGPT*
3. *INSERT statements with clean data*

DEMO 3 - Prompts Used

SYSTEM MESSAGE:

You are a Database admin, and you are tasked with job to take raw data files with a space as the delimiter and convert each line into an insert statement to put in a table in an Oracle 26ai database. You will be provided with the schema for the customers table, as well as the raw data file that has customer data.

DEMO 3 - Prompts Used

PROMPT:

Below is the structure of the raw data file:

1. transaction ID

2. transaction date

3. Store ID

4. customer title (optional)

5. customer first name

6. customer middle name (optional)

7. customer last name

8. full street name

9. city

10. state

11. zip code

DEMO 3 - Prompts Used

PROMPT:

The raw data file contains more fields than the customers table so discard any fields that don't apply such as the transaction ID, transaction date, Store ID, customer title, and customer middle name

DEMO 3 - Prompts Used

PROMPT:

below is the schema for the CUSTOMERS table

###

CREATE TABLE CUSTOMERS (CUSTOMER_ID NUMBER, FIRSTNAME VARCHAR2(255), LASTNAME VARCHAR2(255), ADDRESS_LINE1 VARCHAR2(255), ADDRESS_LINE2 VARCHAR2(255), ADDRESS_LINE3 VARCHAR2(255), CITY VARCHAR2(100), STATE VARCHAR2(2), ZIP_CODE VARCHAR2(20), COUNTRY VARCHAR2(100), PHONE_NUMBER VARCHAR2(20));

DEMO 3 - Prompts Used

PROMPT:

L9ERQJEGOS2Z 2010-01-01 SEA01 Dr. Maryam Huda Ali 9599 Market St SW Seattle WA 98103 721278404753 X5H2EQYN5L93 2010-01-02 SEA01 - Alessandro - Bianchi 6707 Aurora Ave N Seattle WA 98154 647257079771 IR2JX5SE1Z3U 2010-01-03 SEA01 Mrs. Chiara - Ferrari 4527 Yesler Way W Seattle WA 98134 192852864975 Y4MECRIIC0OY 2010-01-04 SEA01 - Wei Jian Li 310 1st Ave SE Seattle WA 98107 816335859875 54WTZA2XREVC 2010-01-05 SEA01 Dr. Stefan Karl Fischer 3374 3rd Ave SW Apt 495 Seattle WA 98121 422559911151 XEUDK9LEBGX9 2010-01-06 SEA01 Dr. Marco Giovanni Ricci 508 Pine St S Apt 119 Seattle WA 98103 252845458726 GGXN6XU21G3F 2010-01-07 SEA01 - Mary Dorothy Smith 6026 3rd Ave N Seattle WA 98154 959888996969

Ramifications

Ramifications

- You are able to extract data from horribly **dirty data** sources

Ramifications

- You are able to extract data from horribly **dirty data** sources
- The demo used Oracle, but the technique can be applied to **any database**

Questions?

7 min Break

PART 4

Managing Unstructured Data:
LOB datatypes

Pearson

What have we accomplished so far?

- We can take a prompt and a schema with dirty data and clean it up for ETL flows

- We can work with **semi-structured** dirty data

- All the information was on a single line

What can be improved upon?

- Let’s see how to work with unstructured data
- BLOB datatypes
- Typically binary files

Problem: Marketing and Sales Need Help

Problem: Marketing and Sales Need Help

- You’re the IT Director for an innovative **Smart Watch startup**

Problem: Marketing and Sales Need Help

- You're the IT Director for an innovative **Smart Watch startup**
- The best way to attract new business is to attend **conferences**

Pearson

Problem: Marketing and Sales Need Help

- You’re the IT Director for an innovative **Smart Watch startup**

- The best way to attract new business is to attend **conferences**

- It’s great for sales, but also good to attract developers on the platform

Pearson

Problem: Marketing and Sales Need Help

- Large events like CES and MWC easily have **over 100k** attendees

Pearson

Problem: Marketing and Sales Need Help

- Large events like CES and MWC easily have **over 100k** attendees

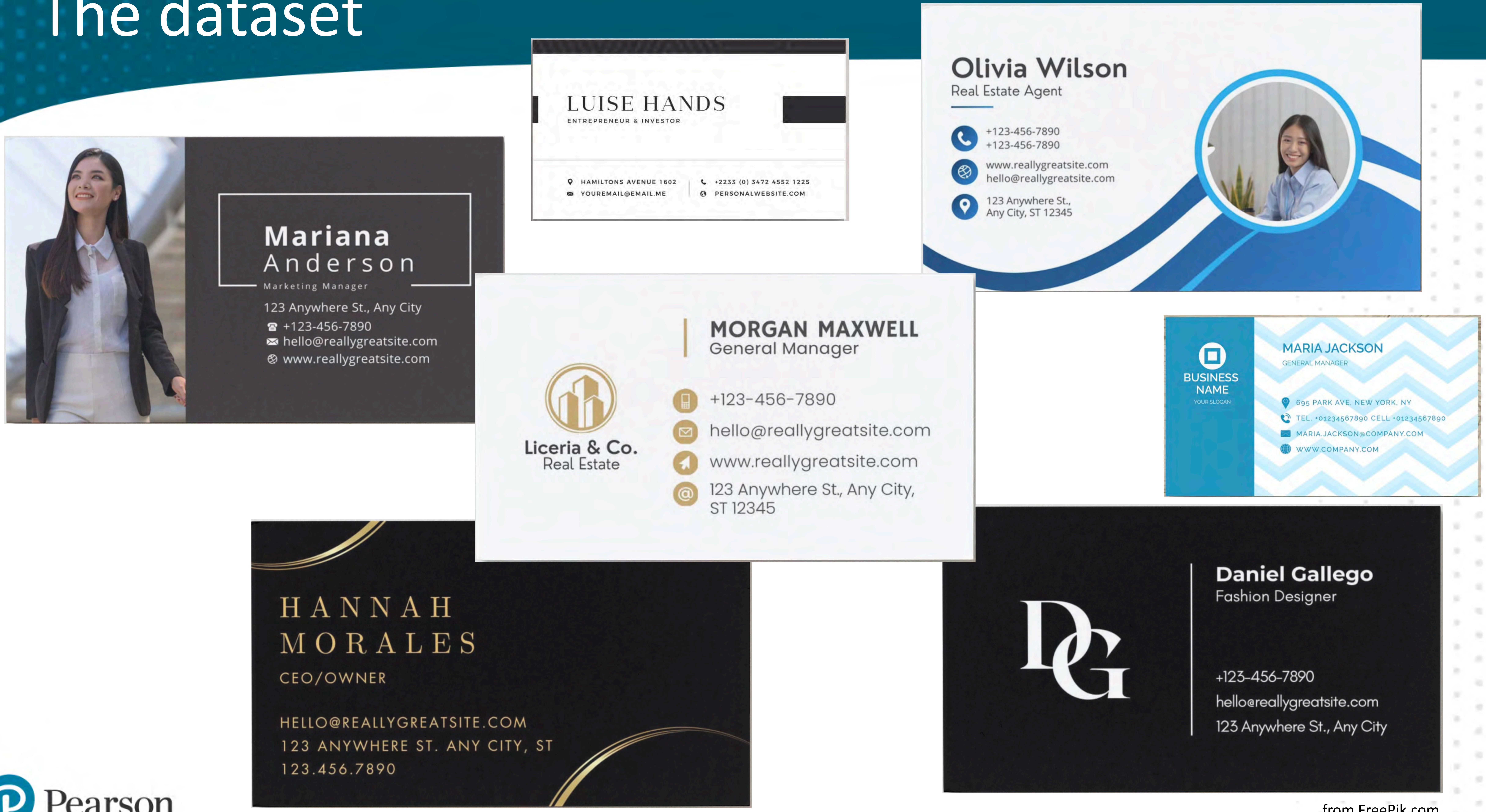
- At the end of the show, the Marketing team has a **stack of business cards** to sort through

Problem: Marketing and Sales Need Help

- Large events like CES and MWC easily have **over 100k** attendees
- At the end of the show, the Marketing team has a **stack of business cards** to sort through
- Time is of the essence! All your **competitors** also attend these events

Pearson

The dataset



from FreePik.com

Problem: Marketing and Sales Need Help

- The **old way** to handle after-event followups:

Problem: Marketing and Sales Need Help

- The **old way** to handle after-event followups:

- The team **manually types** in the business card info into Excel or the CRM

Pearson

Problem: Marketing and Sales Need Help

- The **old way** to handle after-event followups:
- The team **manually types** in the business card info into Excel or the CRM
- This is **slow and error prone**. The old business cards are kept in case of typos

Pearson

Solution

- Use ChatGPT to convert image data (business cards) to table rows

- For our purposes, we'll create a prompt to automate the process for batch processing

Converting Business Card Images into Excel rows

DEMO 4

Converting Business Card Images into Excel rows

DEMO 4

GOAL:

1. *Give instructions to ChatGPT*
2. *Give business cards*
3. *Excel Spreadsheet card data extracted*

Pearson

DEMO 4 - Prompts Used

SYSTEM MESSAGE:

You are a data administrator. In this conversation you will be provided with image scans of business cards. Your role in this conversation is to extract the following items from each business card:

1. *full name*
2. *business name*
3. *Email address*
4. *Complete phone number*
5. *Website URL*
6. *Mailing address*

DEMO 4 - Prompts Used

PROMPT:

You will take each of these items and continually append them to an Excel spreadsheet. At the end of the conversation, I will ask you to provide me with the final spreadsheet.

DEMO 4 - Prompts Used

PROMPT:

Please give me the spreadsheet now

Ramifications

Ramifications

- You are able to extract data from BLOB data sources

- This means that the following items are sources for data:

- PNG, GIF, JPEG, PDF, XLSX, DOCX, PPTX, etc.

Questions?

7 min Break

Pearson

PART 5

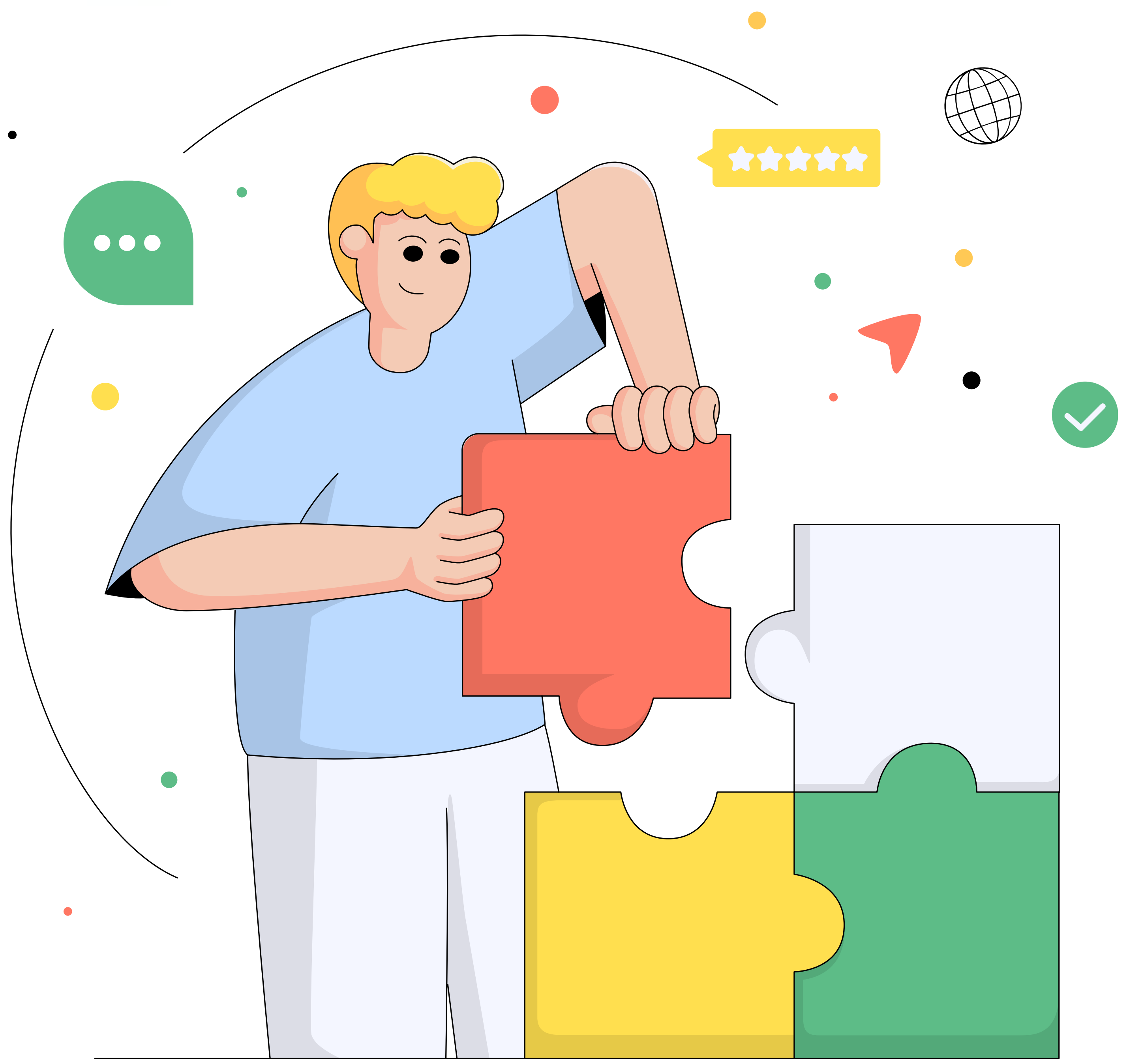
Getting started with **SELECT AI** in Oracle Autonomous Database 26ai

Pearson

What have we accomplished so far?

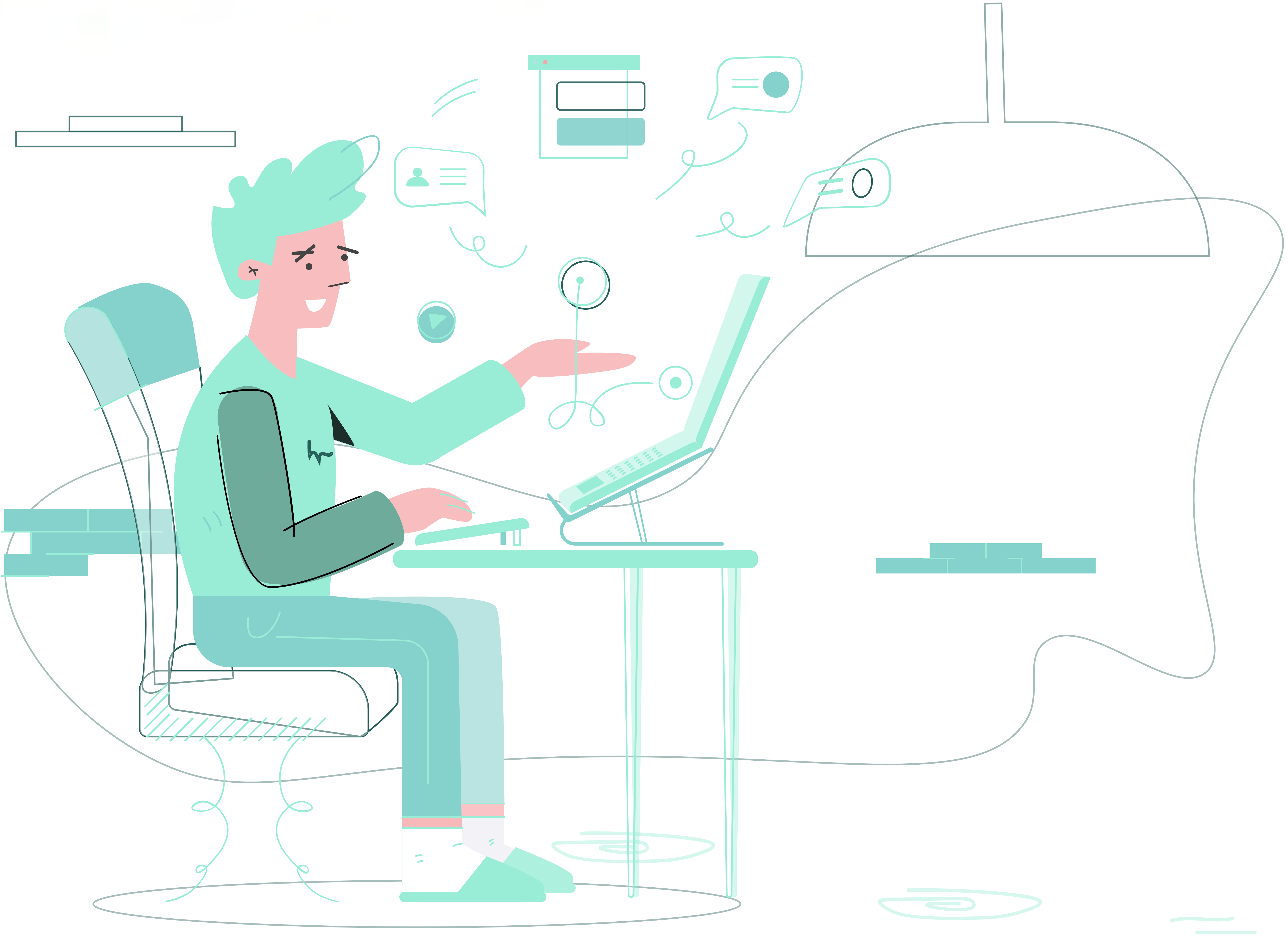
- We are able to extract data from **semi-structured** sources
- We are able to extract data from **un-structured** sources

What can be improved upon?



- Everything we've done can be applied to any database
- What are the features unique to Oracle ADB 26ai?
- **SELECT AI**

Audience Poll Question #5



Pearson

Audience Poll Question #5

Have you accessed ChatGPT (or any other AI model) programmatically using an API key? (Single response)

- Yes, with Python, JavaScript, Java (or some other language)
- No

SELECT AI Overview

- **SELECT AI**

Pearson

SELECT AI Overview

- **SELECT AI**

- Query your DB in pure Natural Language

SELECT AI Overview

- **SELECT AI**
- Query your DB in pure Natural Language
- This is a revolutionary concept

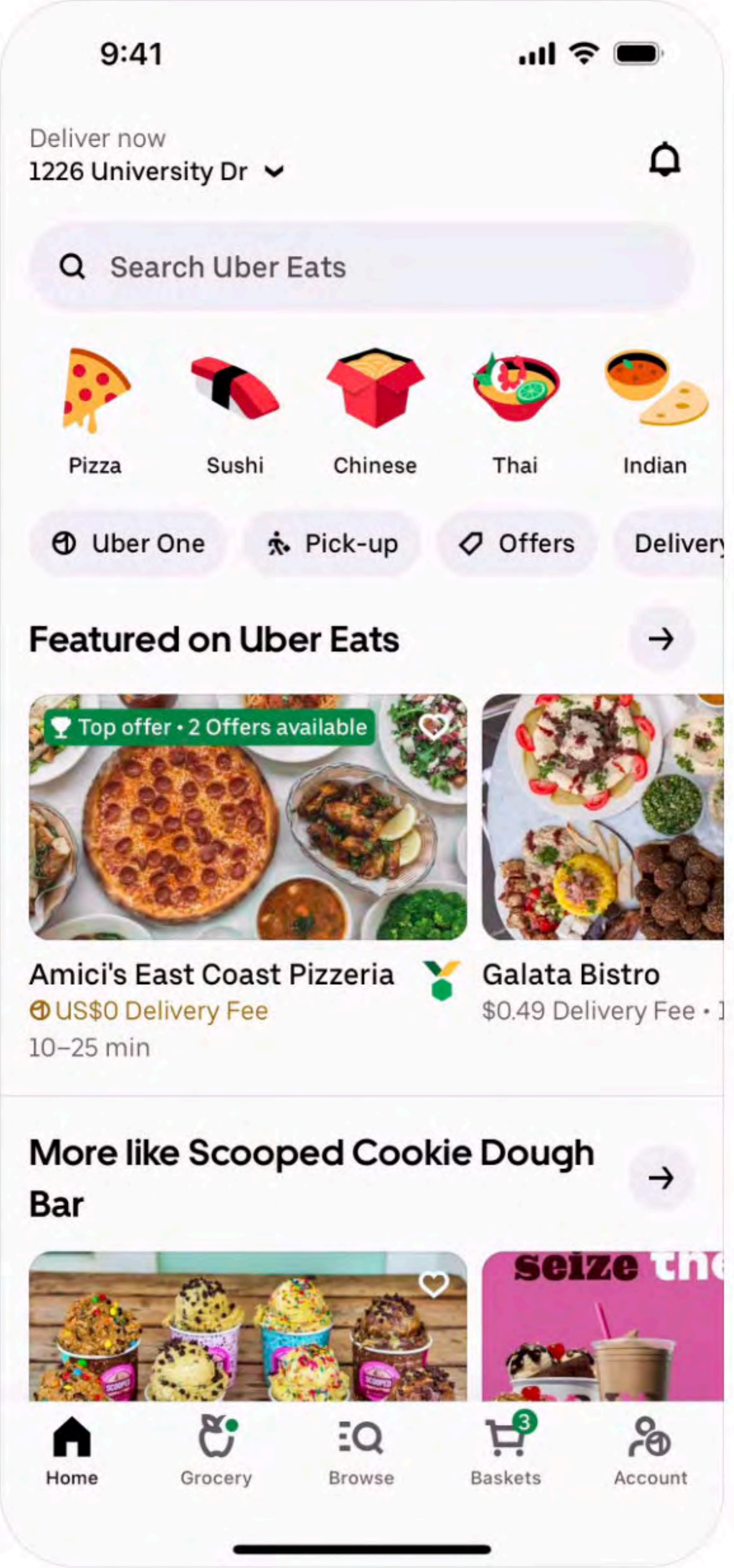
Pearson

SELECT AI Overview

- **SELECT AI**
- Query your DB in pure Natural Language
- This is a revolutionary concept
- SQL is no longer needed

SELECT AI

DEMO 5

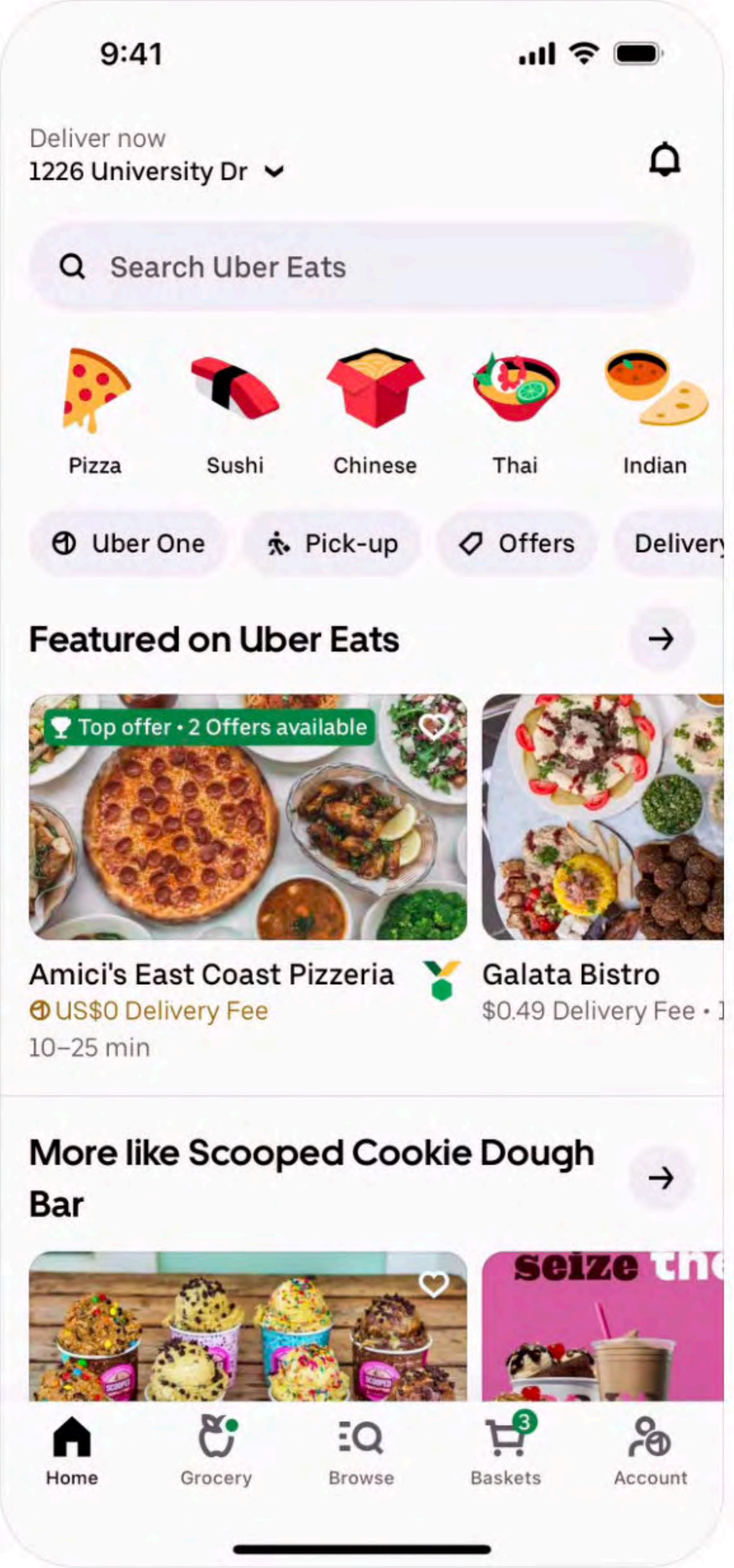


SELECT AI

DEMO 5

PROMPT:

"Who has the cheapest french fries?"



Pearson

Let's learn some more about SELECT AI

- **SELECT AI has 6 actions**
  - `runsql`
  - `showsql`
  - `explainsql`
  - `showprompt`
  - `narrate`
  - `chat`

Supported AI Providers

OpenAI

Gemini

Claude

Microsoft Azure

cohere

HUGGING FACE

OCI Generative AI

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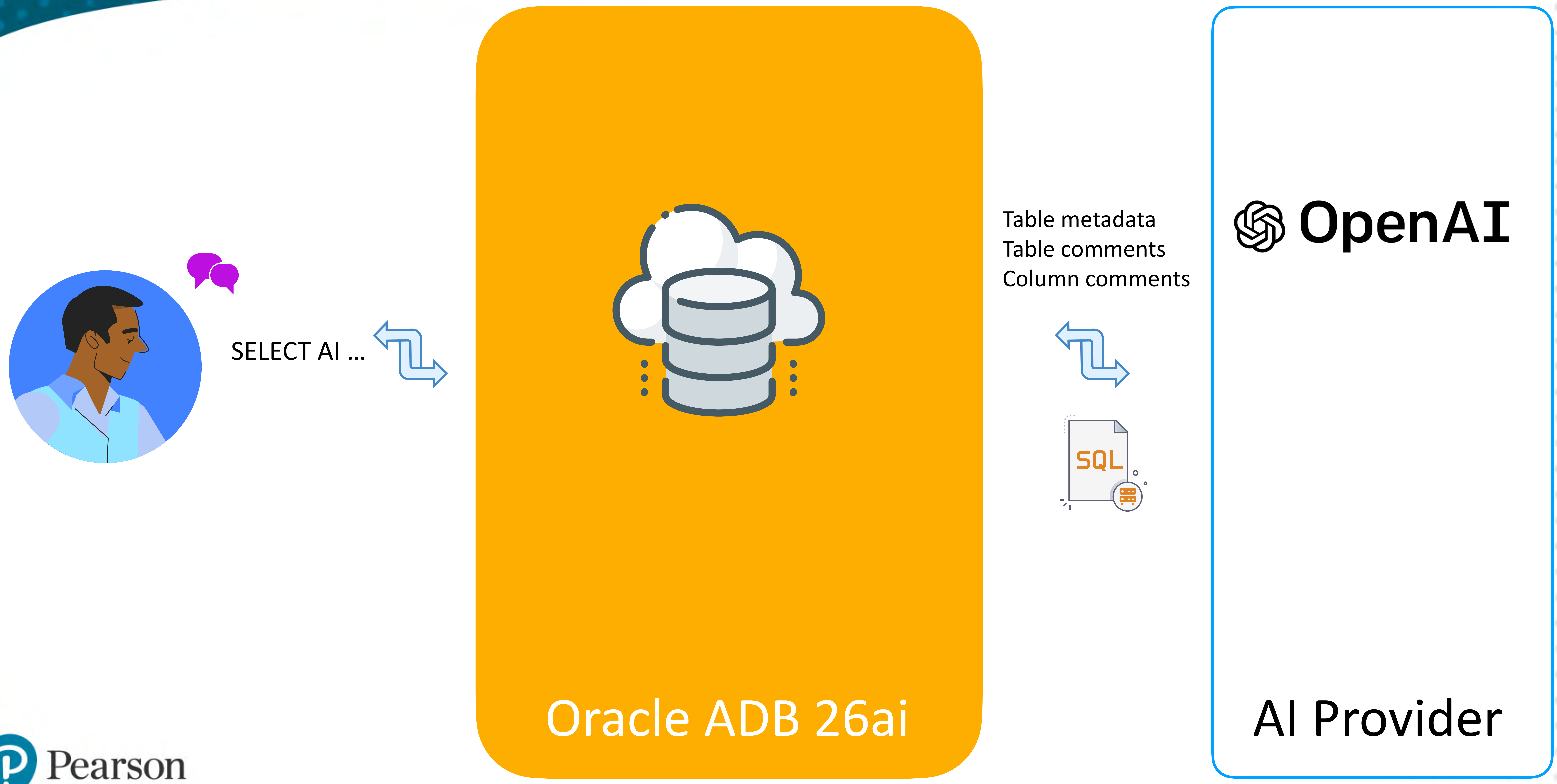
OCI Generative AI

* not available in all regions

SELECT AI - Under the Hood



Pearson

SELECT AI - **Under the Hood**

SELECT AI ...

- `runsql`
- `showsql`
- `explainsql`
- `showprompt`
-
- `chat`

Oracle ADB 26ai

Table metadata
Table comments
Column comments

SSL/TLS

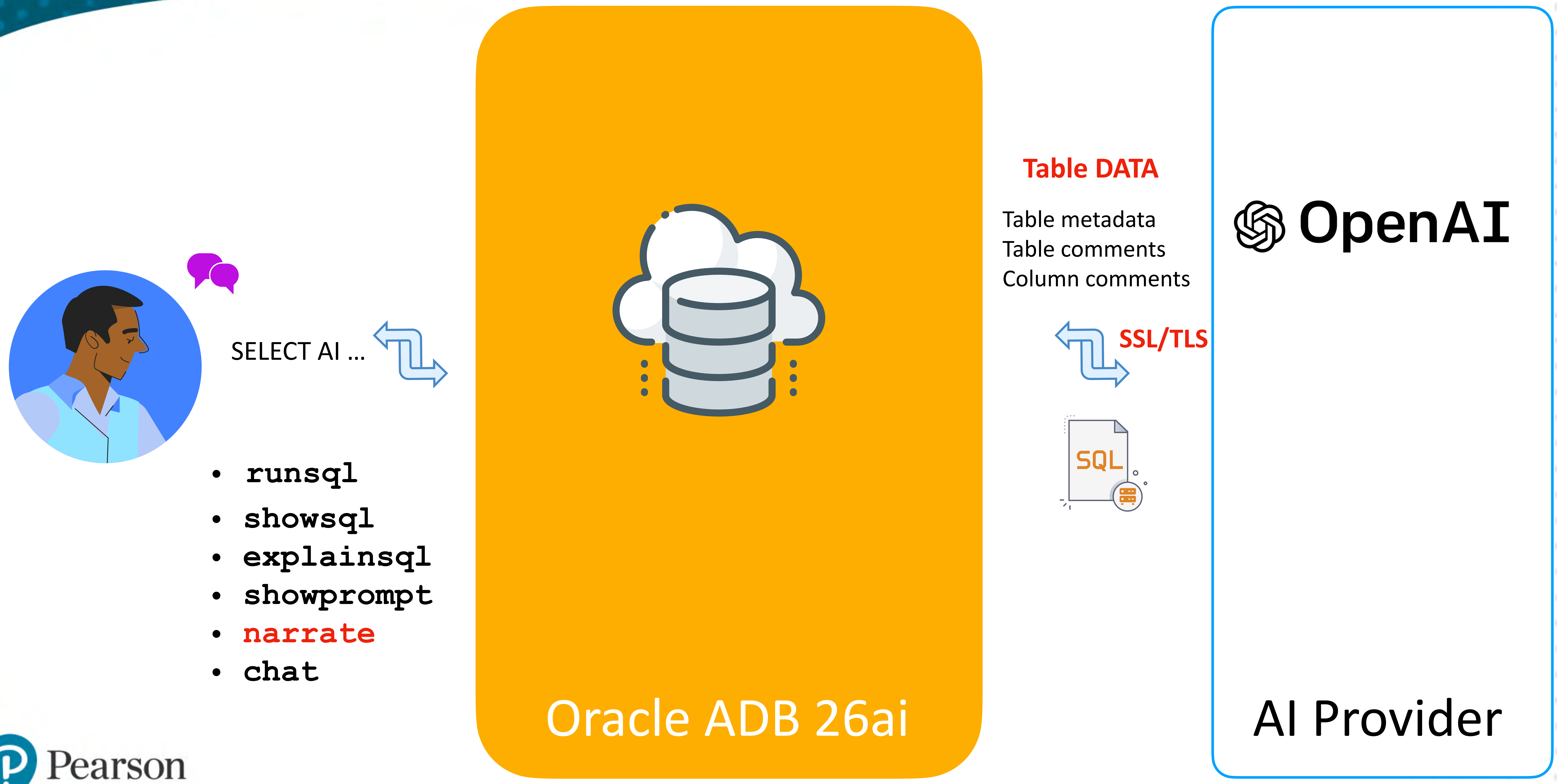
SQL

OpenAI

AI Provider

Pearson

SELECT AI - **Under the Hood**



SELECT AI - **Under the Hood**

SELECT AI …

- `runsql`
- `showsql`
- `explainsql`
- `showprompt`
- `narrate`
- `chat`

Table DATA

Table metadata
Table comments
Column comments

SQL

Oracle ADB 26ai

OCI
Generative AI

AI Provider

Pearson

SELECT AI - runsql

- SELECT AI **`runsql`**

SELECT AI - runsql

- SELECT AI `runsql`
- DEFAULT

SELECT AI - **runsql**

- SELECT AI `runsql`
- DEFAULT
- Takes the Natural Language statement and converts it to SQL

Pearson

SELECT AI - **runsql**

- SELECT AI `runsql`
- DEFAULT
- Takes the Natural Language statement and converts it to SQL
- executes it

SELECT AI - **runsql**

- SELECT AI **`runsql`**
- DEFAULT
- Takes the Natural Language statement and converts it to SQL
- executes it

SELECT AI **`runsql`** *Who has the cheapest french fries*

SELECT AI - **runsql**

```
RESTAURANT_NAME ITEM_NAME                BASE_PRICE
--------------- ------------------------ ----------
Burger King     French Fries (Medium)          2.69
```

Pearson

SELECT AI - showsql

- SELECT AI `showsql`

SELECT AI - **showsql**

- SELECT AI `showsql`

- Takes the Natural Language statement and converts it to SQL

SELECT AI - **showsql**

- SELECT AI `showsql`
- Takes the Natural Language statement and converts it to SQL
- but **DOES NOT** execute it

SELECT AI - **showsql**

- SELECT AI `showsql`
- Takes the Natural Language statement and converts it to SQL
- but **DOES NOT** execute it

SELECT AI `showsql` *Who has the cheapest french fries*

Pearson

SELECT AI - **showsql**

```
SELECT
    r."NAME" AS restaurant_name,
    mi."NAME" AS item_name,
    mi."BASE_PRICE" AS base_price
FROM
    "ADMIN"."MENU_ITEMS" mi
    JOIN "ADMIN"."MENU_SECTIONS" ms ON mi."SECTION_ID" = ms."SECTION_ID"
    JOIN "ADMIN"."RESTAURANTS" r ON ms."RESTAURANT_ID" = r."RESTAURANT_ID"
WHERE
    UPPER(mi."NAME") LIKE '%FRENCH FRIES%'
ORDER BY
    mi."BASE_PRICE" ASC
FETCH FIRST 1 ROW ONLY
```

SELECT AI - **explainsql**

- SELECT AI `explainsql`

SELECT AI - **explainsql**

- SELECT AI `explainsql`

- Takes the Natural Language statement and converts it to SQL

SELECT AI - **explainsql**

- SELECT AI `explainsql`

- Takes the Natural Language statement and converts it to SQL

- But **DOES NOT** execute it

Pearson

SELECT AI - **explainsql**

- SELECT AI `explainsql`
- Takes the Natural Language statement and converts it to SQL
- But **DOES NOT** execute it
- Explains the code

SELECT AI - **explainsql**

- SELECT AI `explainsql`
- Takes the Natural Language statement and converts it to SQL
- But **DOES NOT** execute it
- Explains the code

SELECT AI `explainsql` *Who has the cheapest french fries*

Pearson

SELECT AI - **explainsql**

```
RESPONSE

------------------------------------------------------------
### Oracle SQL Query

```sql
SELECT
    r."NAME" AS restaurant_name,
    mi."NAME" AS item_name,
    mi."BASE_PRICE" AS base_price
FROM
    "ADMIN"."MENU_ITEMS" mi
    JOIN "ADMIN"."MENU_SECTIONS" ms ON mi."SECTION_ID" = ms."SECTION_ID"
    JOIN "ADMIN"."RESTAURANTS" r ON ms."RESTAURANT_ID" = r."RESTAURANT_ID"
WHERE
    UPPER(mi."NAME") LIKE '%FRENCH FRIES%'
ORDER BY
    mi."BASE_PRICE" ASC
FETCH FIRST 1 ROW ONLY
```
```

SELECT AI - **explainsql**

````
...

### Explanation

1. **Table Aliases**:
   - `mi` for `"ADMIN"."MENU_ITEMS"`
   - `ms` for `"ADMIN"."MENU_SECTIONS"`
   - `r` for `"ADMIN"."RESTAURANTS"`

2. **Joins**:
   - Join menu items to menu sections (`mi` to `ms`) using `"SECTION_ID"`.
   - Join menu sections to restaurants (`ms` to `r`) using `"RESTAURANT_ID"`.

3. **WHERE Clause**:
   - The question asks for "french fries" (not in double quotes), so use case-insensitive comparison:
     ```sql
     UPPER(mi."NAME") LIKE '%FRENCH FRIES%'
     ```
   - This finds menu items whose name contains "french fries" in any case.

4. **ORDER BY and LIMIT**:
   - Order by `"BASE_PRICE"` ascending to get the cheapest.
   - `FETCH FIRST 1 ROW ONLY` returns only the cheapest one.

5. **Selected Columns**:
   - Restaurant name, item name, and base price for clarity.

---

**Result:**
This query returns the restaurant and menu item with the lowest price for "french fries" across all restaurants.
````

SELECT AI - **showprompt**

- SELECT AI `showprompt`

SELECT AI - **showprompt**

- SELECT AI `showprompt`

- It shows you everything that would have been sent to the AI model

SELECT AI - **showprompt**

- SELECT AI **`showprompt`**
- It shows you everything that would have been sent to the AI model

SELECT AI **`showprompt`** *Who has the cheapest french fries*

SELECT AI - **showprompt**

```
PROMPT

[
  {
    "role" : "system",
    "content" : "### Oracle SQL tables with their properties:"
  },
  {
    "role" : "system",
    "content" : " # CREATE TABLE \"ADMIN\".
\"RESTAURANTS\" (\"RESTAURANT_ID\" NUMBER, \"UPDATED_AT\" DATE, \"CREATED_AT\" DATE, \"PHONE_NUMBER\" VARCHAR2(20), \"COUNTRY\" V
ARCHAR2(100), \"ZIP_CODE\" VARCHAR2(20), \"STATE\" VARCHAR2(100), \"CITY\" VARCHAR2(100), \"ADDRESS_LINE3\" VARCHAR2(255), \"ADDR
ESS_LINE2\" VARCHAR2(255), \"ADDRESS_LINE1\" VARCHAR2(255), \"IS_UBER_ONE_ELIGIBLE\" CHAR(1), \"TOTAL_REVIEWS\" NUMBER, \"AVERAGE
_RATING\" NUMBER(2,1), \"NAME\" VARCHAR2(255))"
  },
  {
    "role" : "system",
    "content" : " # CREATE TABLE \"ADMIN\".
\"RESTAURANT_SCHEDULES\" (\"DAY_OF_WEEK\" VARCHAR2(9), \"START_TIME\" VARCHAR2(5), \"END_TIME\" VARCHAR2(5), \"RESTAURANT_ID\" NU
MBER, \"SCHEDULE_ID\" NUMBER, \"MEAL_TYPE\" VARCHAR2(50))"
  },
  {
    "role" : "system",
    "content" : " # CREATE TABLE \"ADMIN\".
\"PROMOTIONS\" (\"START_DATE\" DATE, \"MIN_PURCHASE_AMOUNT\" NUMBER(6,2), \"DESCRIPTION\" VARCHAR2(32767), \"PROMOTION_ID\" NUMBE
R, \"END_DATE\" DATE, \"RESTAURANT_ID\" NUMBER, \"DISCOUNT_AMOUNT\" NUMBER(6,2))"
  },
  {
    "role" : "system",
    "content" : " # CREATE TABLE \"ADMIN\".
\"MENU_SECTIONS\" (\"SECTION_ID\" NUMBER, \"SECTION_NAME\" VARCHAR2(255), \"RESTAURANT_ID\" NUMBER)"
  },
```

SELECT AI - **showprompt**


```
  {
    "role" : "system",
    "content" : " # CREATE TABLE \"ADMIN\".
\"ITEM_OPTIONS\" (\"ORDERED_COUNT\" NUMBER, \"ADDITIONAL_COST\" NUMBER(6,2), \"OPTION_ID\" NUMBER, \"OPTION_NAME\" VARCHAR2
(255), \"POPULARITY_LABEL\" VARCHAR2(50), \"ITEM_ID\" NUMBER, \"CALORIE_CHANGE\" NUMBER(5,0))"
  },
  {
    "role" : "system",
    "content" : " # CREATE TABLE \"ADMIN\".
\"FEATURED_ITEMS\" (\"FEATURED_ITEM_ID\" NUMBER, \"ITEM_ID\" NUMBER, \"RESTAURANT_ID\" NUMBER, \"FEATURED_RANK\" NUMBER(3,0
))"
  },
  {
    "role" : "system",
    "content" : " # CREATE TABLE \"ADMIN\".
\"DELIVERY_OPTIONS\" (\"ETA_LOWER_BOUND\" NUMBER(3,0), \"ETA_UPPER_BOUND\" NUMBER(3,0), \"MIN_PURCHASE\" NUMBER(6,2), \"DEL
IVERY_OPTION_ID\" NUMBER, \"FEE\" NUMBER(6,2), \"OPTION_TYPE\" VARCHAR2(50), \"RESTAURANT_ID\" NUMBER)"
  },
  {
    "role" : "user",
    "content" : "Always use table alias and easy to read column names, consider table name, schema name and column name to
be case sensitive and enclose in double quotes only type Oracle SQL and nothing else.
\n\nFor string comparisons in WHERE clause, CAREFULLY check if any string in the question is in DOUBLE QUOTES, and follow t
he rules: \n - If a string is in DOUBLE QUOTES, use case SENSITIVE comparisons with NO UPPER() function.\n -
 If a string is not in DOUBLE QUOTES, use case INSENSITIVE comparisons by using UPPER() function around both operands of th
e string comparison.
\nNote: These rules apply strictly to string comparisons in the WHERE clause and do not affect column names, table names, o
r other query components.\n\nQuestion: Whose french fries are cheapest"
  }
]
```

SELECT AI - **narrate**

- SELECT AI **`narrate`**

SELECT AI - **narrate**

- SELECT AI `narrate`

- Takes the Natural Language statement and converts it to SQL

Pearson

SELECT AI - **narrate**

- SELECT AI `narrate`

- Takes the Natural Language statement and converts it to SQL

- executes it

SELECT AI - **narrate**

- SELECT AI `narrate`

- Takes the Natural Language statement and converts it to SQL

- executes it

- Takes and result and sends it to the AI model for explanation

Pearson

SELECT AI - **narrate**

- SELECT AI **`narrate`**
- Takes the Natural Language statement and converts it to SQL
- executes it
- Takes and result and sends it to the AI model for explanation

SELECT AI **`narrate`** *Who has the cheapest french fries*

Pearson

SELECT AI - **narrate**

```
RESPONSE
--------------------------------------------------------------------------
Burger King has the cheapest French Fries (Medium) at a price of $2.69.
```

SELECT AI - **chat**

- SELECT AI `chat`

SELECT AI - **chat**

- SELECT AI `chat`

- This is no different from using ChatGPT

SELECT AI - **chat**

- SELECT AI `chat`

- This is no different from using ChatGPT

- This is simply talking to the AI model (outside of the context of your database)

Pearson

SELECT AI - **chat**

- SELECT AI `chat`
- This is no different from using ChatGPT
- This is simply talking to the AI model (outside of the context of your database)

SELECT AI `chat` *Who has the cheapest french fries*

Pearson

SELECT AI - **chat**

```
RESPONSE


The price of French fries can vary depending on location, size, and current promotio
ns, but generally, among major fast food chains in the United States, **McDonald's**
 and **Wendy's** are often known for having some of the cheapest French fries, espec
ially for their value or small size options. **Burger King** also frequently offers
competitive prices and deals.

For the absolute lowest price, **McDonald's** is often cited as having the cheapest
small fries, especially with their value menu or app deals. However, prices can chan
ge and may be different in your area, so it's a good idea to check the latest prices
 at your local restaurants or through their mobile apps for the most accurate inform
ation.
```

Ramifications

Ramifications

- You are able to use Natural Language to query your DB
- Your company is able to use Natural Language to query your DB
- Your customers are able to use Natural Language to query your DB

Questions?

PART 6

Hands-on Enabling AI Features in Oracle ADB 26ai

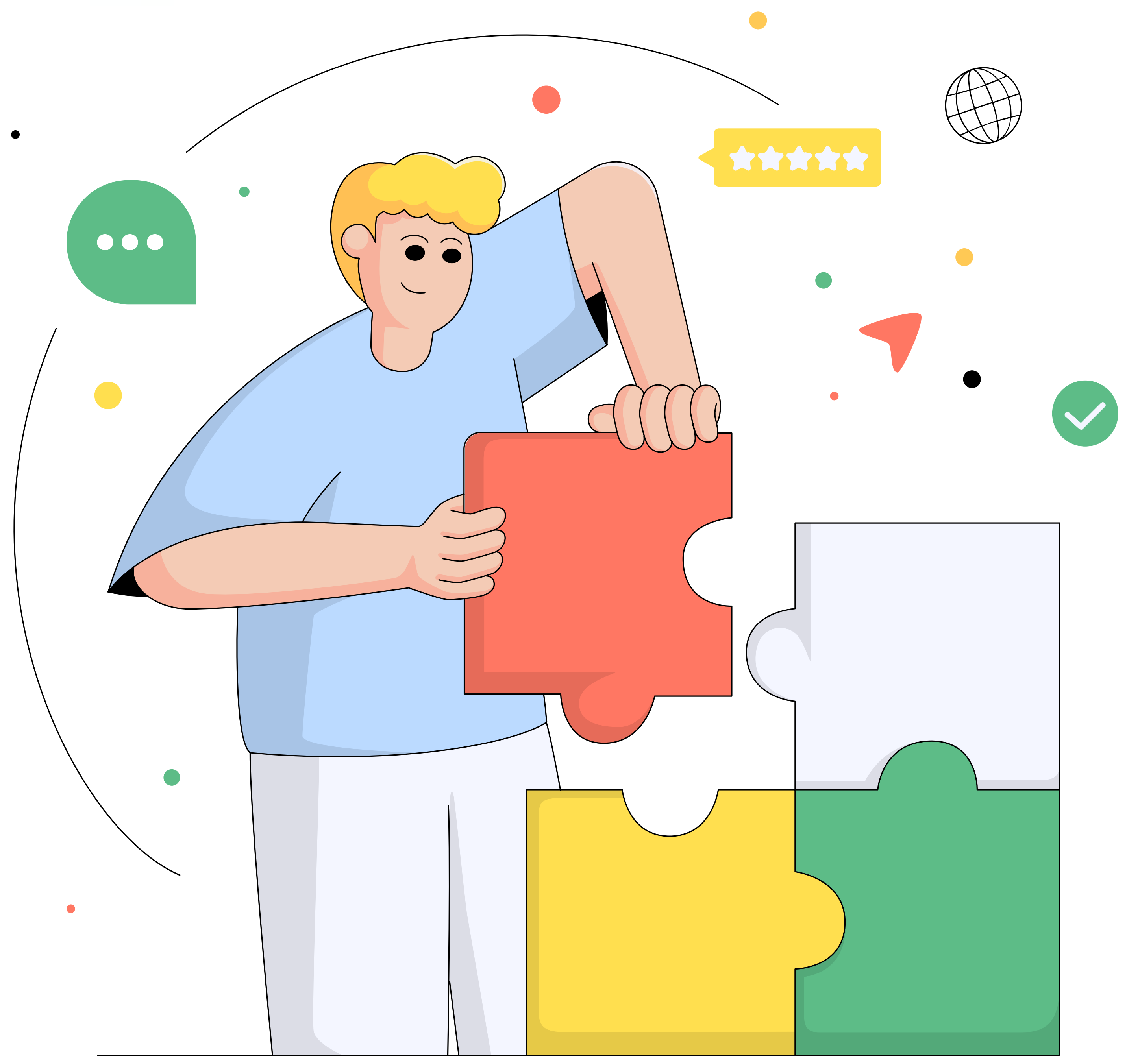
7 min Break

What have we accomplished so far?

- We saw how to use all the capabilities of SELECT AI
- We can query our DB using Natural Language

Pearson

What can be improved upon?



- I showed you SELECT AI running on my own account
- What are the steps necessary to get it working on yours?
- Let’s create a new user and grant all the permissions

Oracle Packages to Get Familiar with

- DBMS_NETWORK_ACL_ADMIN

Oracle Packages to Get Familiar with

- DBMS_NETWORK_ACL_ADMIN
- DBMS_CLOUD

Oracle Packages to Get Familiar with

- DBMS_NETWORK_ACL_ADMIN
- DBMS_CLOUD
- DBMS_CLOUD_AI

Pearson

Oracle Packages to Get Familiar with

- DBMS_NETWORK_ACL_ADMIN
- DBMS_CLOUD
- DBMS_CLOUD_AI
- These contain all of the procedures necessary to enable SELECT AI

Steps Involved to Enable SELECT AI

- (1) Create the new user

Steps Involved to Enable SELECT AI

- (1) Create the new user

- (2) Allow the user to access the internet

Steps Involved to Enable SELECT AI

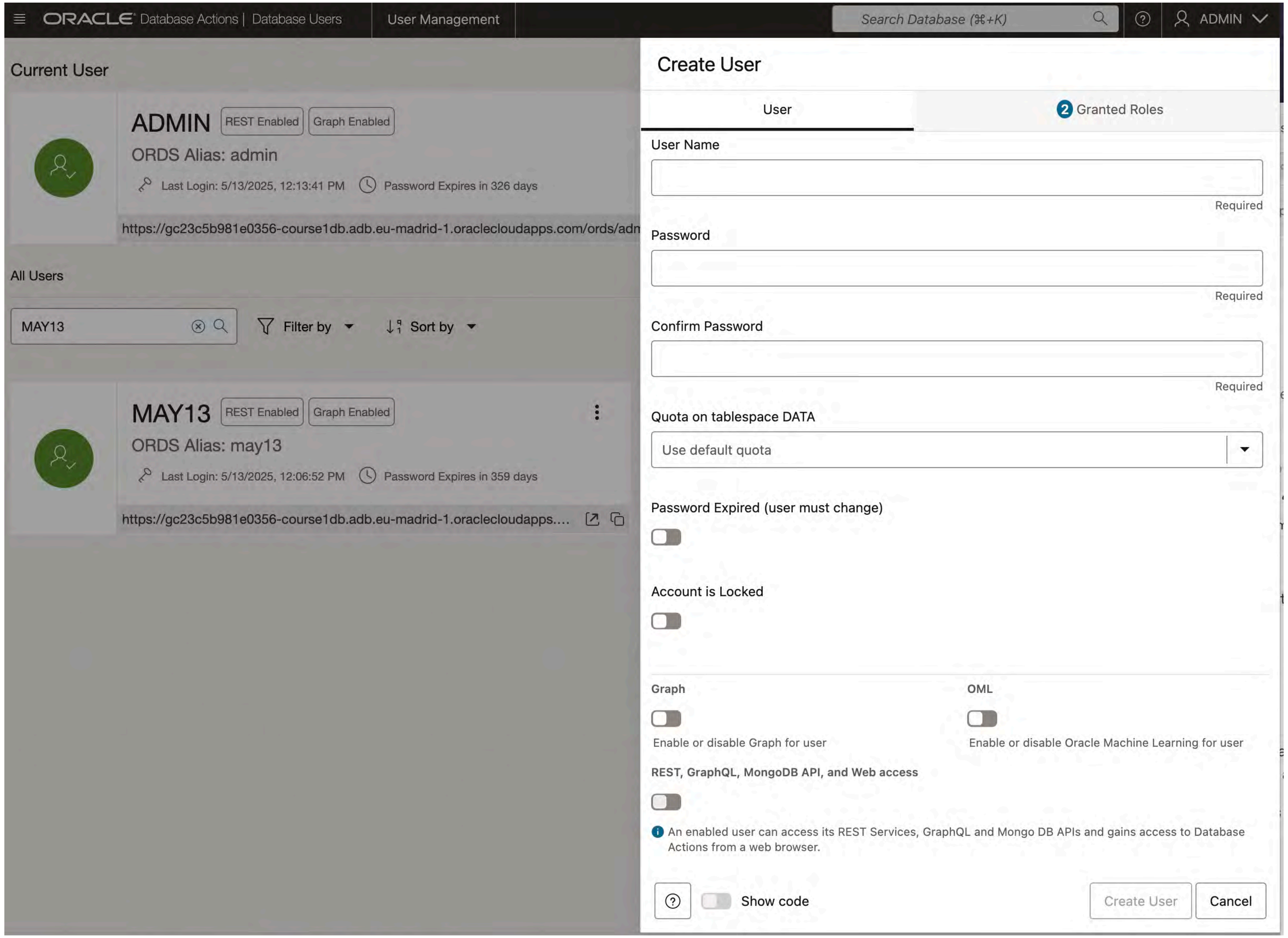
- (1) Create the new user
- (2) Allow the user to access the internet
- (3) Create a credential for the AI provider

Steps Involved to Enable SELECT AI

- (1) Create the new user
- (2) Allow the user to access the internet
- (3) Create a credential for the AI provider
- (4) Create an AI profile for the user that specifies the credential

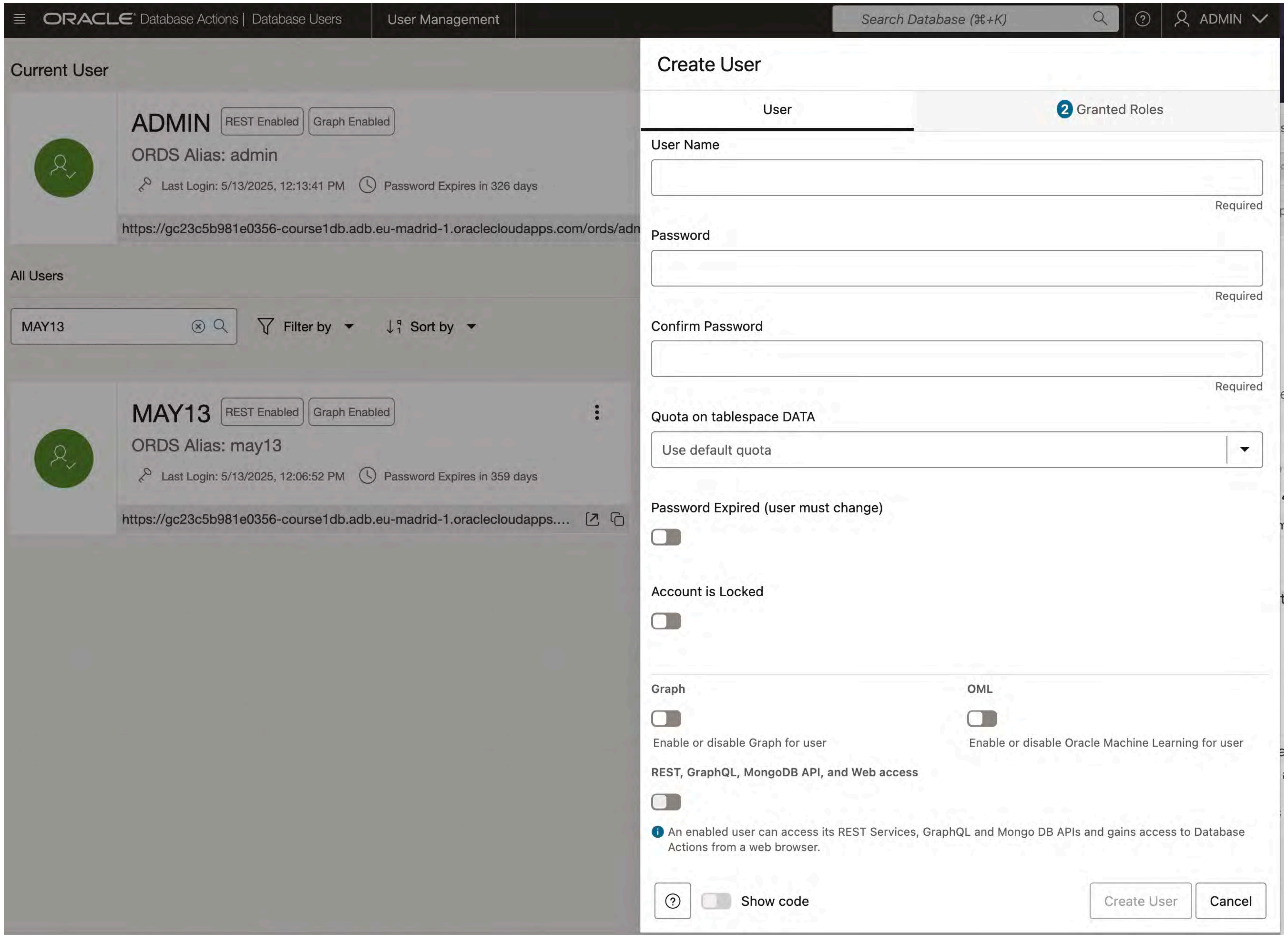
Step 1: Create the new user

Step 1: Create the new user



Pearson

Step 1: Create the new user



user_oreilly_nov25

Pearson

Create the new user

DEMO 6.1

Pearson

Step 2 - Allow the user to access the internet

Step 2 - Allow the user to access the internet

```
BEGIN
    DBMS_NETWORK_ACL_ADMIN.append_host_ace(
        HOST    => 'api.openai.com',
        ace     =>
            xs$ace_type(privilege_list   => XS$name_list('http'),
                        principal_name   => 'user_oreilly_nov25',
                        principal_type   => xs_acl.ptype_db));
END;
```

Step 2 - Allow the user to access the internet

```sql
SET DEFINE OFF;
GRANT RESOURCE TO user_oreilly_nov25;
GRANT CREATE SESSION TO user_oreilly_nov25;
GRANT CREATE VIEW TO user_oreilly_nov25;
GRANT CREATE TABLE TO user_oreilly_nov25;
GRANT CONNECT TO user_oreilly_nov25;
GRANT ALTER SYSTEM TO user_oreilly_nov25;
GRANT ALTER USER TO user_oreilly_nov25;
ALTER USER user_oreilly_nov25 QUOTA 10M ON temp;
GRANT CONSOLE_DEVELOPER TO user_oreilly_nov25;
GRANT DWROLE TO user_oreilly_nov25;
GRANT EXECUTE ON DBMS_CLOUD TO user_oreilly_nov25;
GRANT EXECUTE ON DBMS_CLOUD_AI TO user_oreilly_nov25;
```

Allow Access to the Internet and Grants

DEMO 6.2

Pearson

Step 3 - Create a credential for the AI provider

- What is a “credential”?

Step 3 - Create a credential for the AI provider

- What is a “credential”?

- It’s essentially a PL/SQL wrapper for your API key

Step 3 - Create a credential for the AI provider

- What is a "credential"?

- It's essentially a PL/SQL wrapper for your API key

- When you access **external** AI services programmatically, you will need an API key

Pearson

Supported AI Providers

OpenAI

Gemini

Claude

Microsoft Azure

cohere

HUGGING FACE

Supported AI Providers

OpenAI

Gemini

Claude

Microsoft Azure

cohere

HUGGING FACE

OCI Generative AI

* not available in all regions

Step 3 - Create a credential for the AI provider

Step 3 - Create a credential for the AI provider

```
BEGIN
  DBMS_CLOUD.create_credential(
    credential_name => 'CREDENTIAL_OPENAI',
    username        => 'OPENAI',
    password        => 'sk-proj-ZfJoYrFQsUyGNA5xxxxx'
  );
END;
```

Step 3 - Create a credential for the AI provider

```
BEGIN
  DBMS_CLOUD.create_credential(
    credential_name => 'CREDENTIAL_OPENAI',
    username        => 'OPENAI',
    password        => 'sk-proj-ZfJoYrFQsUyGNA5xxxxx'
  );
END;
```

Documentation:

https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/select-ai-manage-profiles.html

Step 4 - Create an AI Profile

- What is an "AI Profile"?
- It contains 2 key items:

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AI Profile

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Step 4 - Create an AI Profile

- What is an “AI Profile”?
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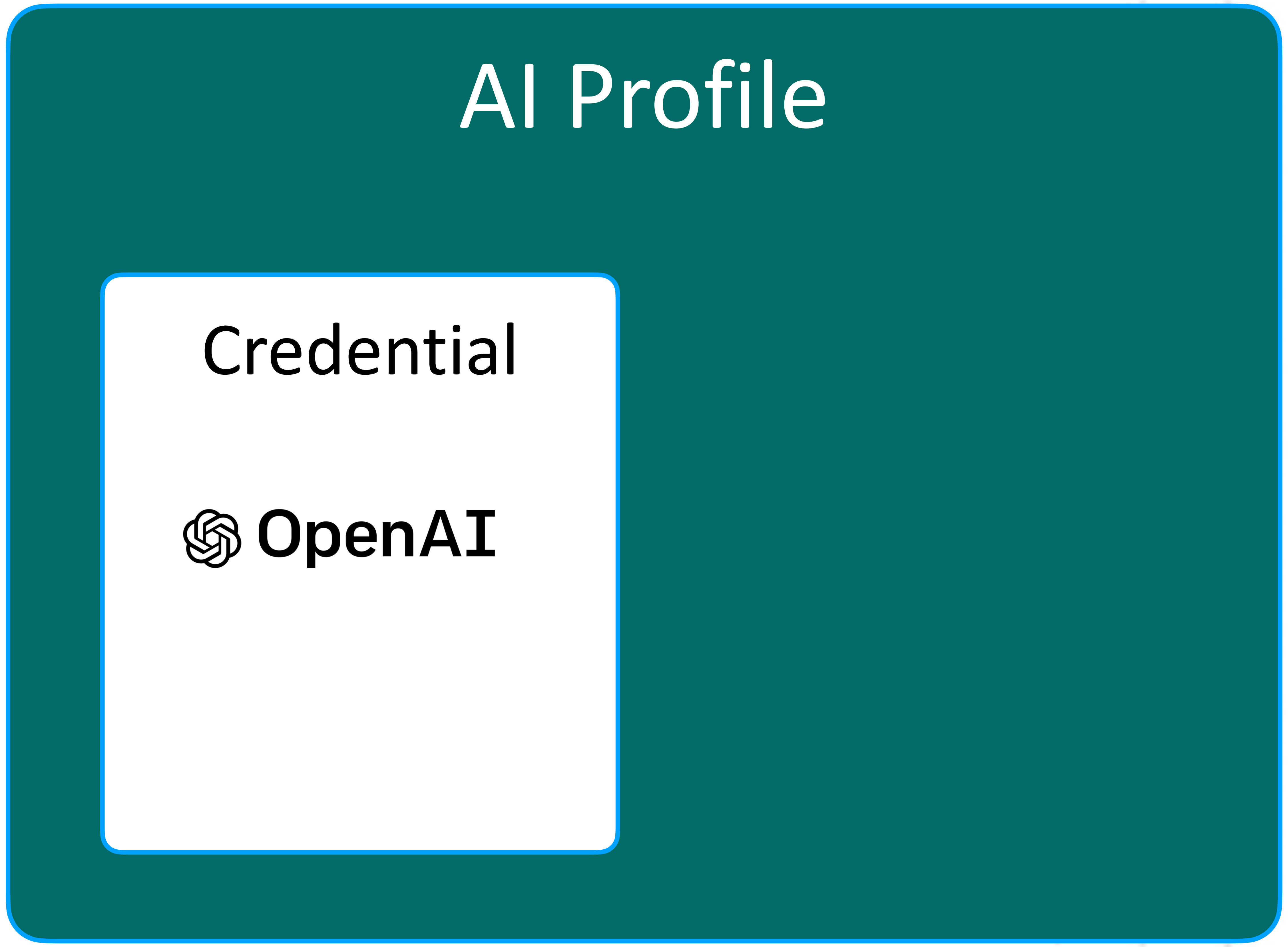
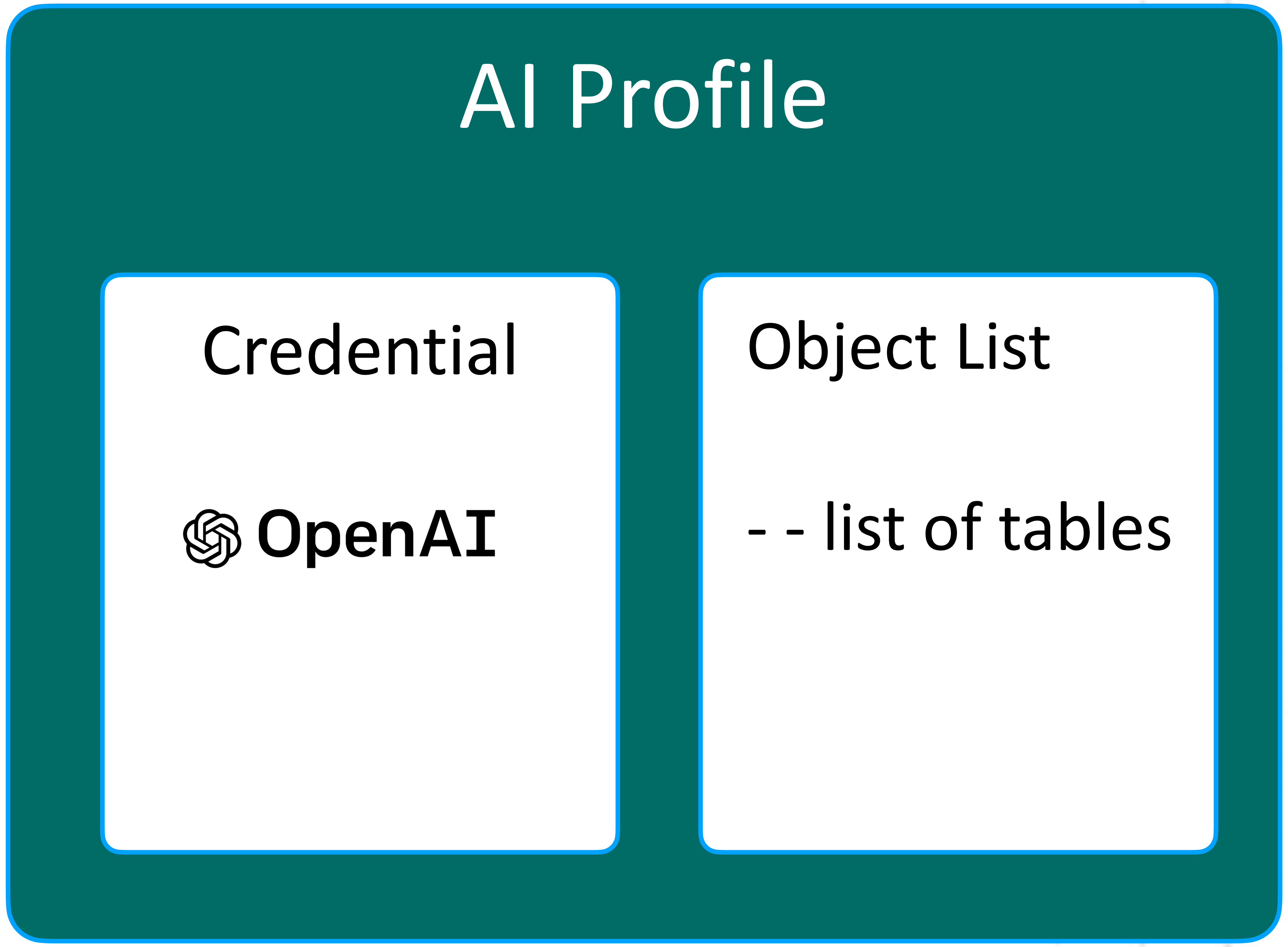


Step 4 - Create an AI Profile

- What is an "AI Profile"?
- It contains 2 key items:
- Your credential to the AI provider
- The list of tables your want the AI provider to use



Step 4 - Create an AI Profile

```
BEGIN

    DBMS_CLOUD_AI.create_profile('PROFILE_OPENAI_EATS', '{
        "provider": "openai",
        "model": "gpt-4o",
        "credential_name": "CREDENTIAL_OPENAI",
        "object_list": [{"owner": "user_oreilly_nov25", "name": "RESTAURANTS"},
                        {"owner": "user_oreilly_nov25", "name": "DELIVERY_OPTIONS"},
                        {"owner": "user_oreilly_nov25", "name": "RESTAURANT_SCHEDULES"},
                        {"owner": "user_oreilly_nov25", "name": "MENU_ITEMS"},
                        {"owner": "user_oreilly_nov25", "name": "PROMOTIONS"},
                        {"owner": "user_oreilly_nov25", "name": "ITEM_OPTIONS"}],
        "conversation": "true"
       }');
END;
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```

Documentation:

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Let's briefly talk about AI models

Compare models

o4-mini	o3	GPT-4.1
o4-mini	o3	GPT-4.1
Faster, more affordable reasoning model	Our most powerful reasoning model	Flagship GPT model for complex tasks
Learn more / Playground	Learn more / Playground	Learn more / Playground

o4-mini		o3		GPT-4.1	
Reasoning	●●●●	Reasoning	●●●●●	Intelligence	●●●●
Speed	⚡⚡⚡	Speed	⚡	Speed	⚡⚡⚡
Input		Input		Input	
Output		Output		Output	
Reasoning tokens	✓	Reasoning tokens	✓	Reasoning tokens	✗
PRICING	PER 1M TOKENS	PRICING	PER 1M TOKENS	PRICING	PER 1M TOKENS
Input	$1.10	Input	$10.00	Input	$2.00
Cached Input	$0.28	Cached Input	$2.50	Cached Input	$0.50
Output	$4.40	Output	$40.00	Output	$8.00

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o4-mini

Faster, more affordable reasoning model

Learn more | Playground

Reasoning	
Speed	
Input	
Output	
Reasoning tokens	

PRICING	PER 1M TOKENS
Input	$1.10

o3

Our most powerful reasoning model

Learn more | Playground

Reasoning	
Speed	
Input	
Output	
Reasoning tokens	

PRICING	PER 1M TOKENS
Input	$10.00

GPT-4.1

Flagship GPT model for complex tasks

Learn more | Playground

Intelligence	
Speed	
Input	
Output	
Reasoning tokens	

PRICING	PER 1M TOKENS
Input	$2.00

Create Credential and AI Profile

DEMO 6.3

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Questions?

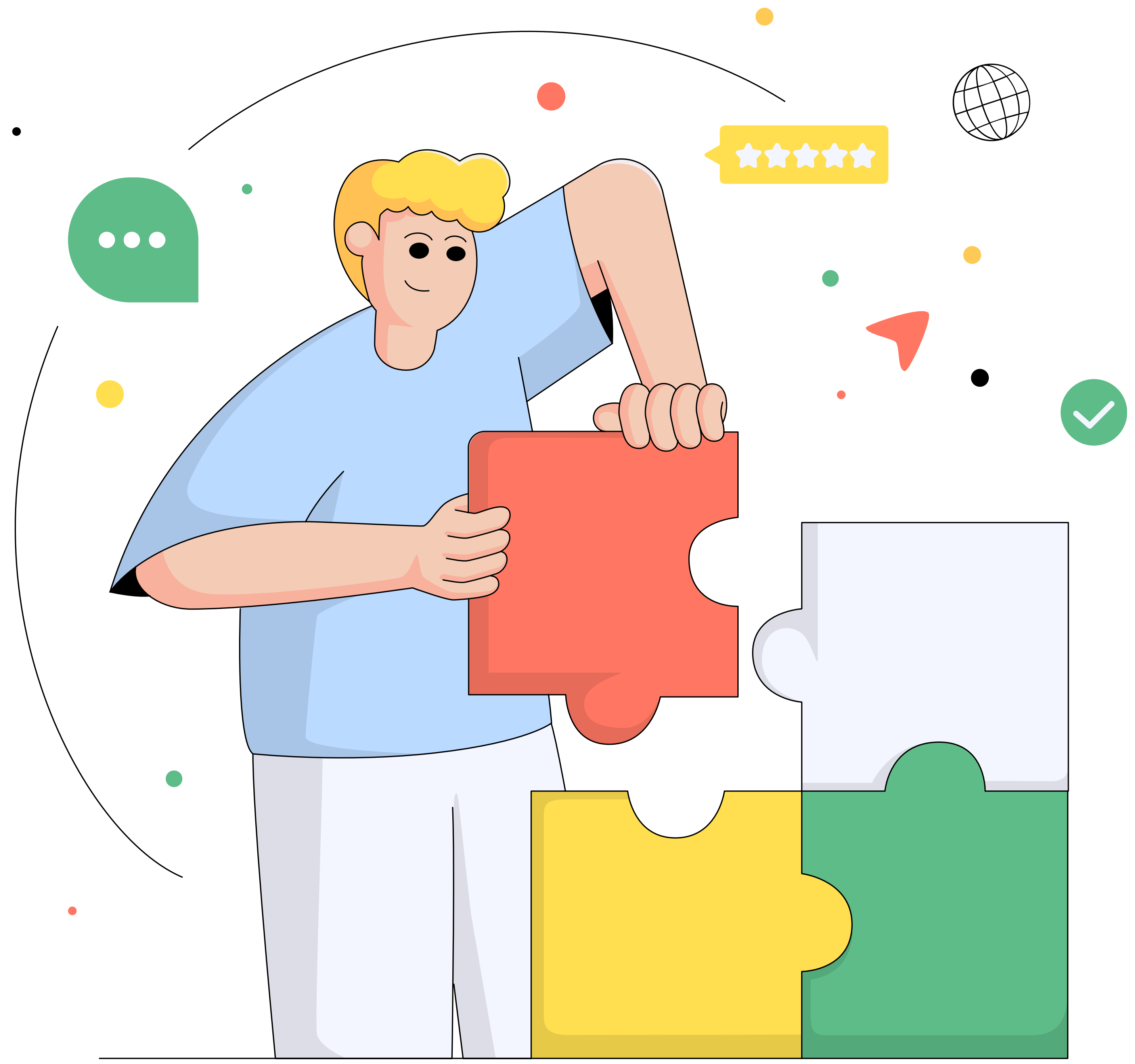
PART 7

Using MCP with the **PostgreSQL Database**

What have we accomplished so far?

- We are very familiar configuring MCP clients
- We know how build MCP servers in 3 languages: Python, Java, Node.js

What can be improved upon?



- One of the major points about MCP, is that we don't always have to build everything ourselves

- Let's learn how to accomplish things with one of the ~~700~~ the **1000+** MCP servers available

You Work for a Shoe Retailer

Problem: Your CIO is Tired Custom SW Projects

- Let’s talk about real numbers

Problem: Your CIO is Tired Custom SW Projects

- Let's talk about real numbers
- **31.1% of internal software** projects are canceled before completion

Problem: Your CIO is Tired Custom SW Projects

- Let's talk about real numbers
- **31.1% of internal software** projects are canceled before completion
- **52.7% exceed** their original budgets

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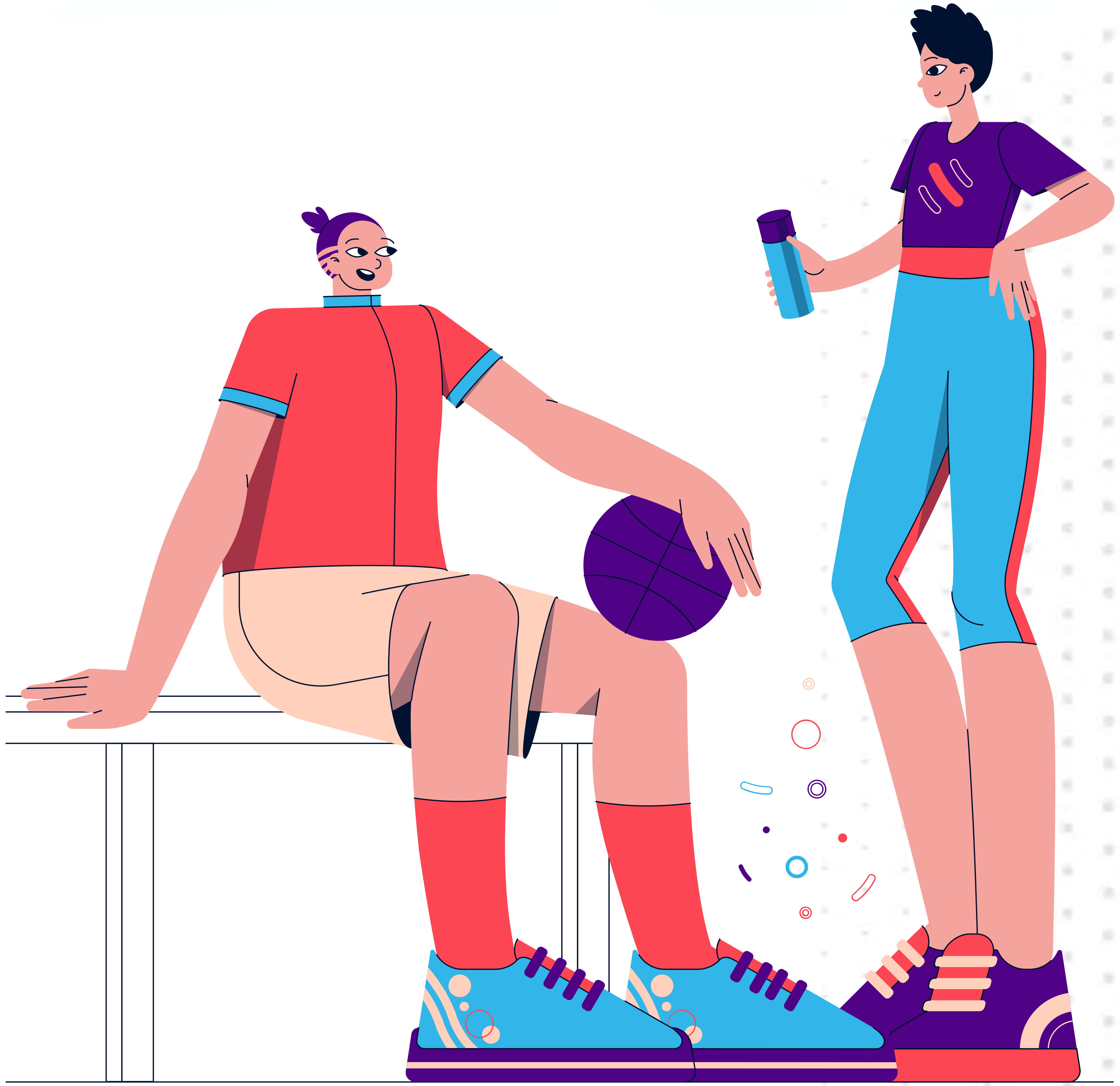
Management Has a Question

What % of customers shop online vs retail?



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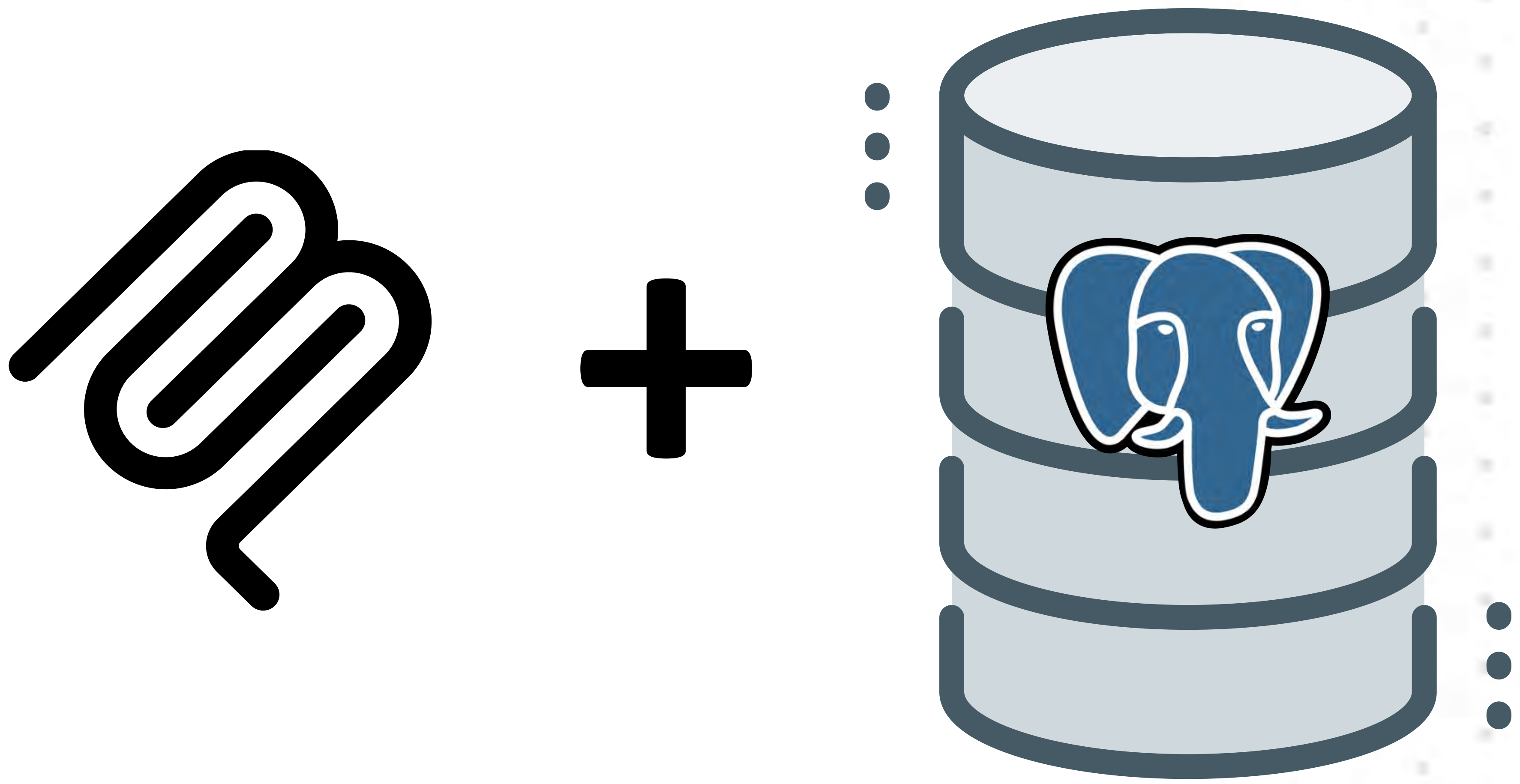
Solution: Off the Shelf MCP Server

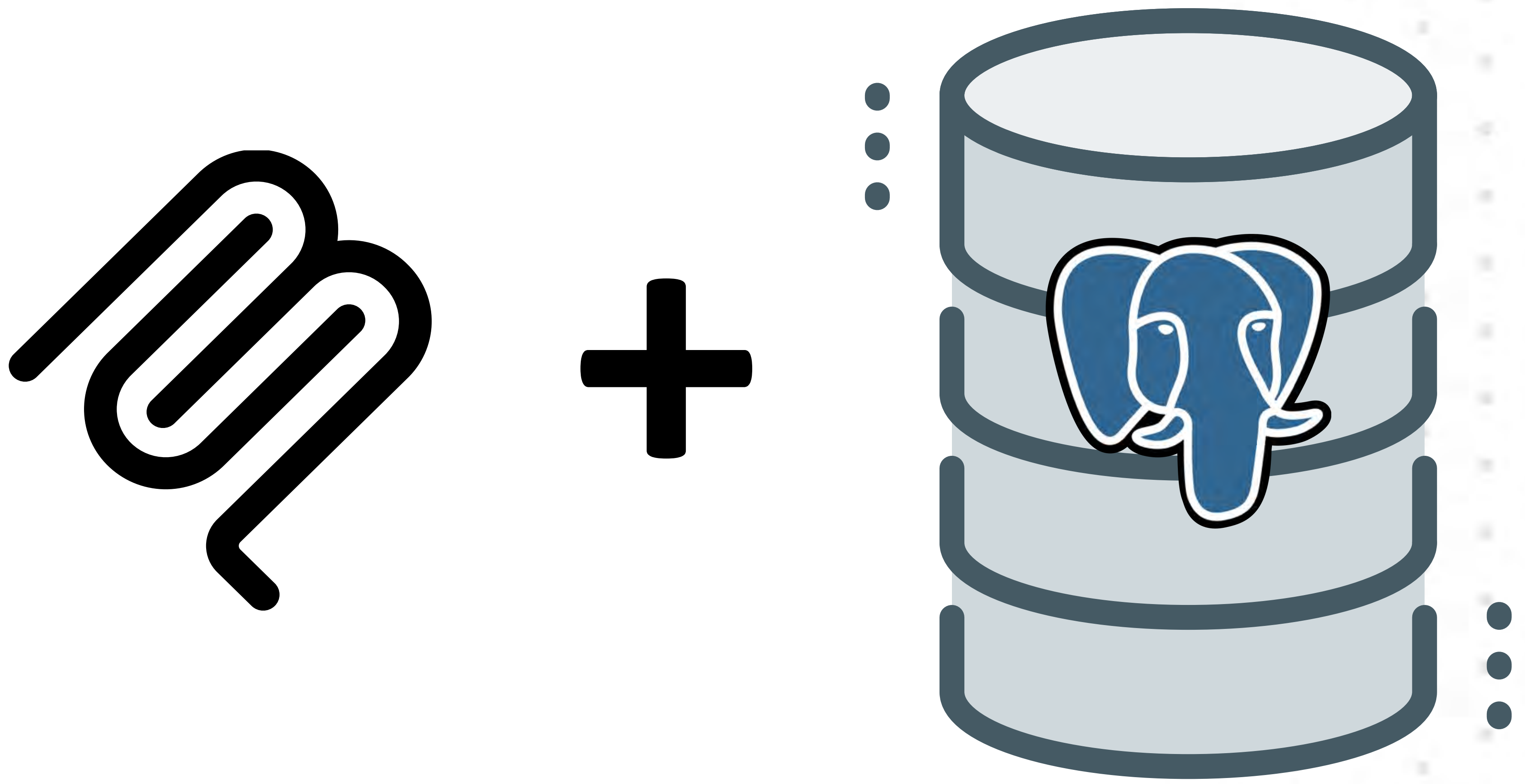
- Let’s use an MCP Server to interface with our database



Solution: Off the Shelf MCP Server

- Let's use an MCP Server to interface with our database

- Instead of building a custom application, let our managers talk to the DB themselves

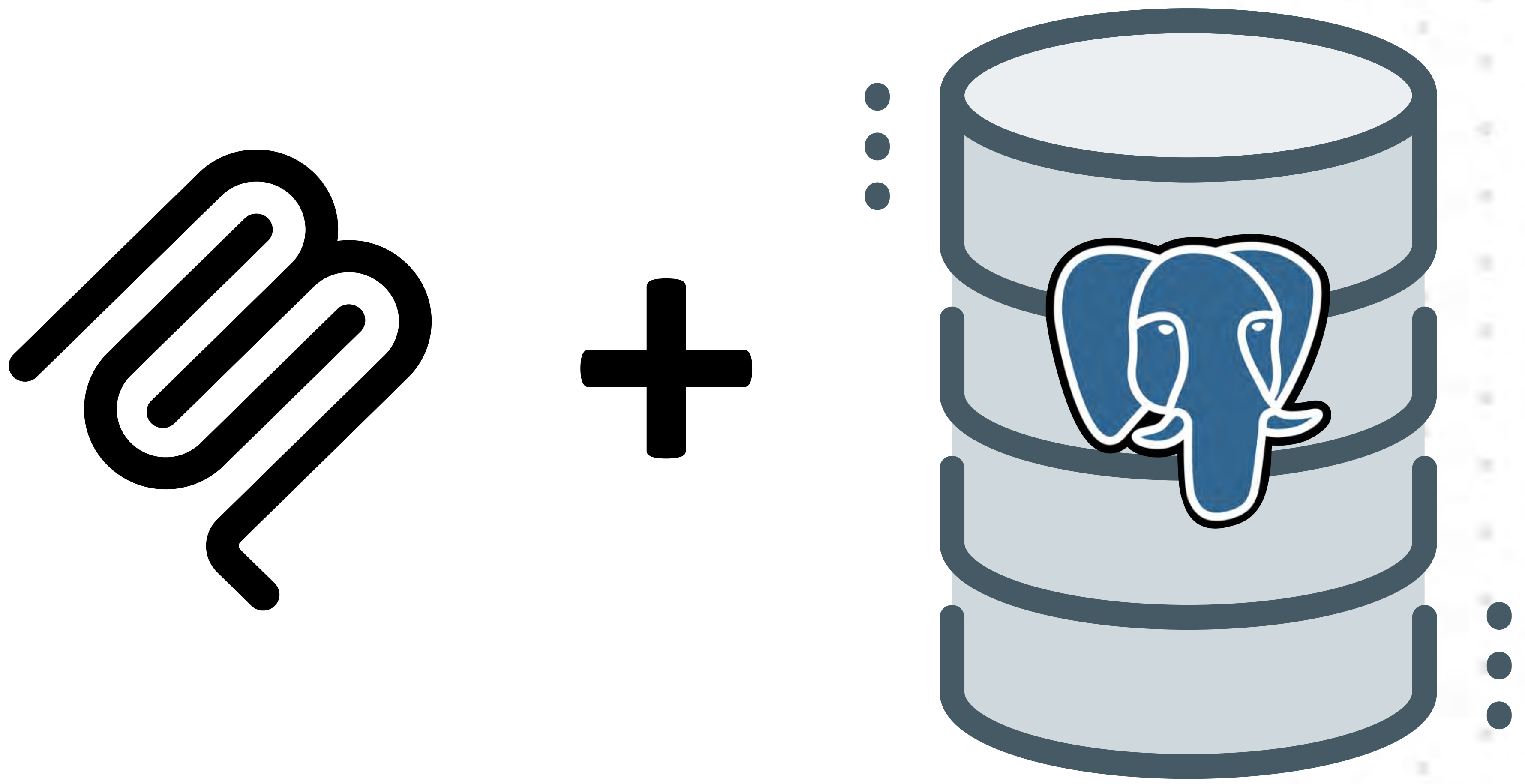


Solution: Off the Shelf MCP Server

- Let's use an MCP Server to interface with our database

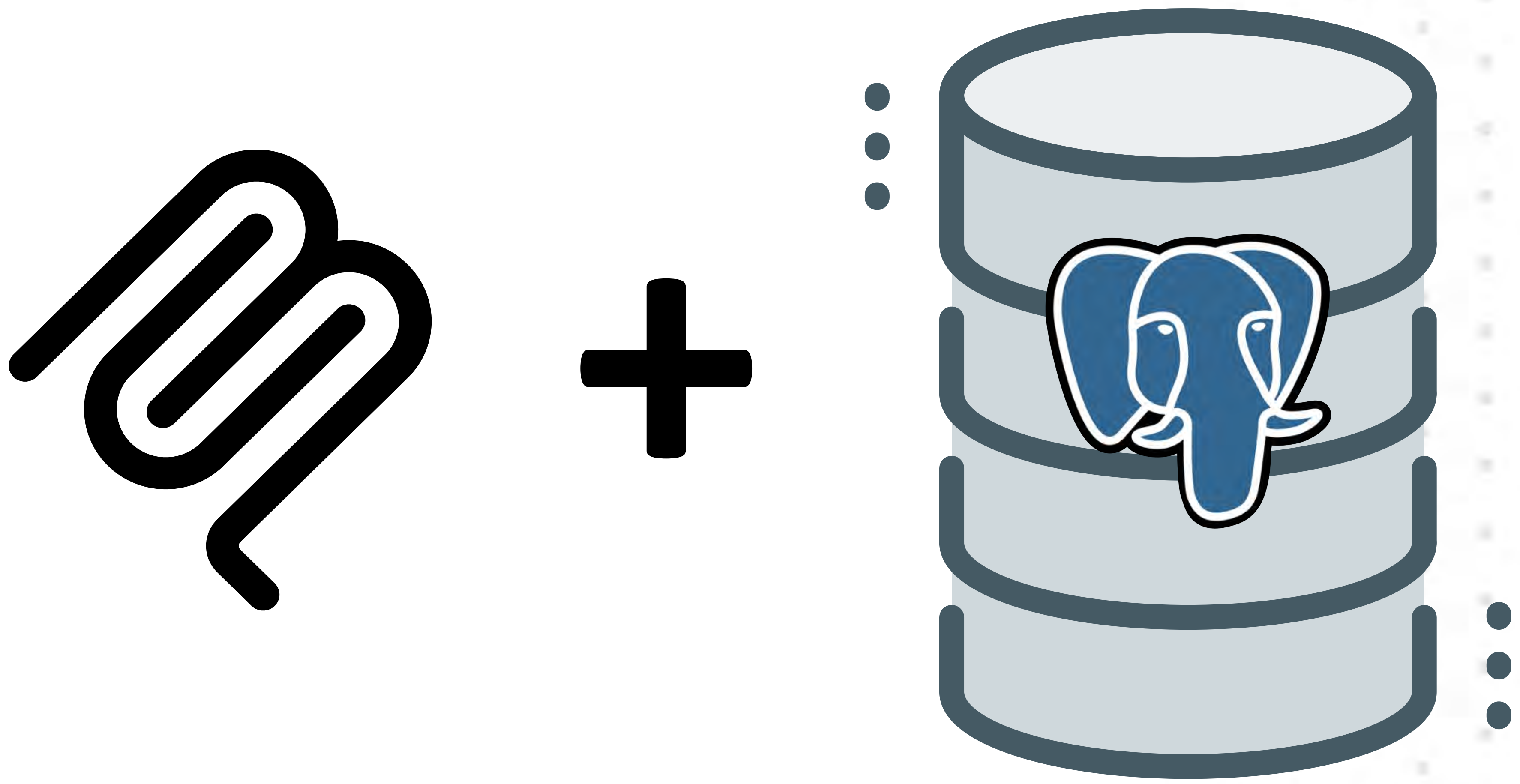
- Instead of building a custom application, let our managers talk to the DB themselves

- We have a few options



PostgreSQL MCP Servers

- Original Anthropic MCP Server: archived https://github.com/modelcontextprotocol/servers-archived/tree/main/src/postgres



PostgreSQL MCP Servers

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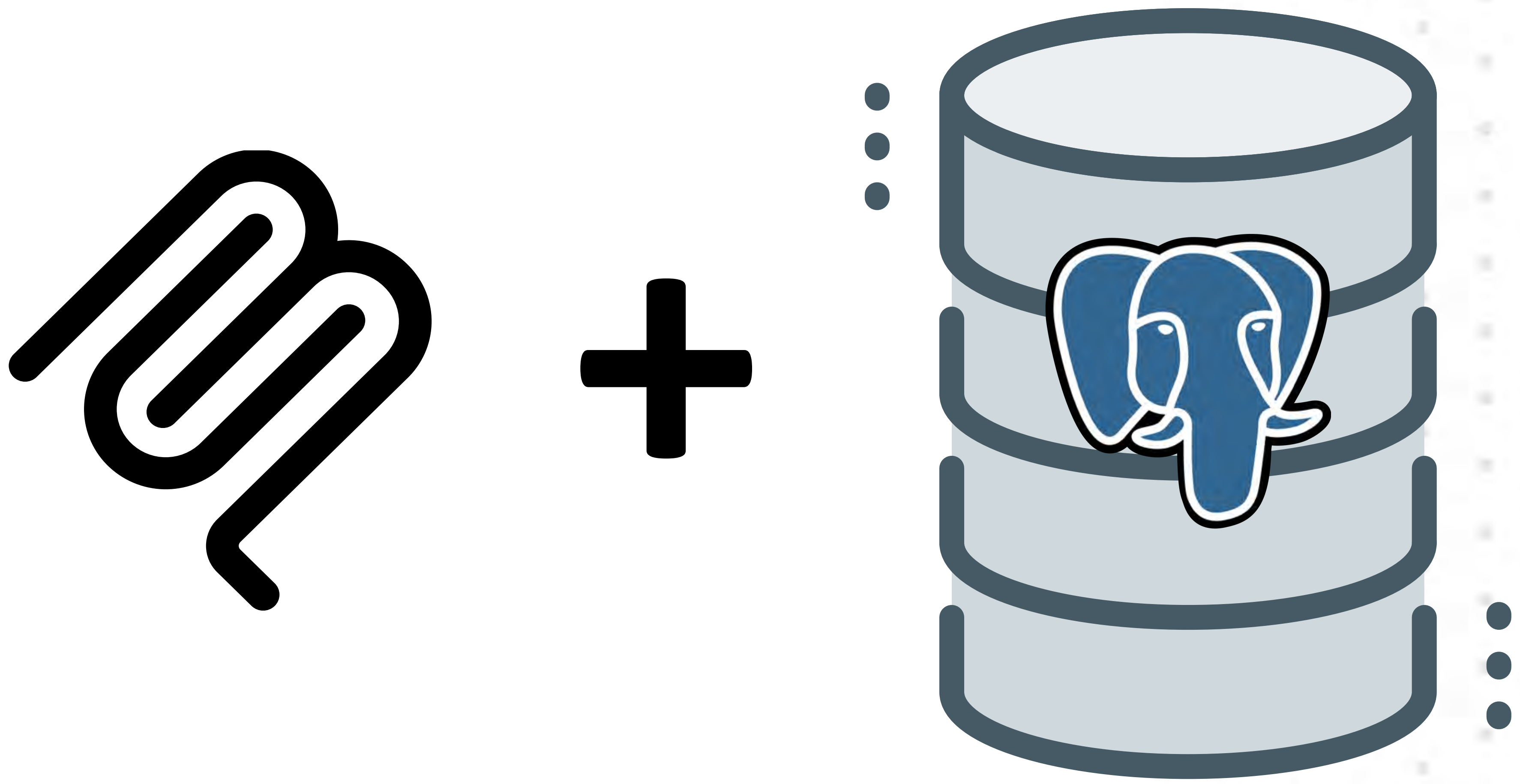
- HenkDz: https://github.com/HenkDz/postgresql-mcp-server

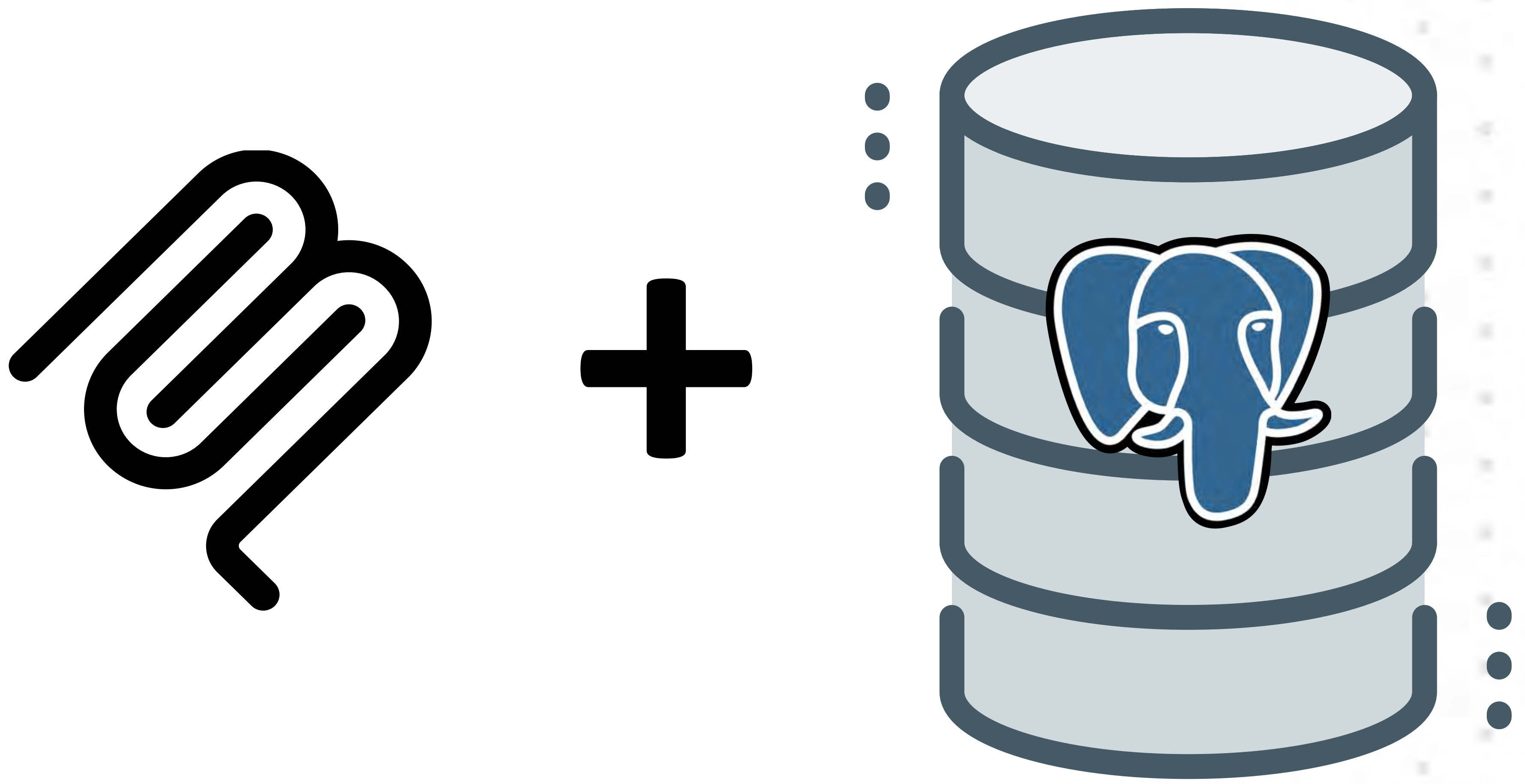


PostgreSQL MCP Servers

- Original Anthropic MCP Server: archived https://github.com/modelcontextprotocol/servers-archived/tree/main/src/postgres

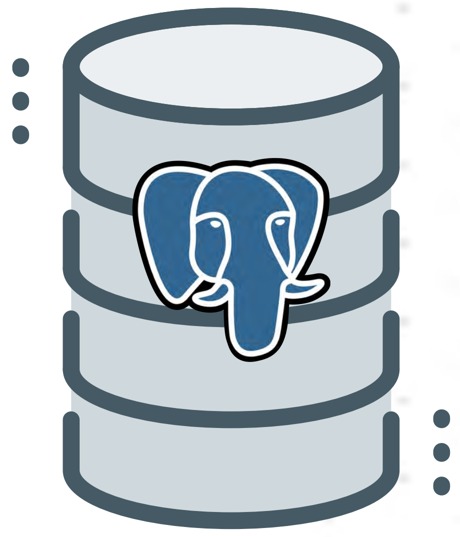
- HenkDz: https://github.com/HenkDz/postgresql-mcp-server

- Crystal DBA: https://github.com/crystaldba/postgres-mcp



How to Setup Your MCP Server with PostgreSQL

- Pre-requisites



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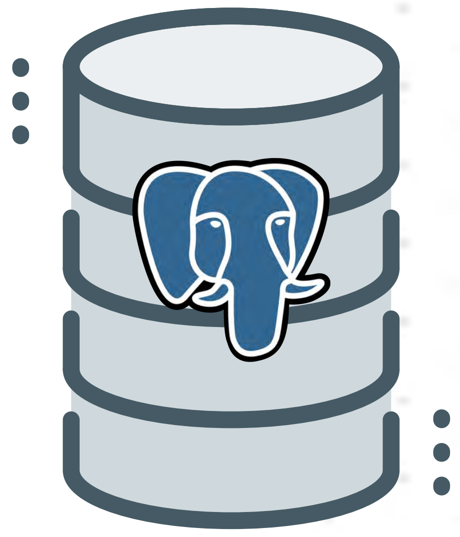
How to Setup Your MCP Server with PostgreSQL

- Pre-requisites

- PostgreSQL DB
  (I’m using v 16)



How to Setup Your MCP Server with PostgreSQL

- Pre-requisites
- PostgreSQL DB
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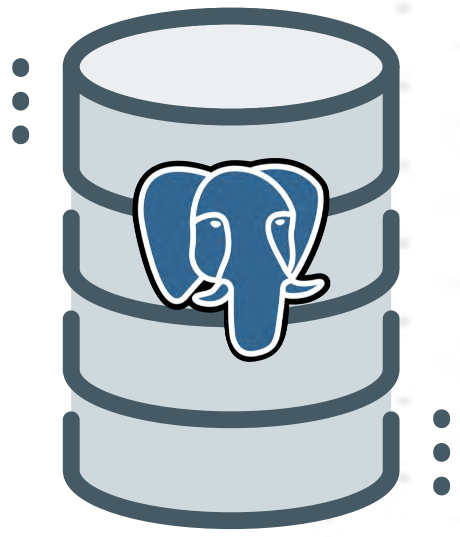

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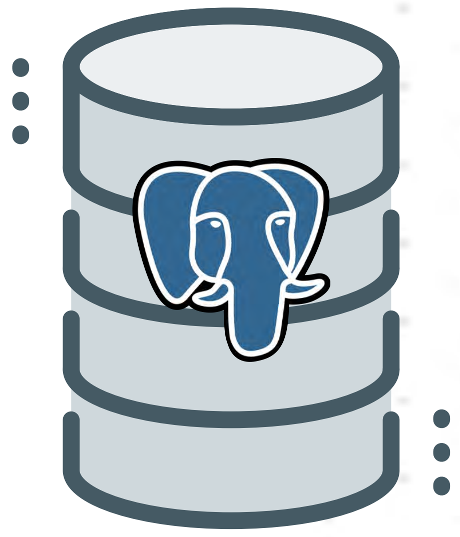
How to Setup Your MCP Server with PostgreSQL

- Pre-requisites

- PostgreSQL DB
  (I’m using v 16)

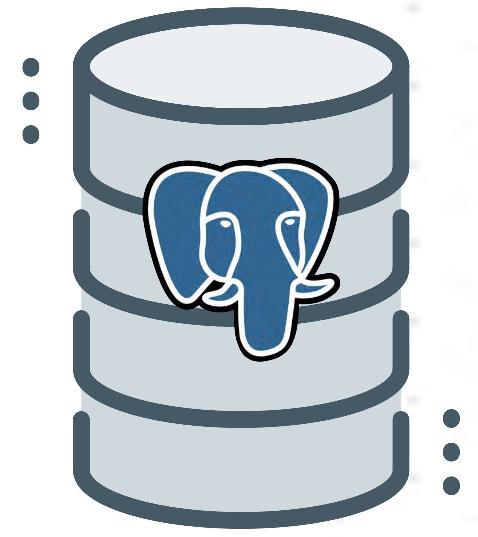
- Node.js

- Claude Desktop



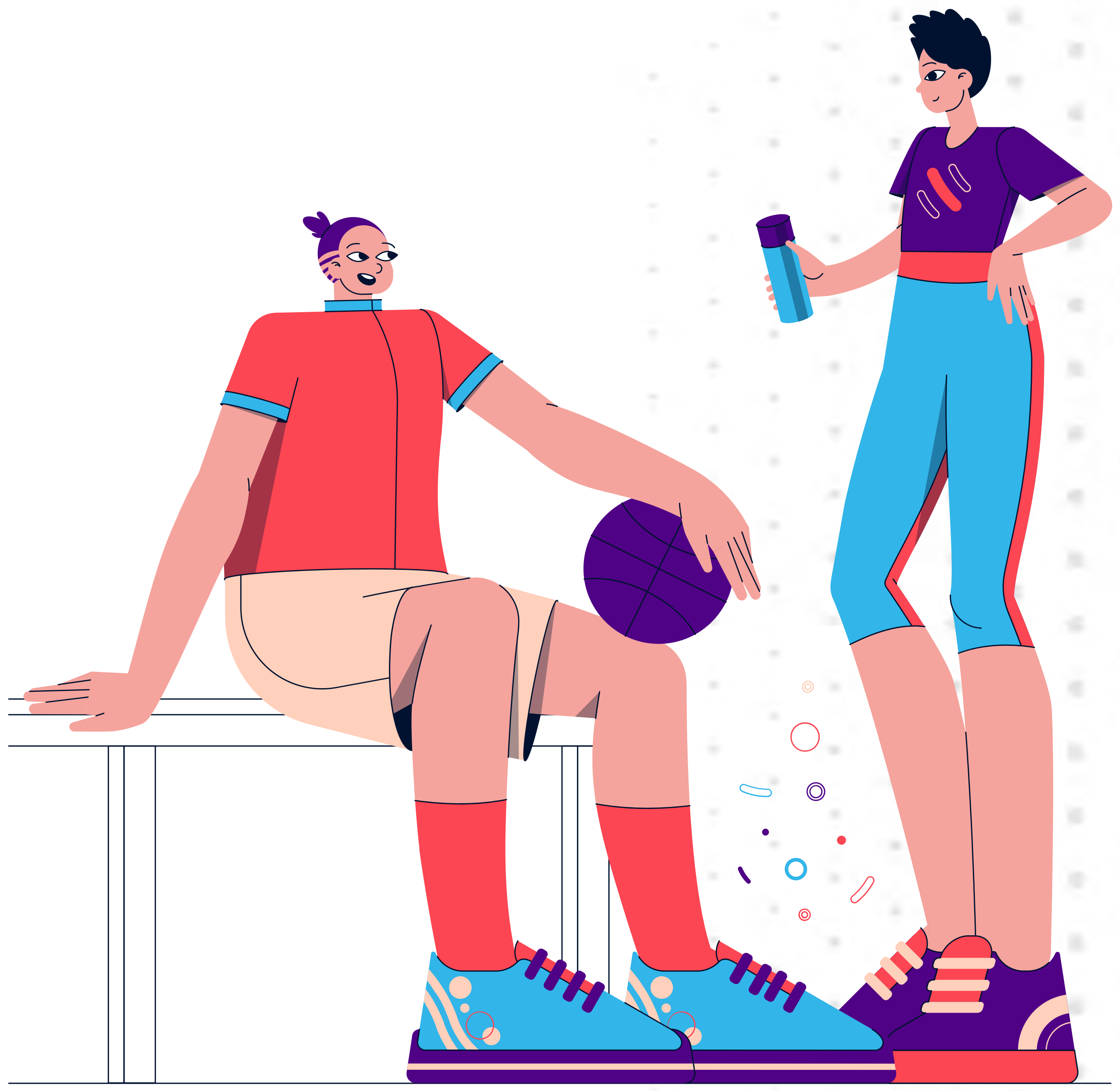
Claude Configuration

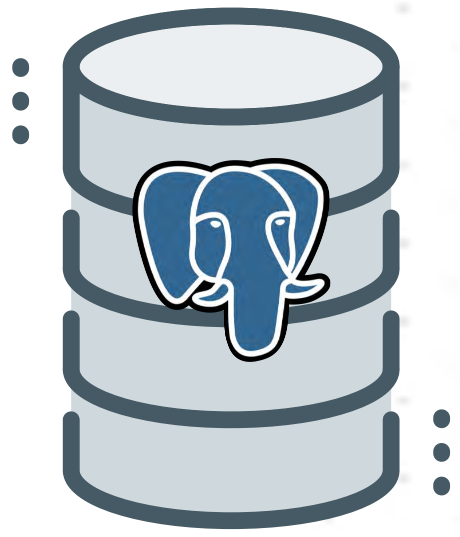
```
{
  "mcpServers": {
    "postgresql-mcp": {
      "command": "npx",
      "args": ["@henkey/postgres-mcp-server"],
      "env": {
        "POSTGRES_CONNECTION_STRING": "postgresql://
postgres:postgres@localhost:5432/shoe_store_analytics"
      }
    }
  }
}
```



Postgres + MCP

DEMO 7





DEMO 6 - Prompts Used

PROMPT:

1. "How many customers prefer online shopping vs retail stores?"
2. "What's the total revenue from online sales compared to retail sales?"
3. "Which products are most popular in retail stores?"
4. "Show me customers who prefer online shopping but have made purchases in retail stores"
5. "What's the average order value for each purchase location?"
6. "Which Nike products have generated the most revenue?"
7. "How many orders were placed in the last 30 days?"
8. "What's the distribution of customers across different states?"
9. "Which retail store has the highest sales volume?"
10. "Show me the top 5 customers by total spending"

Ramifications

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Ramifications

- This will lead to a new generation of applications: NO User Interfaces

Ramifications

- This will lead to a new generation of applications: NO User Interfaces

- The biggest impact: Internal corporate apps

Ramifications

- This will lead to a new generation of applications: NO User Interfaces
- The biggest impact: Internal corporate apps
- The demo used PostgreSQL, but the technique can be applied to **any database**

Questions?

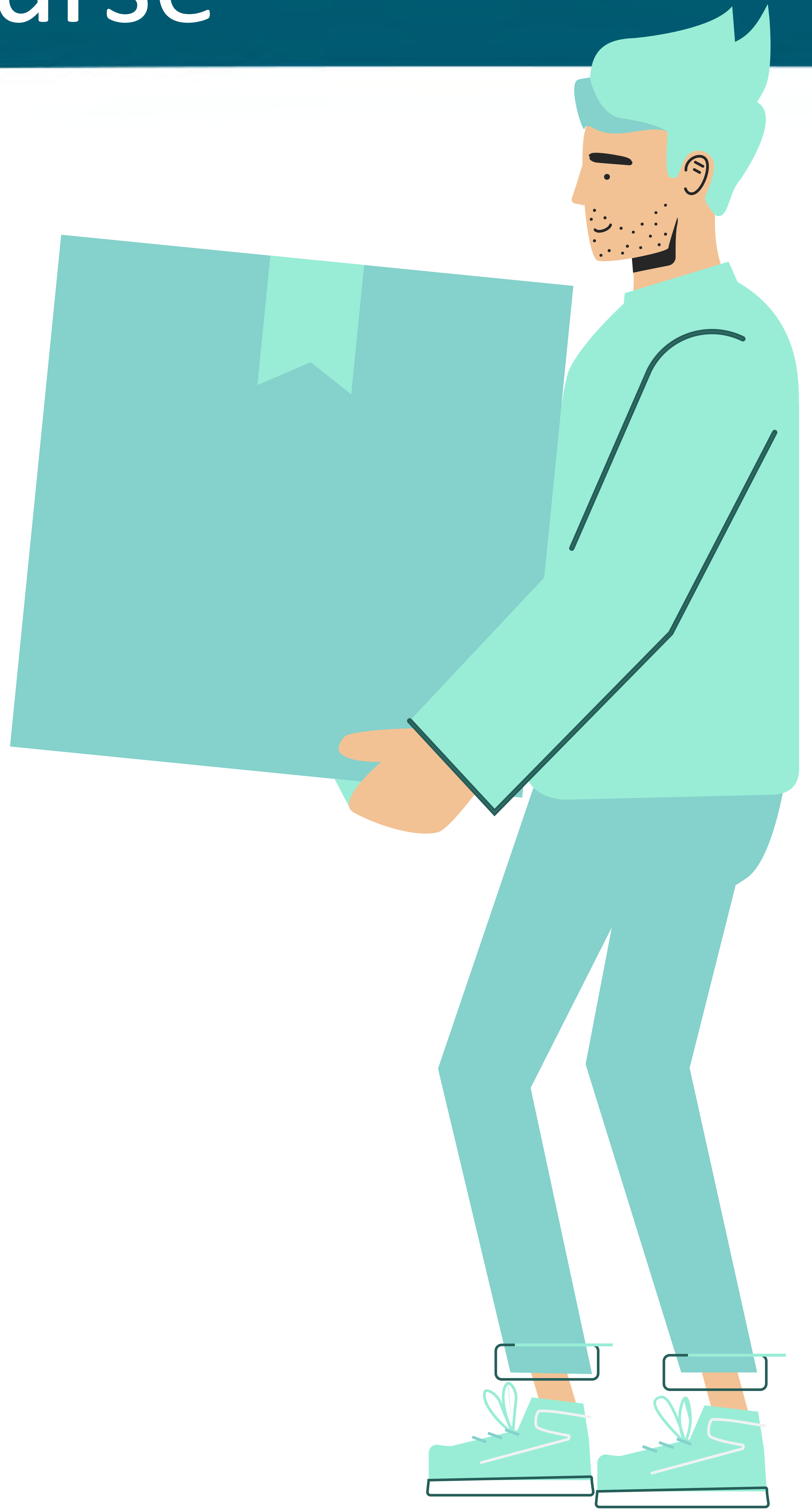
PART 7

Course Wrap-up

Key Takeaways from This Course

PART 1

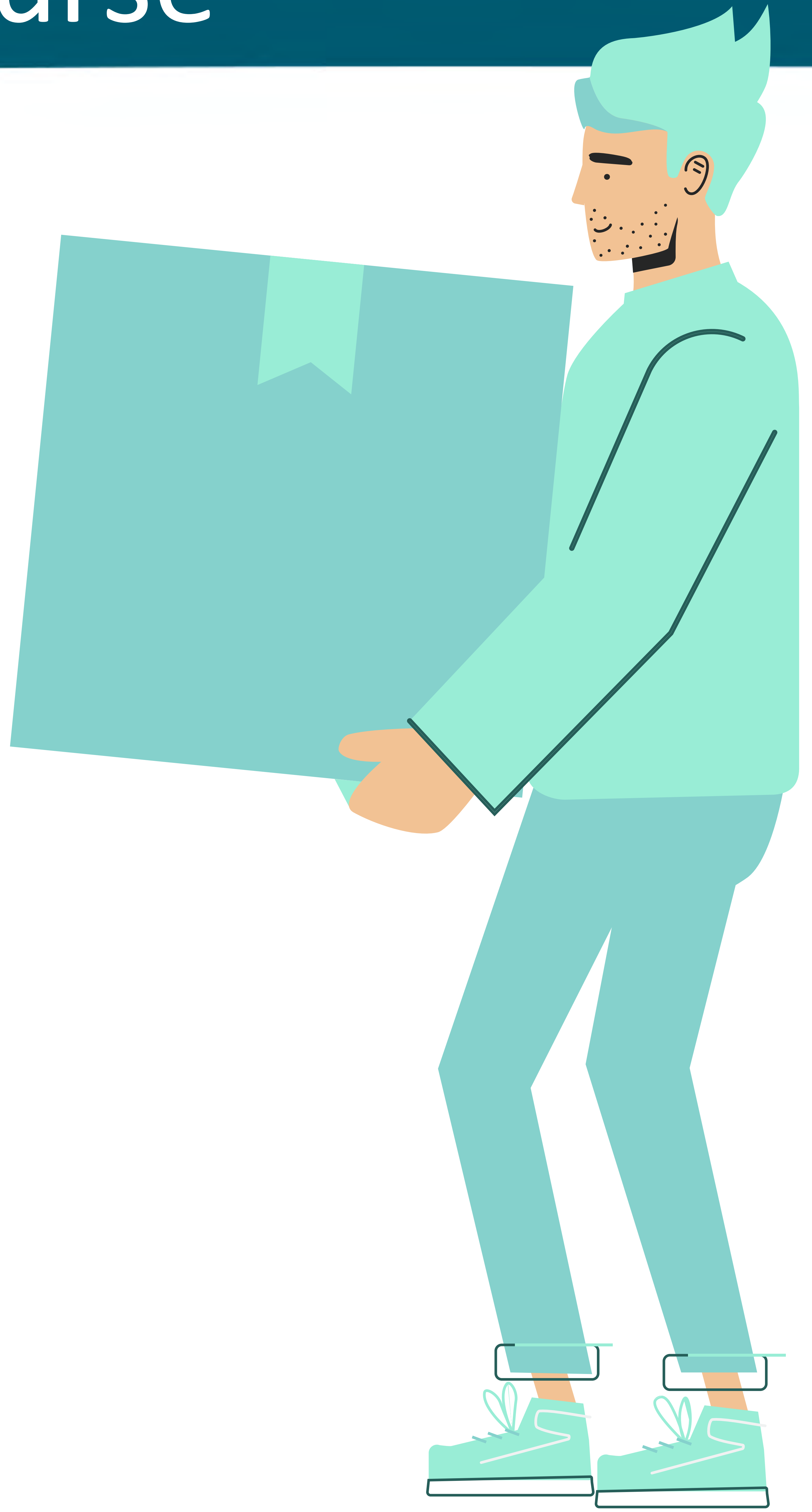
You will be get familiar with the features and capabilities of the Oracle Autonomous Database 26ai



Key Takeaways from This Course

PART 2

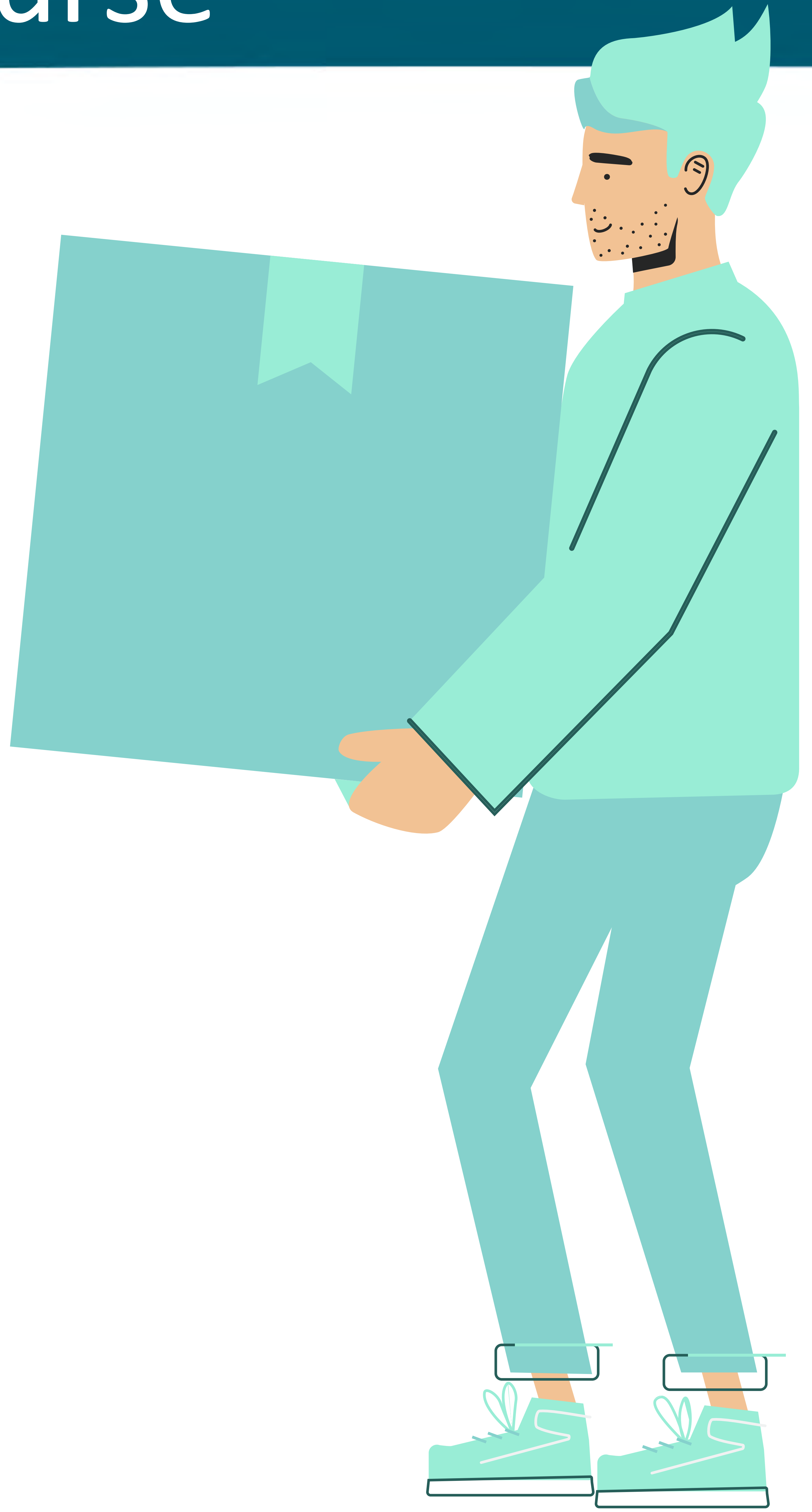
You will learn how to automatically create database schemas for POC type applications



Key Takeaways from This Course

PART 3

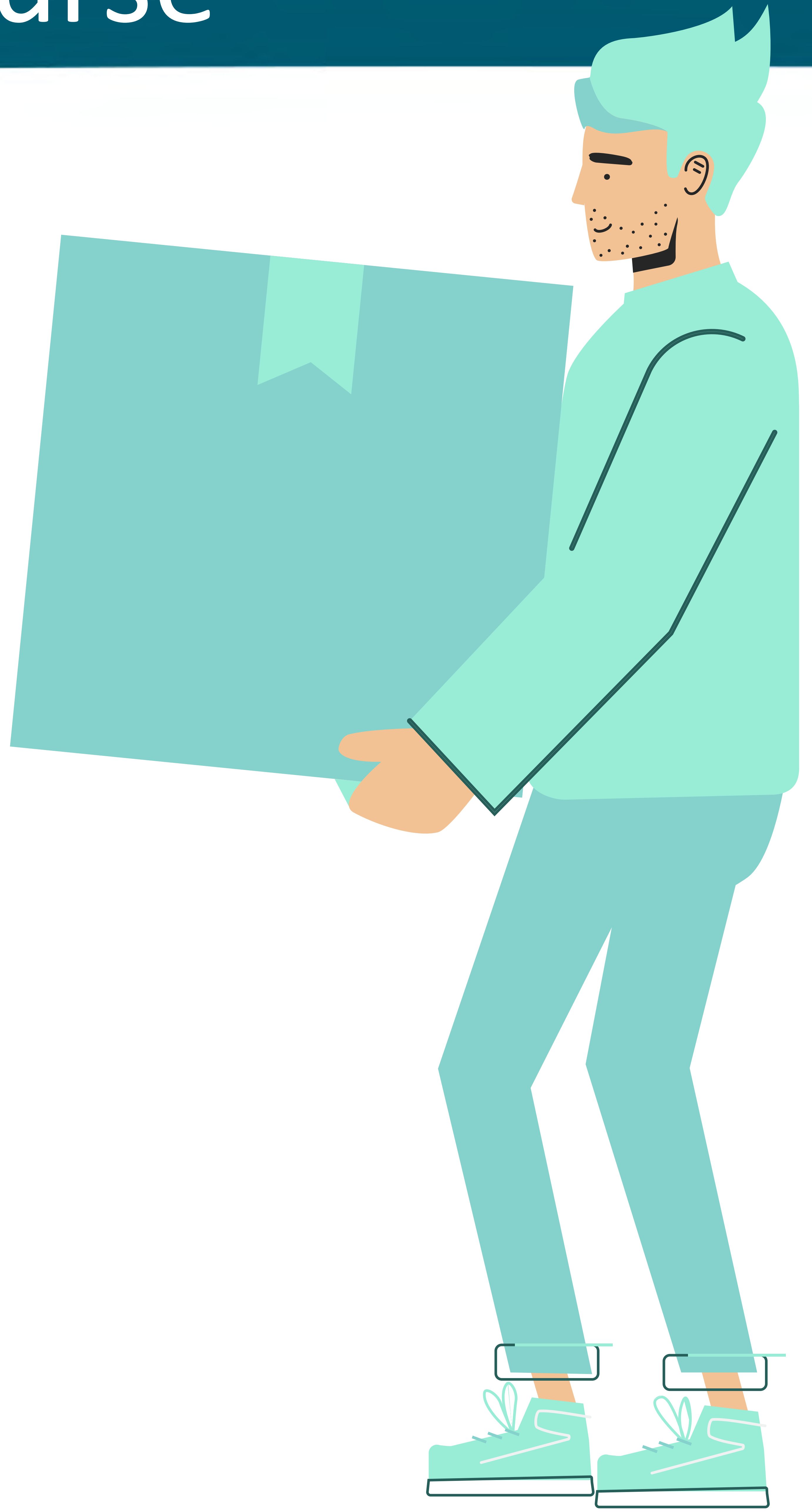
You will be learn how to use AI and ChatGPT effectively in order to extract, transform, and load (ETL) data in your database



Key Takeaways from This Course

PART 4

You will learn how to extract structured data from LOB (Large Object) datatypes



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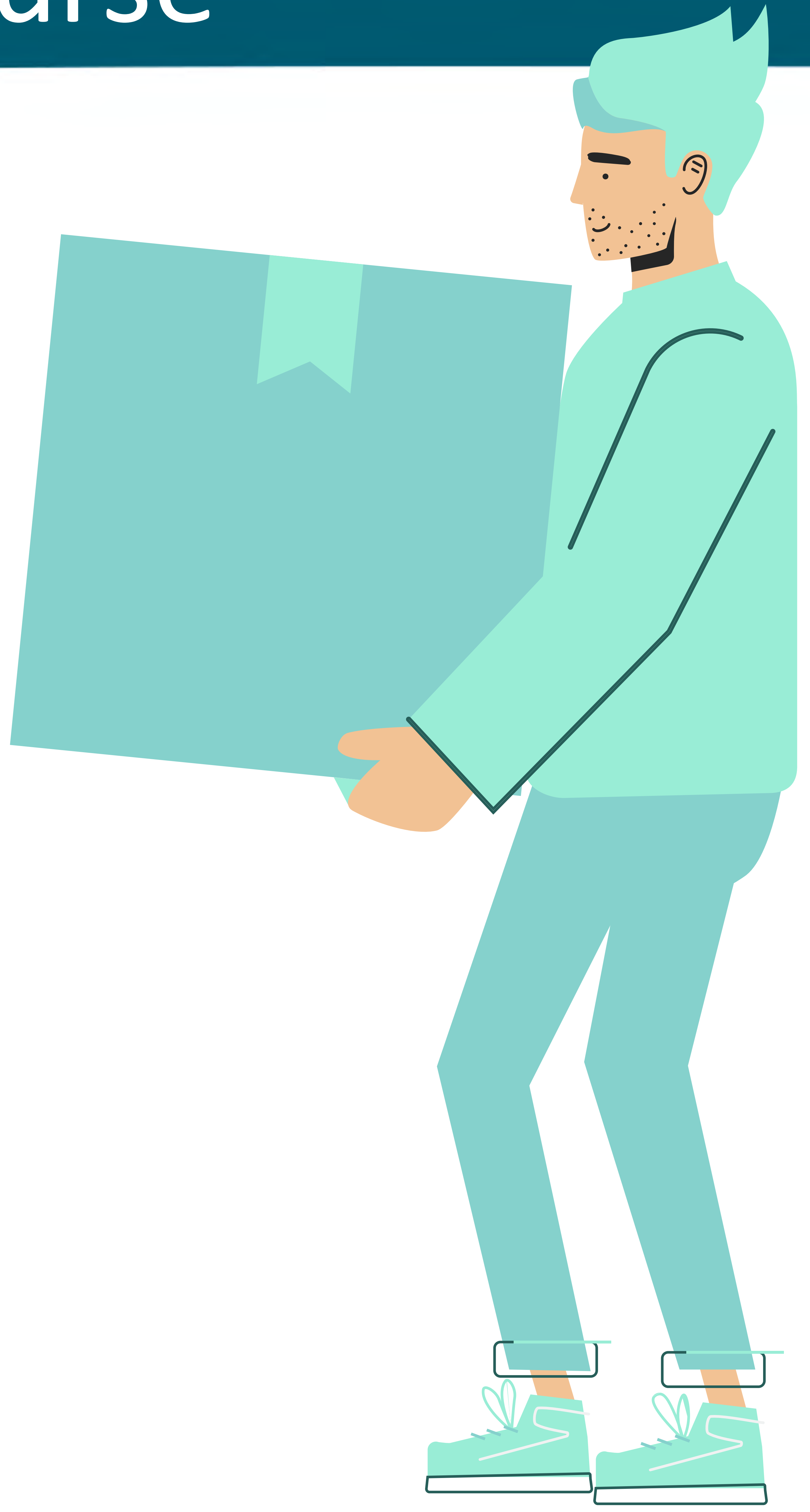
Key Takeaways from This Course

PART 5

You will learn about a powerful new feature of the Oracle ADB 26ai

SELECT AI

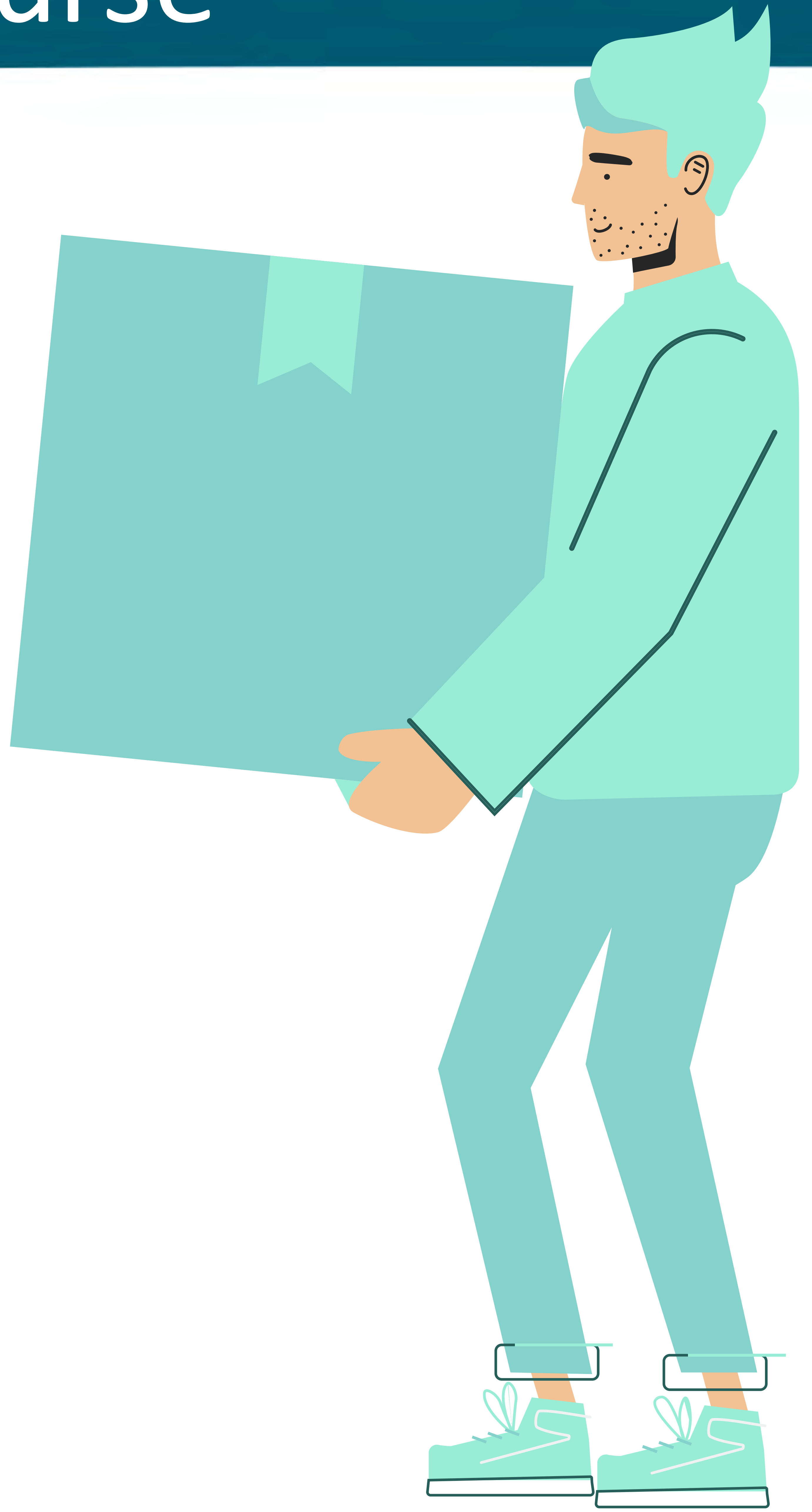
This allows you to execute queries in your database in Natural Language



Key Takeaways from This Course

PART 6

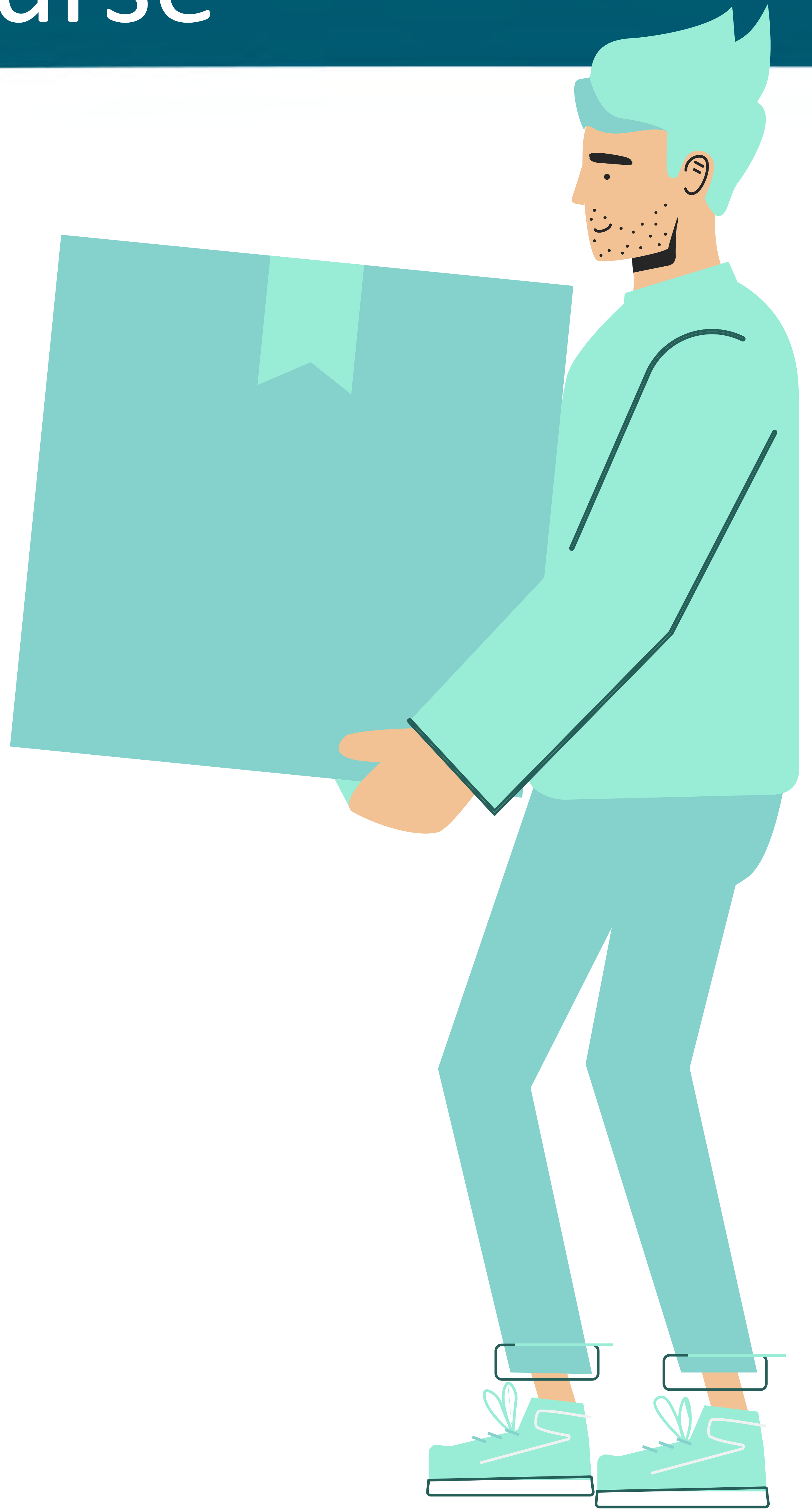
You will learn all the steps necessary to enable SELECT AI within your own database using your preferred AI provider



Key Takeaways from This Course

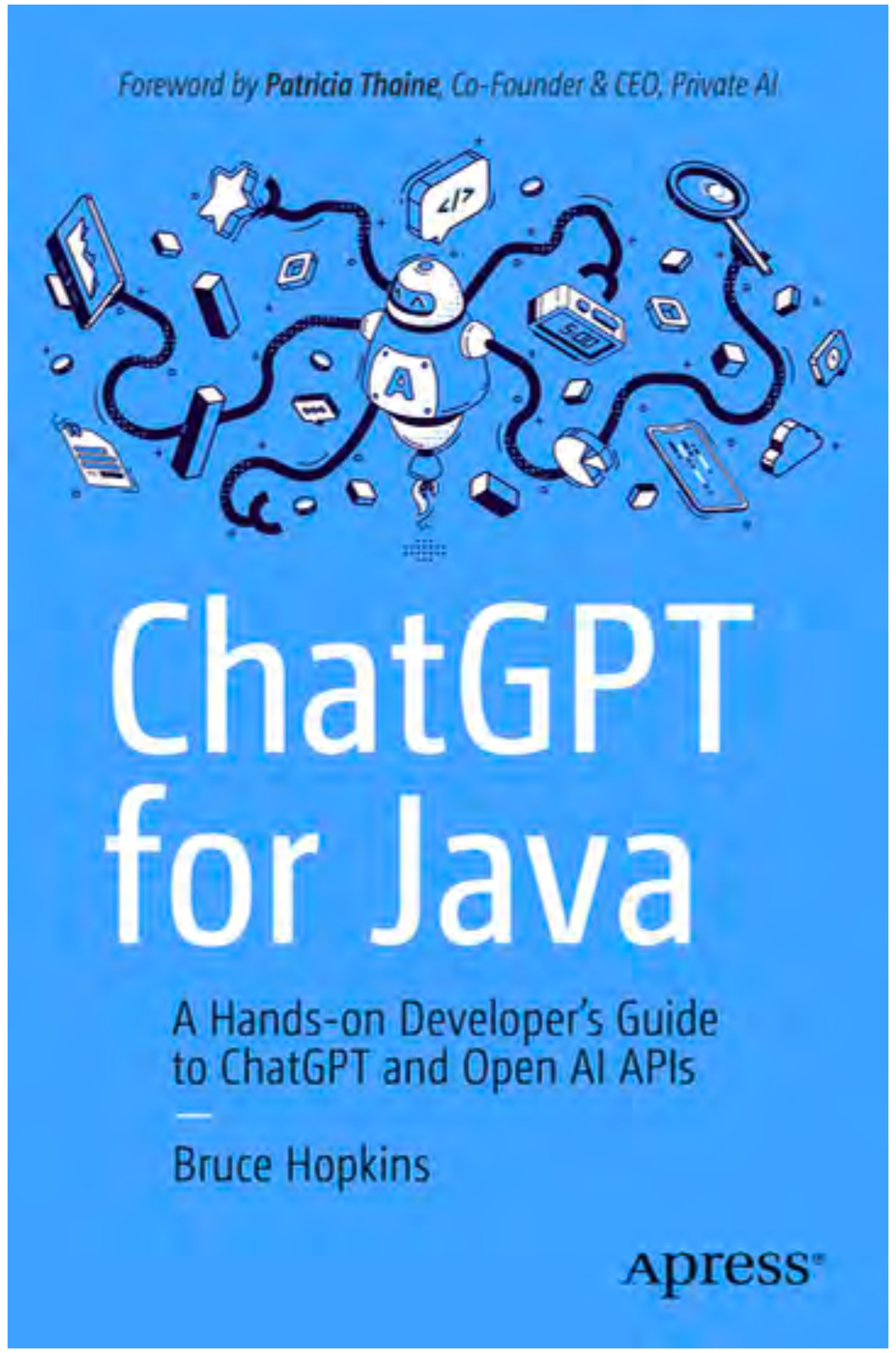
PART 7

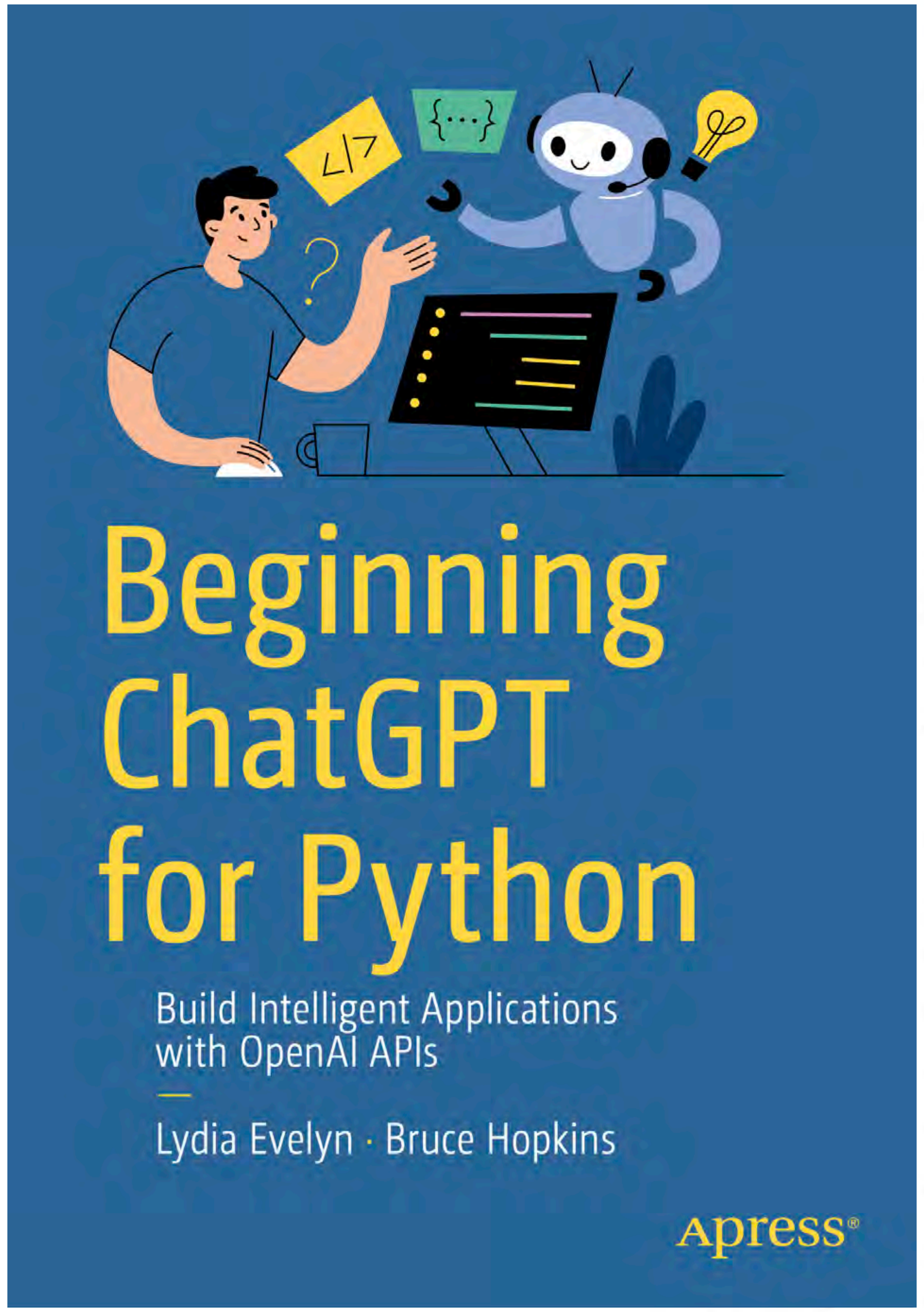
You will learn all use MCP to have a natural language conversation with your database



Contact Info

- Twitter/ X
  @JavaChatGPT
- Bluesky
  @javachatgpt.bsky.social





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- Upcoming Training on **Apr 14**

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Build AI Agents from Existing APIs with MCP

Published by Pearson

Intermediate

Transform Your REST APIs into Intelligent Agents

What you'll learn   Is this live event for you?   Schedule

- Learn the best practices on how to AI-enable your personal or public HTTP REST APIs to become AI Agents using MCP.
- Get hands-on experience using MCP to have Q&A with custom data sources and data repositories. Have a Natural Language conversation with your automobile using its native OBD-II interface.
- Create AI Agents with off-the-shelf hardware to convert dumb appliances into smart appliances for your home.

In this live course, you will see how to take your existing APIs, data sources, and services and learn how to make them into AI Agents that conform to the Model Context Protocol (MCP) standard. By adopting the MCP protocol, your end-users can easily interact with your services using a standard MCP Client application. Practical examples and live demonstrations in this course will also show how to use MCP standard in combination with the automotive OBD-II standard to allow end-users to ask questions and get answers about the status of their vehicles.

Participants will also learn how to create custom MCP Servers that work with off-the-shelf hardware units to convert standard home appliances to become AI-enabled. A live demonstration will show how to use a MCP client to ask a device in the home about its power consumption.

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Bruce Hopkins

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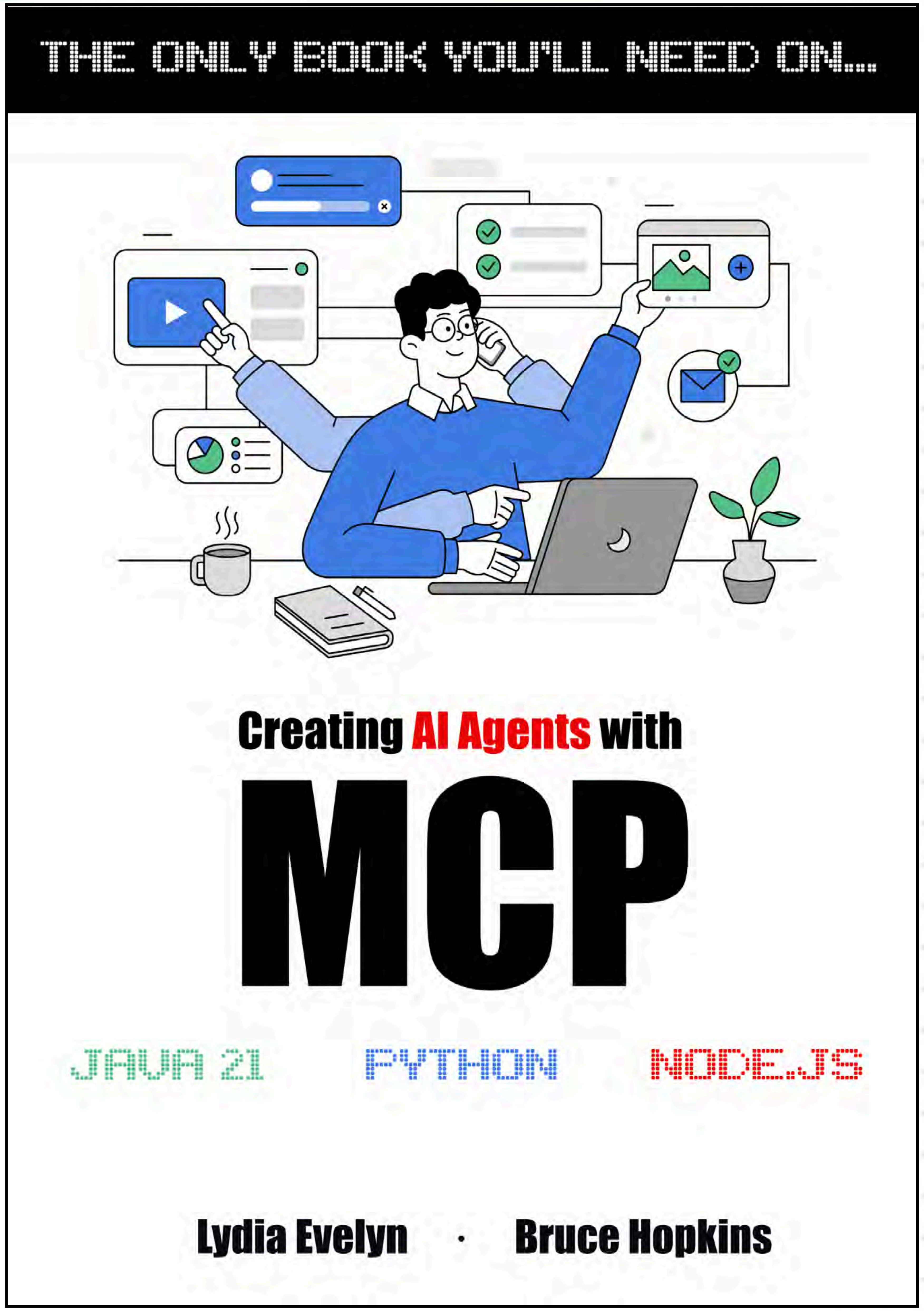

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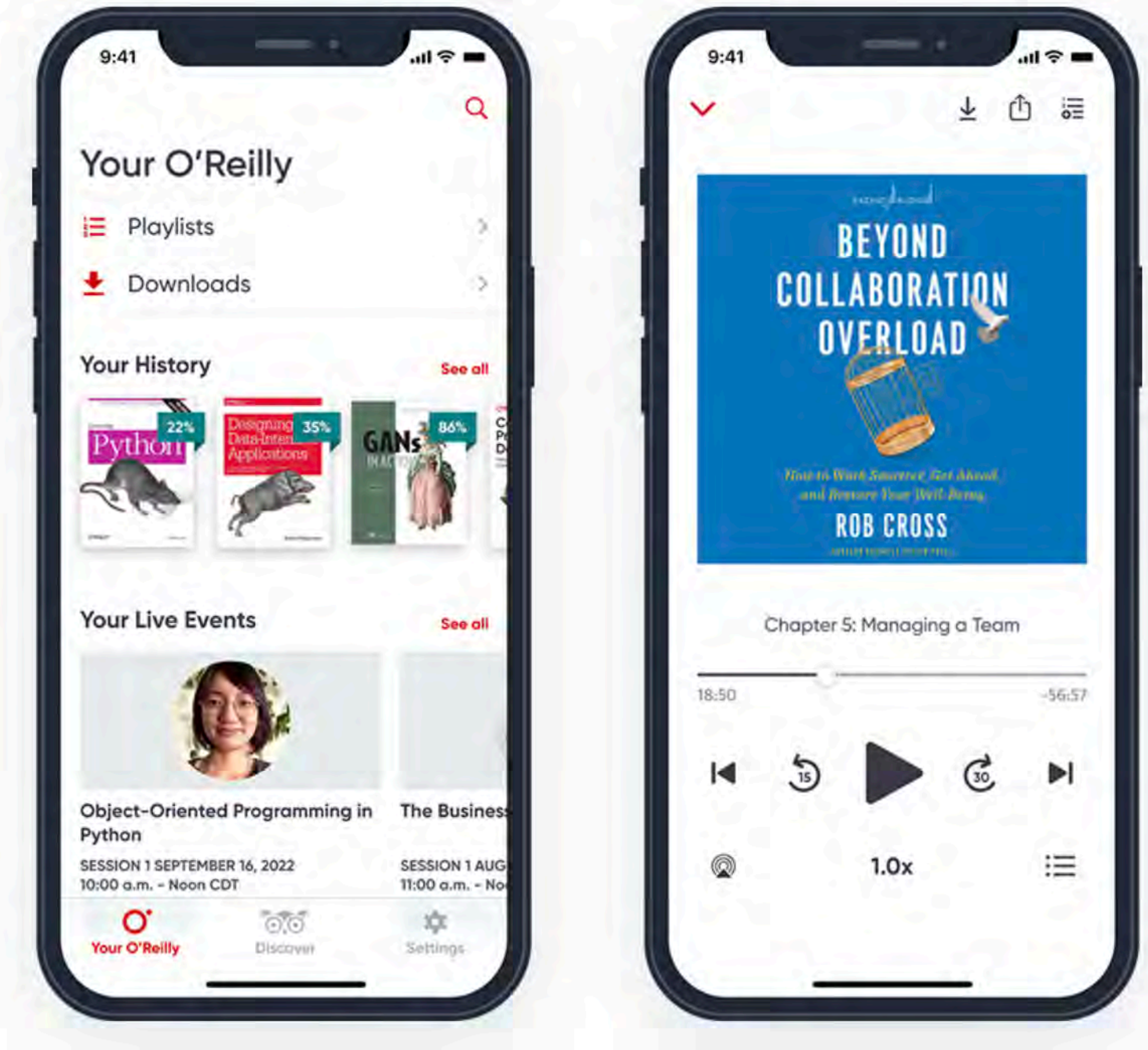

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